

DevOps Project

Problem Statement:

Create an end-to-end CI/CD pipeline in AWS platform using Jenkins as the orchestration tool, GitHub as scm, maven as the build tool, deploy in a docker instance and create a docker image, store the docker image in ECR, Kubernetes deployment using ECR image. Build sample java web app using maven.

CI/CD Pipeline Setup Approach:

Requirements:

- Git (Local Version Control System)
- GitHub (Distributed Version Control System)
- Jenkins (Continuous Integration Tool)
- Maven (Build Tool)
- Docker (Containerization)
- Kubernetes (Container Orchestration)

Step 1: CI/CD with GitHub, Jenkins, Maven & Tomcat

1. Install & Configure Jenkins
2. Install & Configure Maven and Git
3. Set Up Apache Tomcat Server
4. Integrate Jenkins with GitHub, Maven, and Tomcat
5. Create CI/CD Jobs in Jenkins
 - CI Job: Build code using Maven
 - CD Job: Deploy to Tomcat
6. Test Deployment on Tomcat

Step 2: CI/CD with GitHub, Jenkins, Maven & Docker

1. Set Up Docker Environment
2. Create Docker Image and Run Container
3. Integrate Docker with Jenkins

4. Create Jenkins CI/CD Job

- CI Job: Build code with Maven
- CD Job: Package into Docker image and deploy to container

Step 3: CI/CD with GitHub, Jenkins, Maven & Kubernetes

1. Set Up Kubernetes Cluster (e.g., EKS)

2. Write Kubernetes Manifest Files

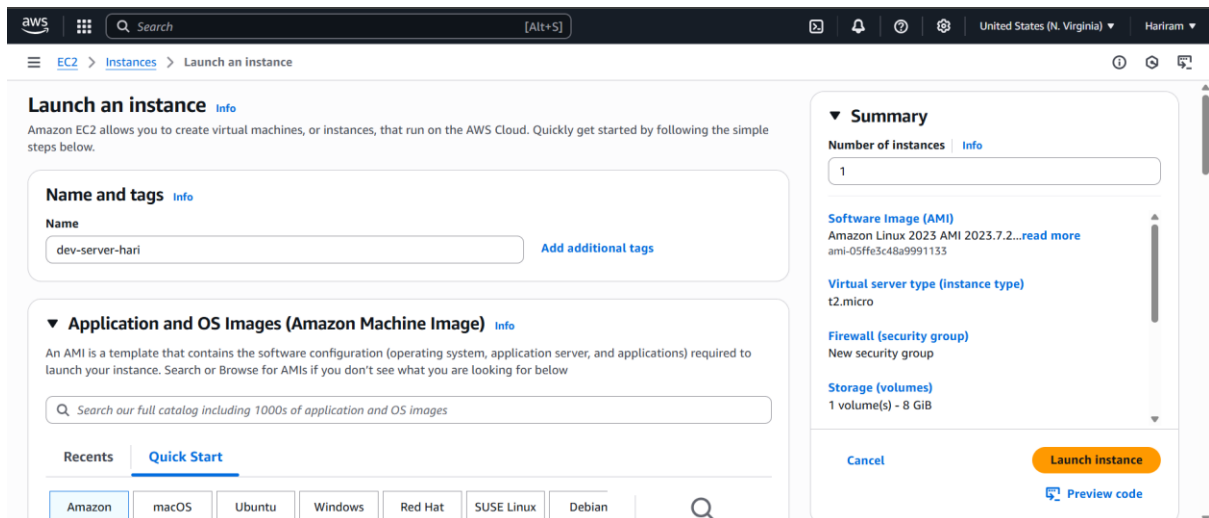
- Deployment
- Service
- Pod specifications

3. Create Jenkins CI/CD Job

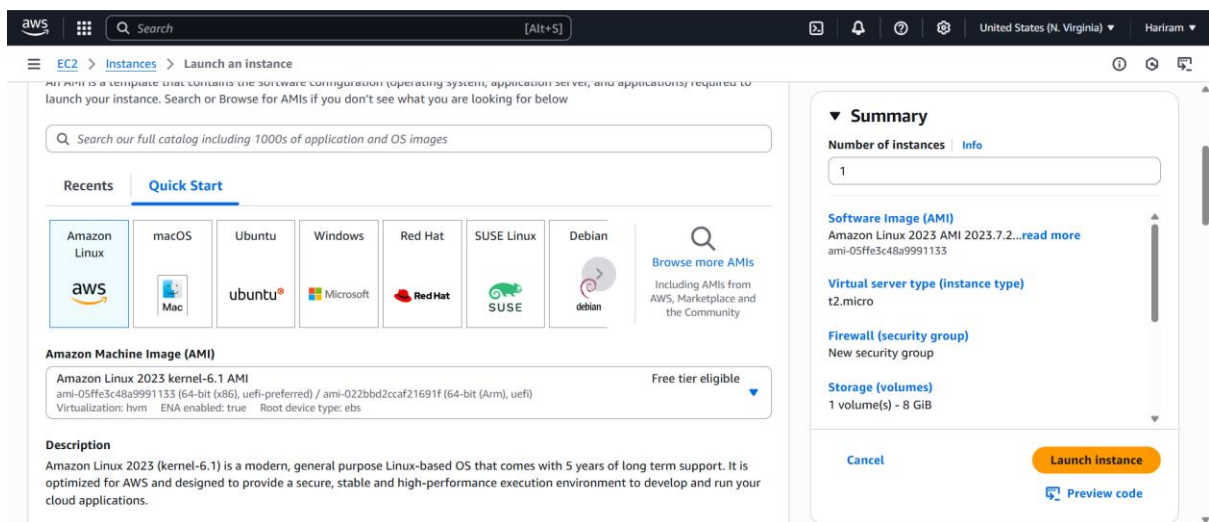
- CI Job: Build code with Maven
- CD Job: Deploy to Kubernetes using kubectl

Step 4: Final Deployment & Validation

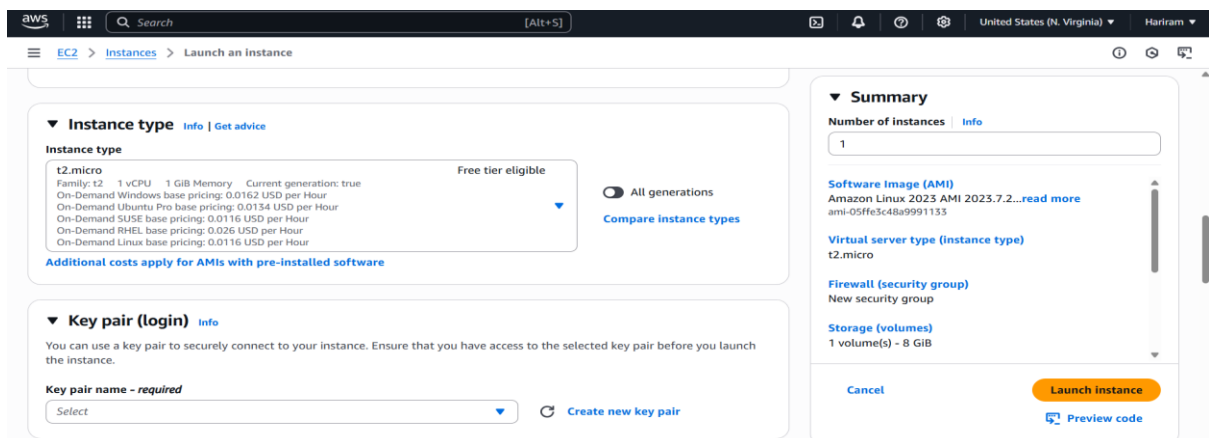
1. Deploy Artifacts to Kubernetes
2. Ensure Docker and Kubernetes Artifacts Contain Required Code
3. Trigger Jenkins Build
4. Verify Pod Creation in Kubernetes
5. Access Application via Kubernetes Service IP
 - Copy service IP
 - Paste in browser to validate output



Creating Dev-server instance .



Selected amazon linux.



T2.micro is selected.

Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

☐ All generations [Compare instance types](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

aws_Ltimindtree [Create new key pair](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.7.2...[read more](#)

ami-05ffe3c48a9991133

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Existing key pair is selected.

Network settings [Info](#)

VPC - *required* [Info](#)

vpc-0c1e9f84aa49d9d9e (default) [Create new VPC](#)

Subnet [Info](#)

subnet-00608da34c56e569d [Create new subnet](#)

VPC: vpc-0c1e9f84aa49d9d9e Owner: 163083073206 Availability Zone: us-east-1a Zone type: Availability Zone IP addresses available: 4091 CIDR: 172.31.16.0/20

Auto-assign public IP [Info](#)

Enable

[Additional charges apply when outside of free tier allowance](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

launch-wizard-1 sg-0cd9a2b14bb27dd1a [Compare security group rules](#)

VPC: vpc-0c1e9f84aa49d9d9e

Security groups that you add or remove here will be added to or removed from all your network interfaces.

[Advanced network configuration](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.7.2...[read more](#)

ami-05ffe3c48a9991133

Virtual server type (instance type)

t2.micro

Firewall (security group)

launch-wizard-1

Storage (volumes)

1 volume(s) - 8 GiB

[Free tier: In your first year of opening an AWS account, you get 750 hours per month of](#)

[Cancel](#) [Launch instance](#) [Preview code](#)

Existing security group is selected with required ports allowed.

Configure storage [Info](#) Advanced

1x 8 GiB gp3 [Root volume, 3000 IOPS, Not encrypted](#)

[Free tier eligible customers can get up to 30 GB of EBS General Purpose \(SSD\) or Magnetic storage](#)

[Add new volume](#)

[Click refresh to view backup information](#)

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File Systems [Edit](#)

[Advanced details](#) [Info](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.7.2...[read more](#)

ami-05ffe3c48a9991133

Virtual server type (instance type)

t2.micro

Firewall (security group)

launch-wizard-1

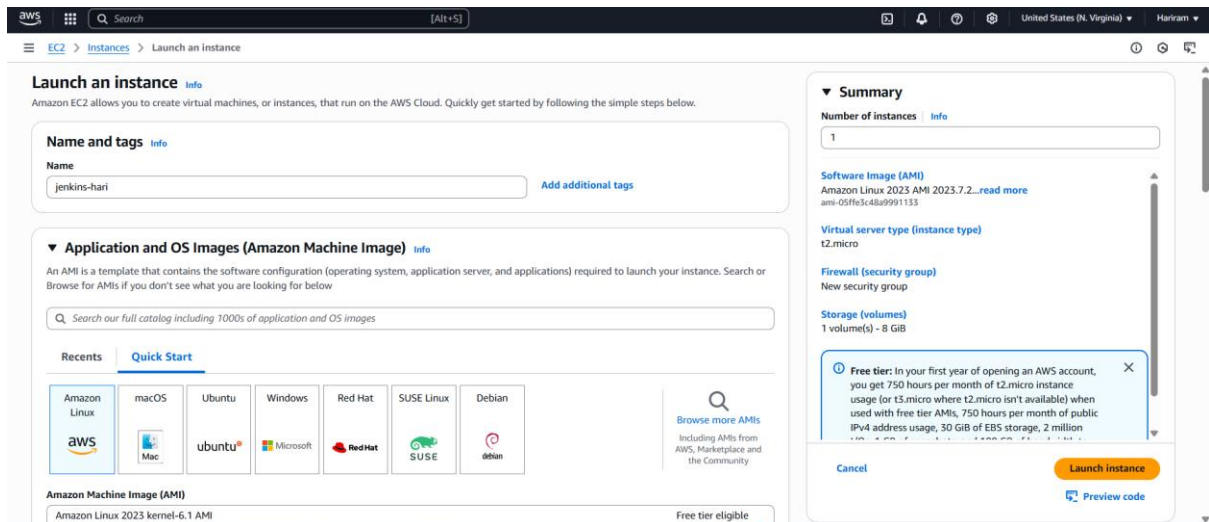
Storage (volumes)

1 volume(s) - 8 GiB

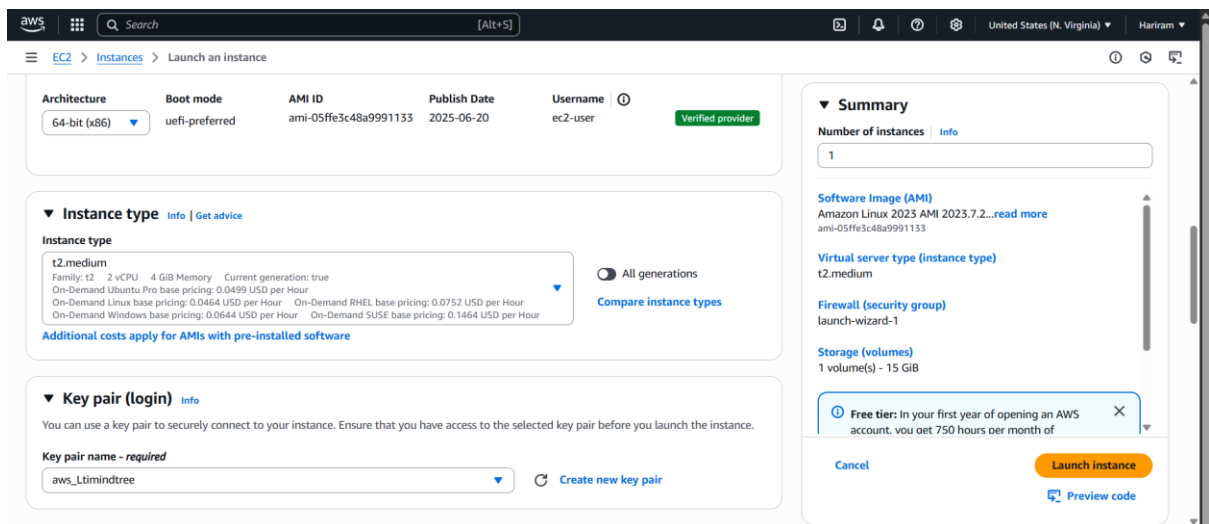
[Free tier: In your first year of opening an AWS account, you get 750 hours per month of](#)

[Cancel](#) [Launch instance](#) [Preview code](#)

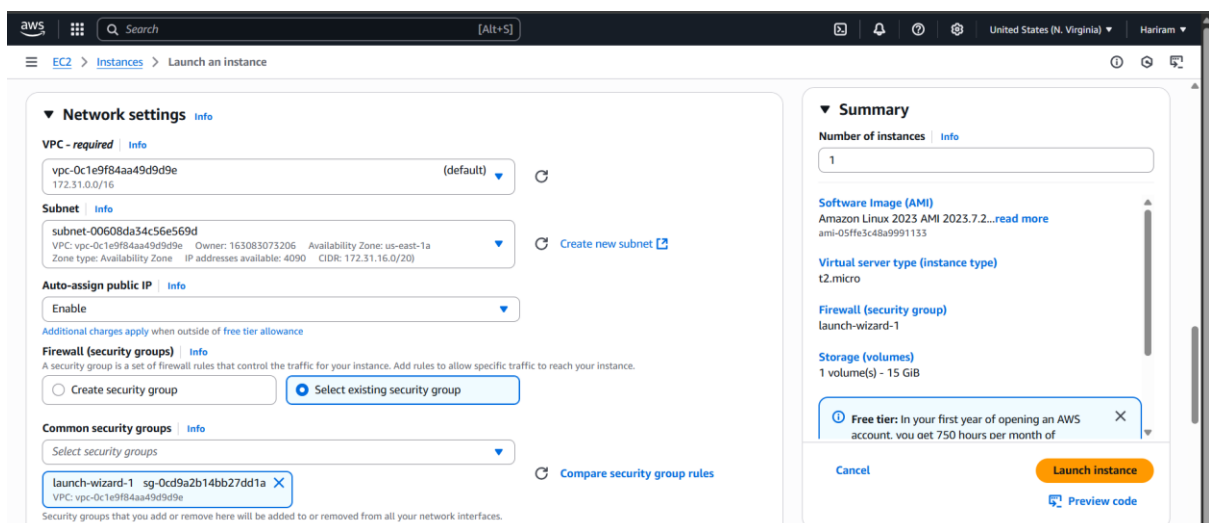
8GB storage and gp3 is selected.



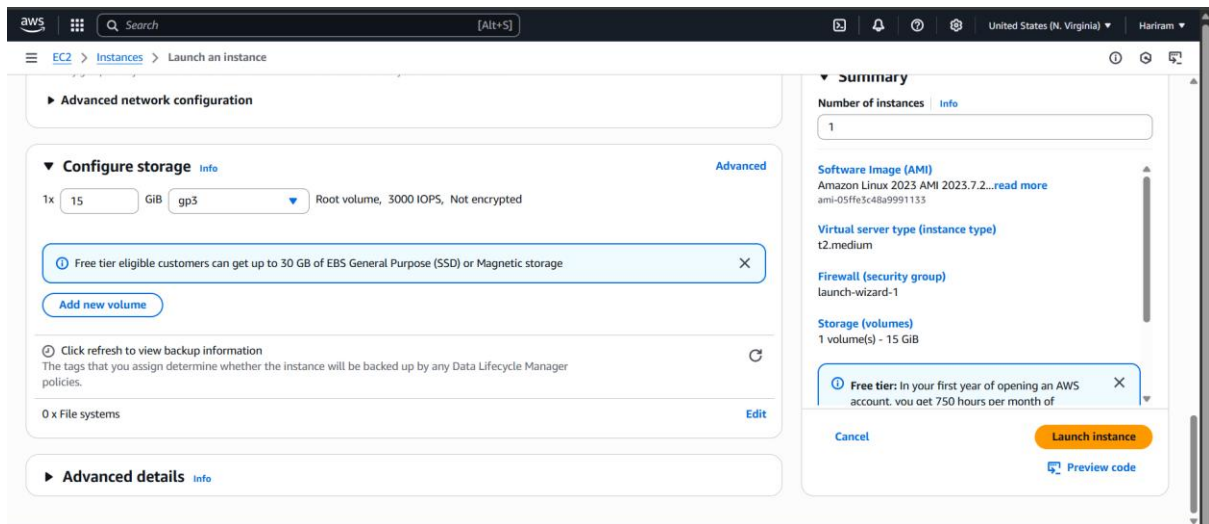
Jenkins server is created with amazon linux.



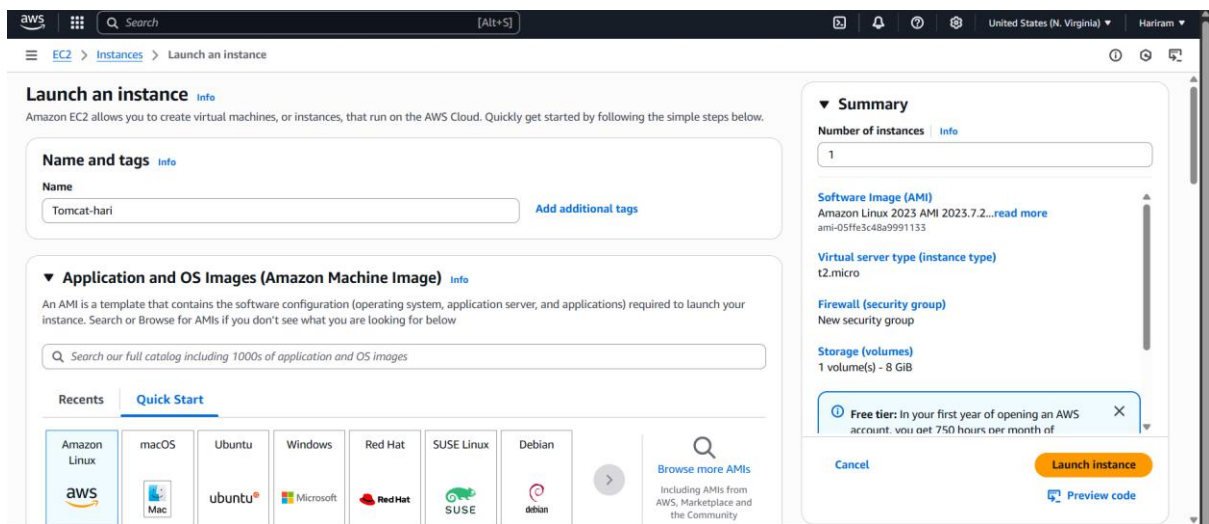
T2.medium is selected as a requirement for Jenkins.



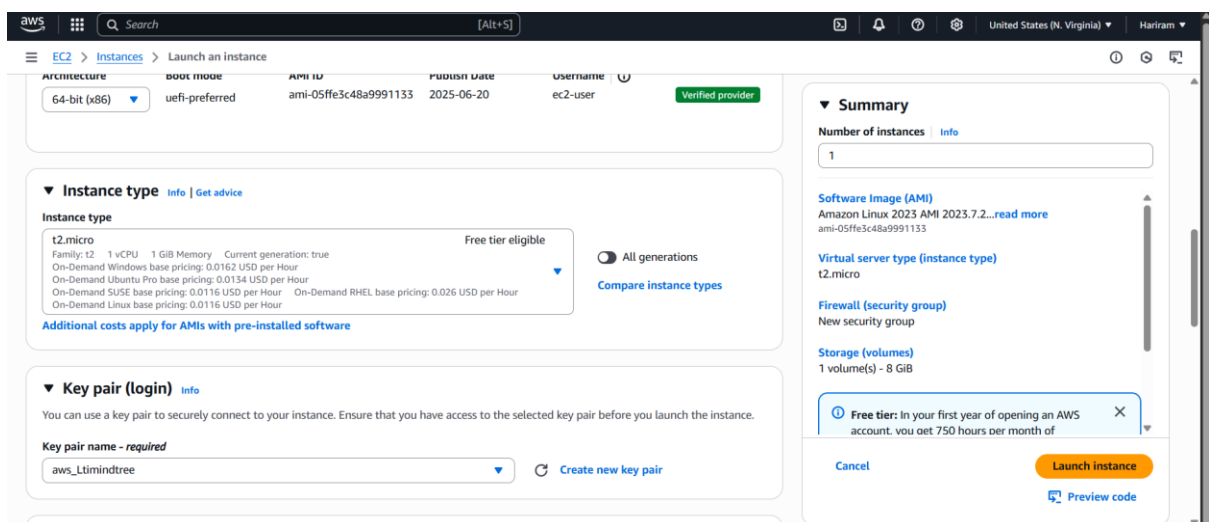
Existing security group is selected.



15GB storage and gp3 is selected.



Tomcat server is created with amazon linux.



T2.micro is selected with existing key apir.

Network settings [Info](#)

VPC - [required](#) | [Info](#)

vpc-0c1e9f84aa49d9d9e (default) [Refresh](#)

Subnet | [Info](#)

subnet-00608da34c56e569d [Refresh](#) [Create new subnet](#)

VPC: vpc-0c1e9f84aa49d9d9e Owner: 163083073206 Availability Zone: us-east-1a
Zone type: Availability Zone IP addresses available: 4089 CIDR: 172.31.16.0/20

Auto-assign public IP | [Info](#)

Enable [Refresh](#)

Additional charges apply when outside of free tier allowance

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups | [Info](#)

Select security groups

launch-wizard-1 sg-0cd9a2b14bb27dd1a [X](#)

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Summary

Number of instances | [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.7.2...[read more](#)
ami-05ffe3c48a9991133

Virtual server type (instance type)

t2.micro

Firewall (security group)

launch-wizard-1

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of

[Cancel](#) [Launch instance](#) [Preview code](#)

Existing security group is selected.

Advanced network configuration

Configure storage [Info](#) [Advanced](#)

1x 10 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier: eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

[Click refresh to view backup information](#)

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems [Edit](#)

Advanced details [Info](#)

Summary

Number of instances | [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.7.2...[read more](#)
ami-05ffe3c48a9991133

Virtual server type (instance type)

t2.micro

Firewall (security group)

launch-wizard-1

Storage (volumes)

1 volume(s) - 10 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of

[Cancel](#) [Launch instance](#) [Preview code](#)

10GB storage and gp3 is selected.

EC2 [Instances](#)

Dashboard
EC2 Global View [Info](#)
Events

Instances

Instances

Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations

Images

AMIs
AMI Catalog

Elastic Block Store

Volumes

Instances (3) [Info](#)

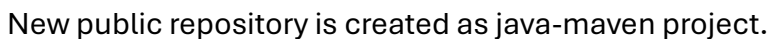
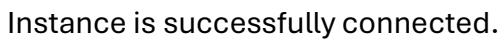
Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

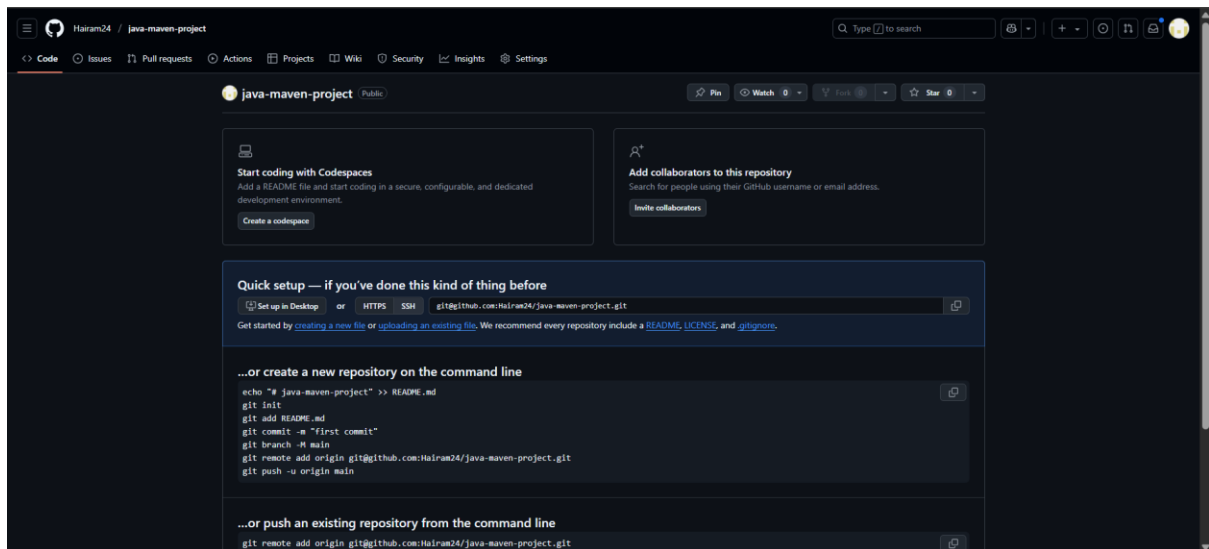
Find Instance by attribute or tag (case-sensitive) All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☐	jenkins-hari	i-01cba33fc82dd9fe1	Running	t2.medium	Initializing	View alarms +	us-east-1a	ec2-3-90-
☐	Tomcat-hari	i-00551ac83ac59787b	Running	t2.micro	Initializing	View alarms +	us-east-1a	ec2-18-2-
☐	dev-server-hari	i-030587200d5d93e2	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-52-9-

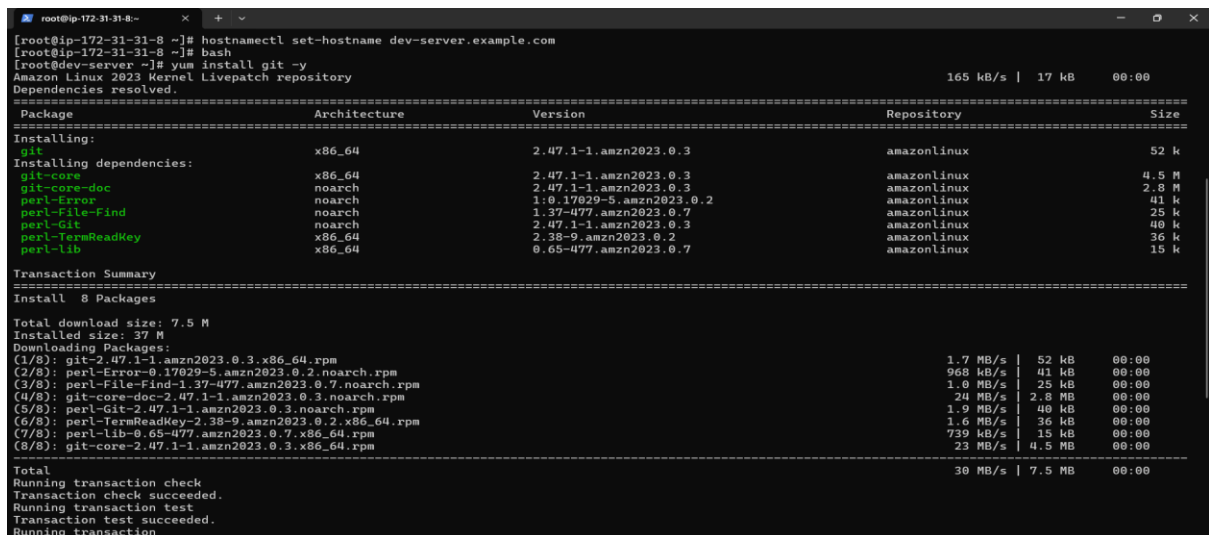
Select an instance

All the three servers are ready to run.

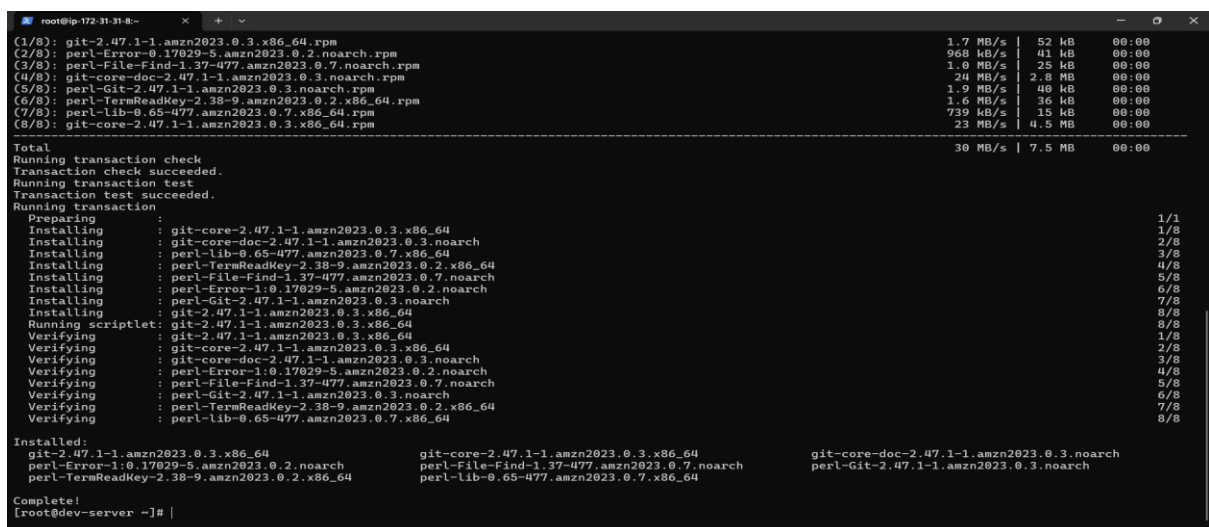




Public repository is created successfully.



Hostname is set and git package is installed.



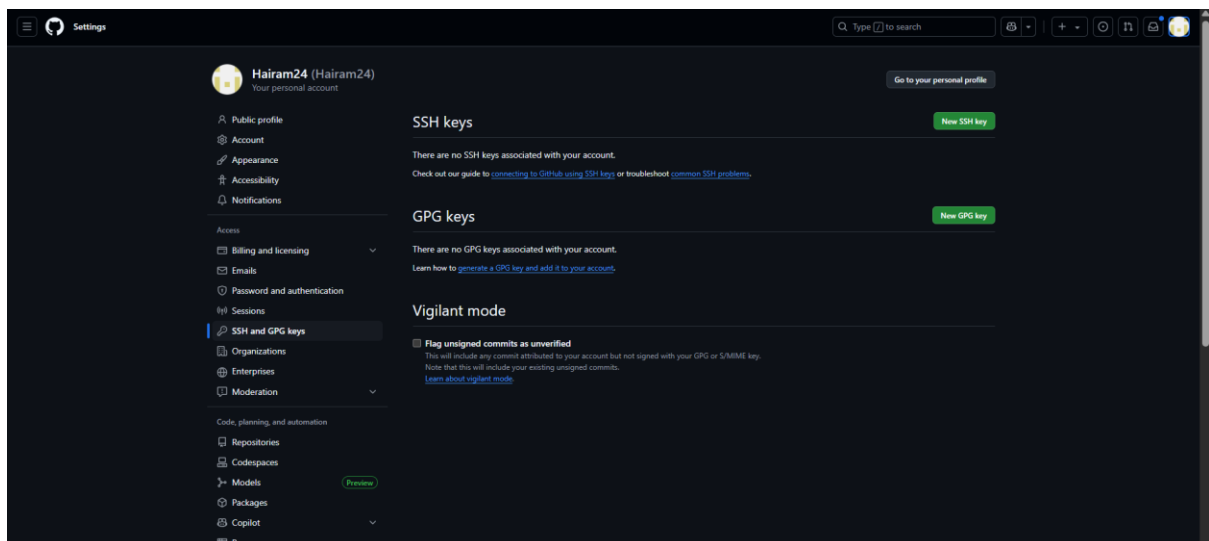
Git package is successfully installed.

```
[root@dev-server ~]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:60RtT/+Vmve741l4RKlbu7xYzinz8dZS2Kfa64vysR8 root@dev-server.example.com
The key's randomart image is:
+----[RSA 3072]-----+
|
|      .
|     S o . . +
|    . o o . +o+
|   o o .EB+
|  o . B@+@
| . o=B%+X
+----[SHA256]-----+
[root@dev-server ~]#
```

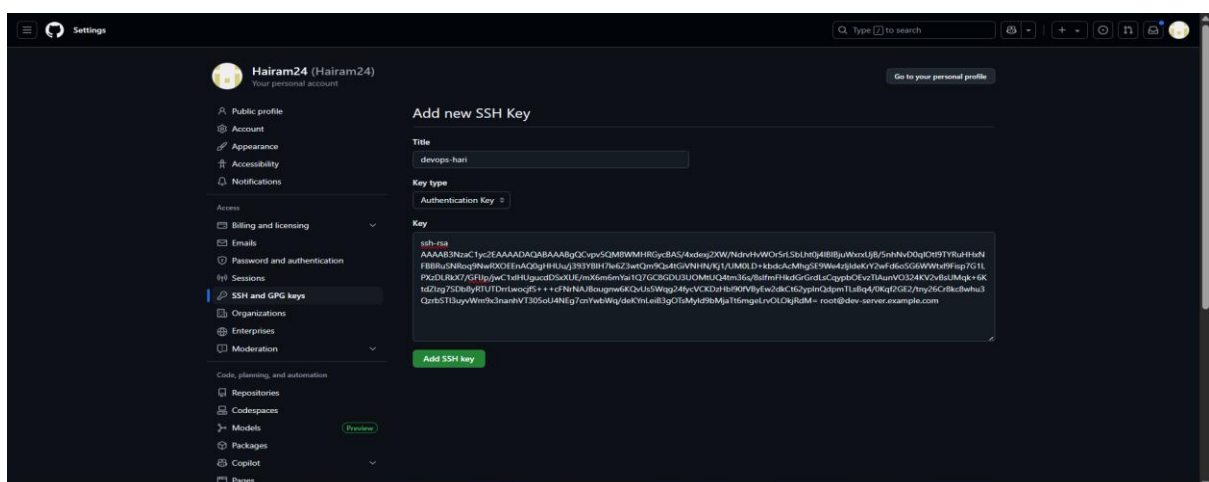
SSH key is generated.

```
[root@dev-server ~]# cd .ssh/
[root@dev-server .ssh]# cat id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGCvpy5QM8WMHRGycBAS/4xdexj2XW/NdrvHVW0r5rL5bLht04I8IBjuWxrxUjB/5nhNvD0qI0tI9TYRuHHxNFB8BRuSNRoq9NwRXOEEaAQ0gHHUu/j393YB
lH7Le6Z3wtQm90s4tGiVNHNN/Kj1/UM0LD+kbdcAcMhgSE9We4zljIdeKrY2Wfd6oSG6WwtXl9Fisp7G1LPzDLRkX7/GFLJp/jwC1xLHugucdD5xXUE/mX6m6mYail07GC8GDU3U0MtU04tm36s/8s1fmFhk
dGrGrdLsCqypb0EvzTLAunV0324KV2vBsUWqk+6KtdZIZg7Sdb8yRTUTDrrLwocjfs+++cFNrNAJBougnw6KQvUs5Wqg24fycVCKDzHbL98fVBYEw2dkCt62ypLnQdpnTLsBq4/0Kqf2GE2/tny26Cr8kc8w
hu3QzrBSTI3uyvWm9x3nanhVT305oU4NEg7cnYwbWq/deKYnLeIB3g0TsMyId9bMjaTt6mgelrv0L0kjRdM= root@dev-server.example.com
[root@dev-server .ssh]#
```

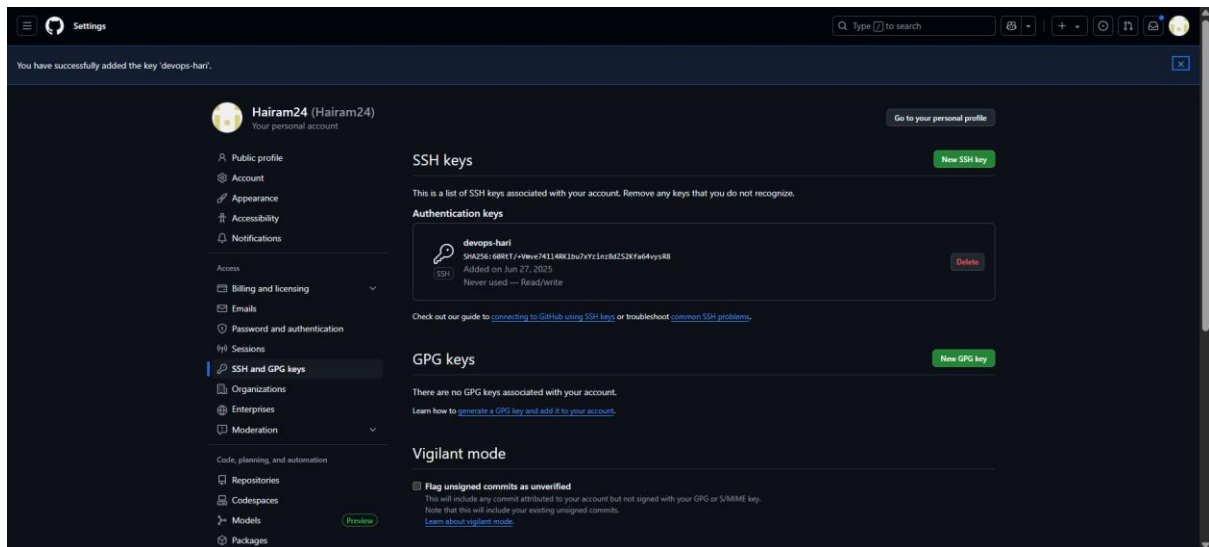
Public key is copied.



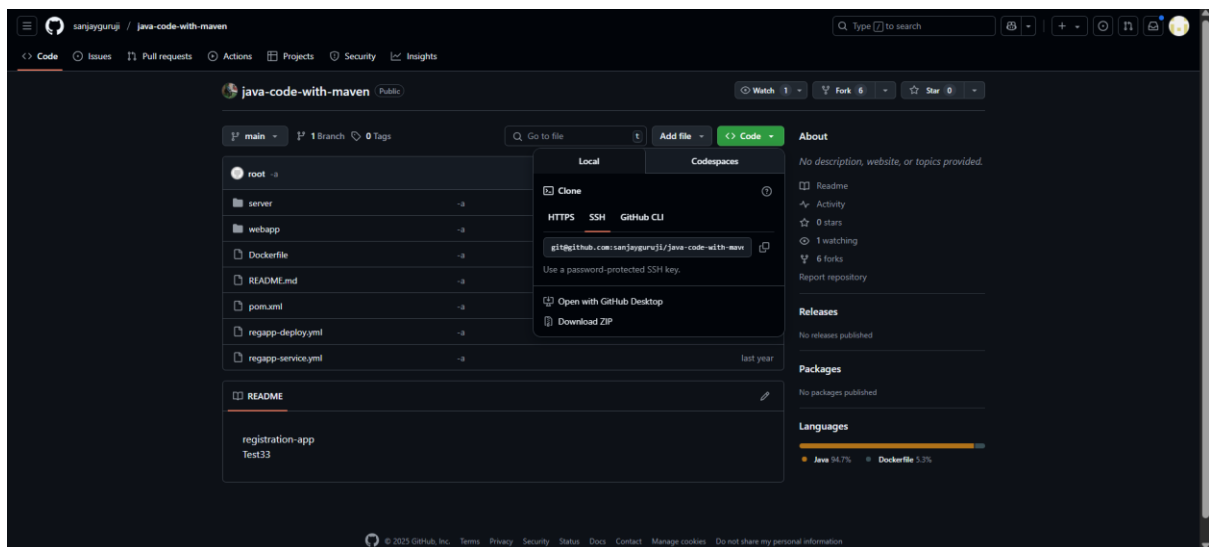
To establish connection to git through ssh key.



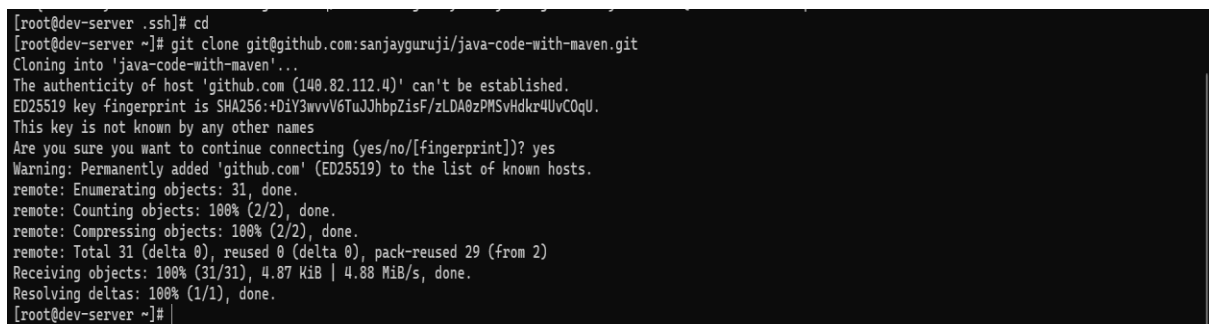
SSH key is added.



SSH key added successfully.



Copied the repo ssh link to clone it.



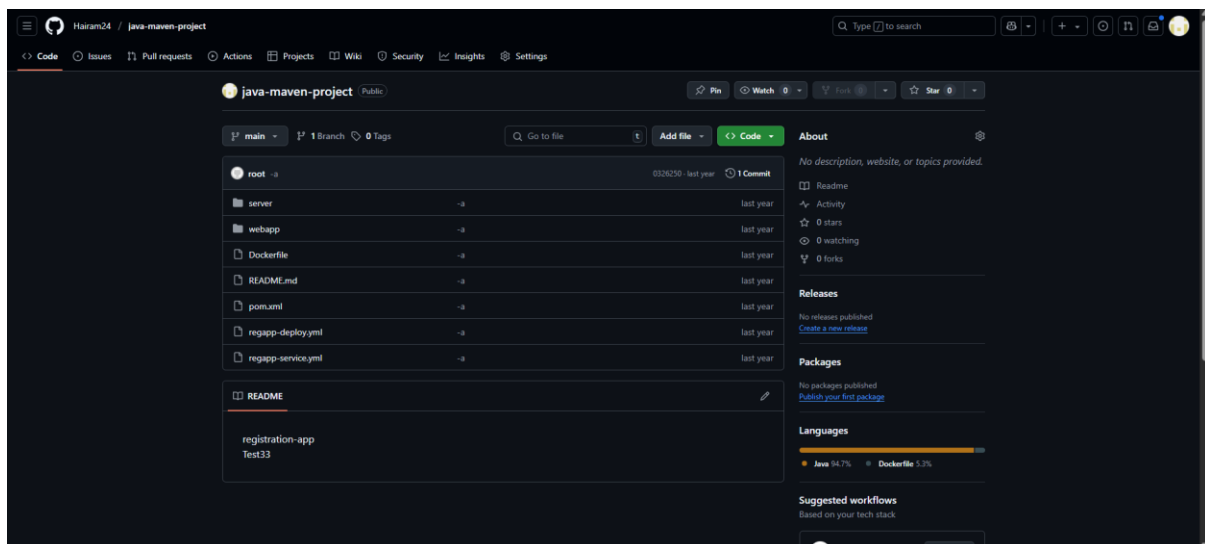
Git repository is cloned successfully.


```
root@ip-172-31-31-8:~/java-c x + v
[root@dev-server ~]# ll
total 0
drwxr-xr-x. 5 root root 147 Jun 27 04:34 java-code-with-maven
[root@dev-server java-code-with-maven]# cd java-code-with-maven
[root@dev-server java-code-with-maven]# git init
Reinitialized existing Git repository in /root/java-code-with-maven/.git/
[root@dev-server java-code-with-maven]# git add .
[root@dev-server java-code-with-maven]# git commit -m "first commit"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
[root@dev-server java-code-with-maven]# git branch -M main
[root@dev-server java-code-with-maven]# git remote add origin git@github.com:Hairam24/java-maven-project.git
error: remote origin already exists.
[root@dev-server java-code-with-maven]# git push -u origin main
ERROR: Permission to sanjaygururji/java-code-with-maven.git denied to Hairam24.
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
[root@dev-server java-code-with-maven]# git remote remove origin
[root@dev-server java-code-with-maven]# git remote add origin git@github.com:Hairam24/java-maven-project.git
[root@dev-server java-code-with-maven]# git push -u origin main
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 2 threads
Compressing objects: 100% (15/15), done.
Writing objects: 100% (31/31), 4.87 MiB | 4.88 MiB/s, done.
Total 31 (delta 1), reused 31 (delta 1), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To github.com:Hairam24/java-maven-project.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
[root@dev-server java-code-with-maven]#
```

Cloned repo is copied to the java-maven project repo.



Successfully add to the project repository.

```
root@ip-172-31-23-209:~ x + v
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\10835505\OneDrive - LTIMindtree\Desktop> ssh -i "aws_ltimindtree.pem" ec2-user@ec2-54-164-19-42.compute-1.amazonaws.com
The authenticity of host 'ec2-54-164-19-42.compute-1.amazonaws.com (54.164.19.42)' can't be established.
ED25519 key fingerprint is SHA256:cls3031siyJ5B23PX+Q8oAY1lK10CU393ZgMO+ftlt8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-164-19-42.compute-1.amazonaws.com' (ED25519) to the list of known hosts.

  ____      _
 / ___|  __| | | |
| |___| |_| | |_| |
 \___|__| |_____|_|

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

  ____      _
 / ___|  __| | | |
| |___| |_| | |_| |
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Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-25-209 ~]$ sudo su -
[root@ip-172-31-25-209 ~]# hostnamectl set-hostname jenkins.example.com
[root@ip-172-31-25-209 ~]# bash
[root@jenkins ~]#
```

Jenkins server is connected and hostname is changed.


```
root@ip-172-31-25-209:~  
[root@jenkins ~]# wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/redhat-stable/jenkins.repo  
--2025-06-27 04:45:34-- https://pkg.jenkins.io/redhat-stable/jenkins.repo  
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.34.133, 2a84:de42:77:645  
Connecting to pkg.jenkins.io (pkg.jenkins.io)[146.75.34.133]:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 85  
Saving to: '/etc/yum.repos.d/jenkins.repo'  
  
/etc/yum.repos.d/jenkins.repo      100%[=====]      85  --.-KB/s   in 0s  
  
2025-06-27 04:45:34 (3.44 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [85/85]  
  
[root@jenkins ~]# rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key  
[root@jenkins ~]# yum upgrade  
Jenkins-stable  
Dependencies resolved.  
Nothing to do.  
Complete!  
[root@jenkins ~]# |
```

Jenkins packages installed.

```
root@ip-172-31-25-209:~  
Verifying      : libICE-1.1.1-3.amzn2023.0.1.x86_64      18/35  
Verifying      : libSM-1.2.4-3.amzn2023.0.1.x86_64      19/35  
Verifying      : libX11-1.8.10-2.amzn2023.0.1.x86_64     20/35  
Verifying      : libX11-common-1.8.10-2.amzn2023.0.1.noarch 21/35  
Verifying      : libXau-1.0.11-6.amzn2023.0.1.x86_64     22/35  
Verifying      : libXext-1.3.6-1.amzn2023.0.1.x86_64     23/35  
Verifying      : libXi-1.8.2-1.amzn2023.0.1.x86_64       24/35  
Verifying      : libXinerama-1.1.5-6.amzn2023.0.1.x86_64  25/35  
Verifying      : libXrandr-1.5.4-3.amzn2023.0.1.x86_64   26/35  
Verifying      : libXrender-0.9.11-6.amzn2023.0.1.x86_64 27/35  
Verifying      : libXt-1.3.0-3.amzn2023.0.1.x86_64       28/35  
Verifying      : libXtst-1.2.5-1.amzn2023.0.1.x86_64     29/35  
Verifying      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64   30/35  
Verifying      : libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64 31/35  
Verifying      : libpng-2.1.6-37-10.amzn2023.0.6.x86_64  32/35  
Verifying      : libxcb-1.17.0-1.amzn2023.0.1.x86_64     33/35  
Verifying      : pixman-0.43.4-1.amzn2023.0.4.x86_64     34/35  
Verifying      : xml-common-0.6.3-56.amzn2023.0.2.noarch  35/35  
  
Installed:  
alsa-lib-1.2.7-2.1.amzn2023.0.2.x86_64  
dejavu-sans-fonts-2.37-16.amzn2023.0.2.noarch  
dejavu-serif-fonts-2.37-16.amzn2023.0.2.noarch  
fonts-filesystem-1.2.0.5-12.amzn2023.0.2.noarch  
glib-5.2.1-9.amzn2023.0.1.x86_64  
google-noto-sans-vf-fonts-20201206-2.amzn2023.0.2.noarch  
harfbuzz-7.0.0-2.amzn2023.0.2.x86_64  
java-17-amazon-corretto-headless-1.17.0.15+6-1.amzn2023.1.x86_64  
langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch  
libSM-1.2.4-3.amzn2023.0.1.x86_64  
libX11-common-1.8.10-2.amzn2023.0.1.noarch  
libXext-1.3.6-1.amzn2023.0.1.x86_64  
libXinerama-1.1.5-6.amzn2023.0.1.x86_64  
libXrender-0.9.11-6.amzn2023.0.1.x86_64  
libXt-1.3.0-3.amzn2023.0.1.x86_64  
libXtst-1.2.5-1.amzn2023.0.1.x86_64  
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64  
libxcb-1.17.0-1.amzn2023.0.1.x86_64  
pixman-0.43.4-1.amzn2023.0.4.x86_64  
xml-common-0.6.3-56.amzn2023.0.2.noarch  
  
Complete!  
[root@jenkins ~]# |
```

Installed successfully.

```
[root@jenkins ~]# yum install jenkins -y  
Last metadata expiration check: 0:01:50 ago on Fri Jun 27 04:45:59 2025.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
jenkins	noarch	2.504.3-1.1	jenkins	90 M

```
=====
```

Transaction Summary

Install 1 Package

Total download size: 90 M
Installed size: 90 M
Downloading Packages:
jenkins-2.504.3-1.1.noarch.rpm 30 MB/s | 90 MB 00:02

Total 30 MB/s | 90 MB 00:02

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Running scriptlet: jenkins-2.504.3-1.1.noarch 1/1
Installing : jenkins-2.504.3-1.1.noarch 1/1
Running scriptlet: jenkins-2.504.3-1.1.noarch 1/1
Verifying : jenkins-2.504.3-1.1.noarch 1/1

Installed:
jenkins-2.504.3-1.1.noarch

Complete!
[root@jenkins ~]# |

Jenkins is installed successfully.

```
[root@jenkins ~]# systemctl enable Jenkins
Failed to enable unit: Unit file Jenkins.service does not exist.
[root@jenkins ~]# systemctl enable jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
[root@jenkins ~]# systemctl start jenkins
[root@jenkins ~]# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Fri 2025-06-27 04:49:44 UTC; 7s ago
     Main PID: 264444 (java)
       Tasks: 48 (Limit: 4656)
      Memory: 714.9M
         CPU: 17.075s
    CGroup: /system.slice/jenkins.service
            └─264444 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jun 27 04:49:40 jenkins.example.com jenkins[264444]: 63458ca172624c49811286630784a40f
Jun 27 04:49:40 jenkins.example.com jenkins[264444]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Jun 27 04:49:40 jenkins.example.com jenkins[264444]: *****
Jun 27 04:49:40 jenkins.example.com jenkins[264444]: *****
Jun 27 04:49:44 jenkins.example.com jenkins[264444]: 2025-06-27 04:49:44.409+0000 [id=31] INFO jenkins.InitReactorRunner$1onAttained: Comple
Jun 27 04:49:44 jenkins.example.com jenkins[264444]: 2025-06-27 04:49:44.420+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is
Jun 27 04:49:44 jenkins.example.com systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Jun 27 04:49:44 jenkins.example.com jenkins[264444]: 2025-06-27 04:49:44.456+0000 [id=40] INFO h.m.DownloadService$Downloadable#load: Obtained
Jun 27 04:49:44 jenkins.example.com jenkins[264444]: 2025-06-27 04:49:44.457+0000 [id=49] INFO hudson.util.Retrier#start: Performed the action
lines 1-20/20 (END)
```

Jenkins is enabled.

```
root@ip-172-31-25-209:~
[root@jenkins ~]# yum install git -y
Last metadata expiration check: 0:04:23 ago on Fri Jun 27 04:45:59 2025.
Dependencies resolved.
=====
Package                                Architecture      Version                                Repository          Size
=====
Installing:
git                                    x86_64            2.47.1-1.amzn2023.0.3                amazonlinux          52 k
Installing dependencies:
git-core                             x86_64            2.47.1-1.amzn2023.0.3                amazonlinux          4.5 M
git-core-doc                         noarch            2.47.1-1.amzn2023.0.3                amazonlinux          2.8 M
perl-Error                           noarch            1.10.17029-5.amzn2023.0.2            amazonlinux          41 k
perl-File-Find                       noarch            1.37-477.amzn2023.0.7                amazonlinux          25 k
perl-Git                             noarch            2.47.1-1.amzn2023.0.3                amazonlinux          40 k
perl-TermReadKey                     x86_64            2.38-9.amzn2023.0.2                 amazonlinux          36 k
perl-lib                             x86_64            0.65-477.amzn2023.0.7                amazonlinux          15 k
Transaction Summary
=====
Install 8 Packages

Total download size: 7.5 M
Installed size: 37 M
Downloading Packages:
(1/8): git-2.47.1-1.amzn2023.0.3.x86_64.rpm                                1.1 MB/s | 52 kB | 00:00
(2/8): perl-Error-0.17029-5.amzn2023.0.2.noarch.rpm                       1.5 MB/s | 41 kB | 00:00
(3/8): git-core-doc-2.47.1-1.amzn2023.0.3.noarch.rpm                      25 MB/s | 2.8 MB | 00:00
(4/8): perl-File-Find-1.37-477.amzn2023.0.7.noarch.rpm                    672 kB/s | 25 kB | 00:00
(5/8): perl-Git-2.47.1-1.amzn2023.0.3.noarch.rpm                          1.8 MB/s | 40 kB | 00:00
(6/8): perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64.rpm                   1.6 MB/s | 36 kB | 00:00
(7/8): perl-lib-0.65-477.amzn2023.0.7.x86_64.rpm                         628 kB/s | 15 kB | 00:00
(8/8): git-core-2.47.1-1.amzn2023.0.3.x86_64.rpm                         25 MB/s | 4.5 MB | 00:00
Total
-----
35 MB/s | 7.5 MB | 00:00

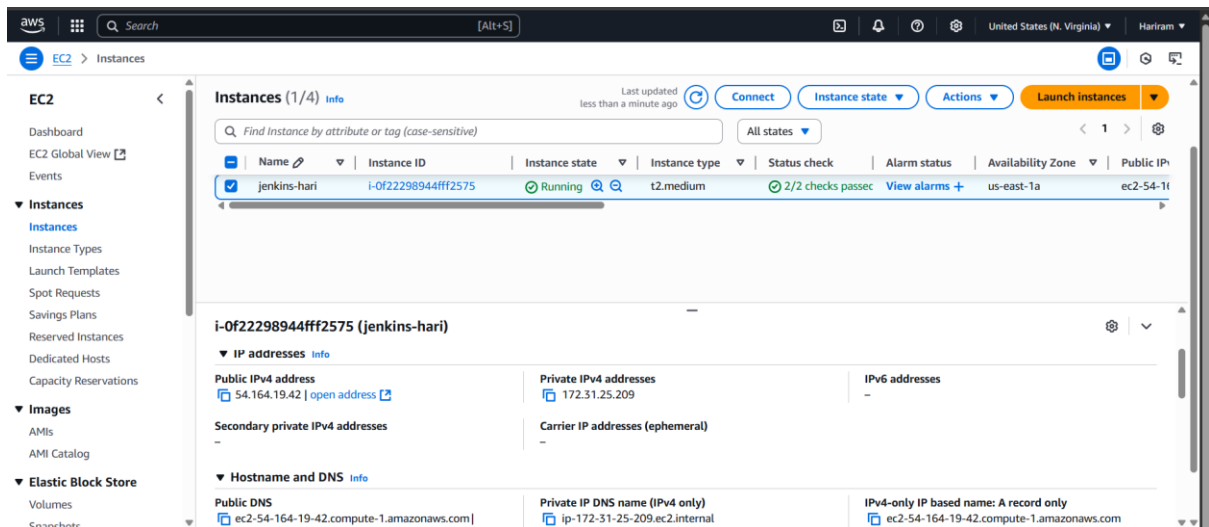
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :
  Installing     : git-core-2.47.1-1.amzn2023.0.3.x86_64
                                     1/1
                                     1/8
```

Git package is installed.

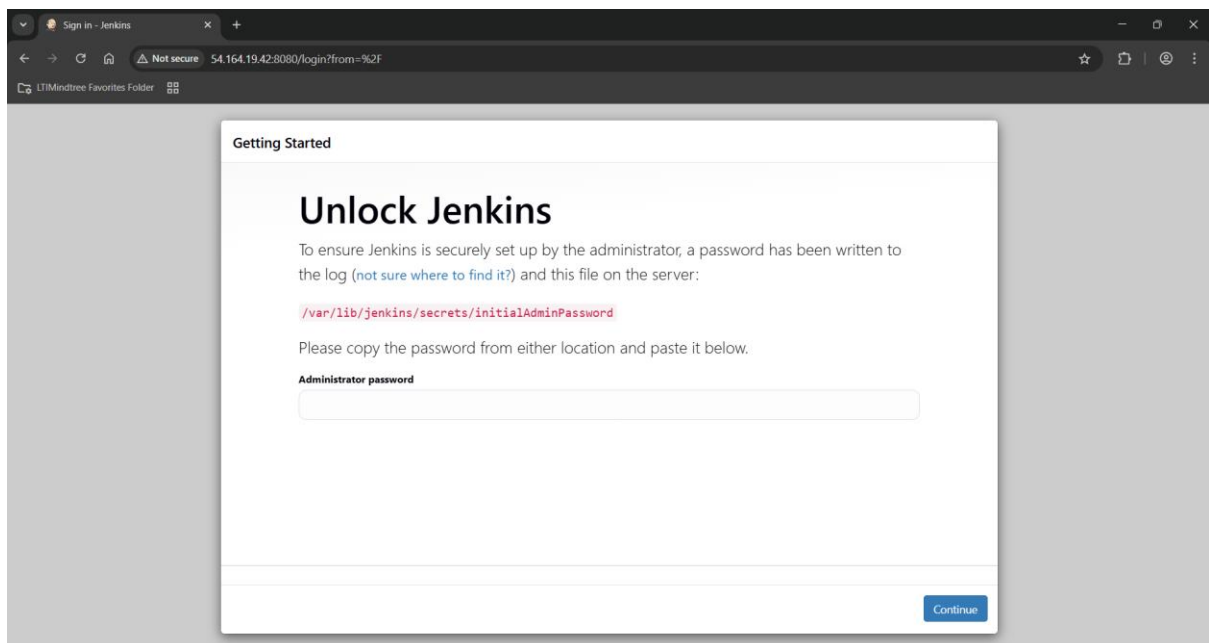
```
root@ip-172-31-25-209:~
[root@jenkins ~]# yum install maven
Last metadata expiration check: 0:05:07 ago on Fri Jun 27 04:45:59 2025.
Dependencies resolved.
=====
Package                                Architecture      Version                                Repository          Size
=====
Installing:
maven                                    noarch            1:3.8.4-3.amzn2023.0.5                amazonlinux          18 k
Installing dependencies:
apache-commons-cli                   noarch            1.5.0-3.amzn2023.0.3                amazonlinux          76 k
apache-commons-codec                 noarch            1.15-6.amzn2023.0.3                amazonlinux          303 k
apache-commons-io                    noarch            1:2.8.0-7.amzn2023.0.5                amazonlinux          203 k
apache-commons-lang3                 noarch            3.12.0-7.amzn2023.0.3                amazonlinux          559 k
atinject                             noarch            1.0.5-3.amzn2023.0.3                amazonlinux          23 k
cdi-api                              noarch            2.0-2-6.amzn2023.0.3                amazonlinux          54 k
google-guice                         noarch            4.2.3-8.amzn2023.0.6                amazonlinux          473 k
guava                                noarch            31.0.1-3.amzn2023.0.6                amazonlinux          2.4 M
httpcomponents-client                noarch            4.5.13-0.amzn2023.0.4                amazonlinux          657 k
httpcomponents-core                  noarch            4.4.13-6.amzn2023.0.3                amazonlinux          632 k
jakarta-annotations                  noarch            1.3.5-13.amzn2023.0.3                amazonlinux          46 k
jansi                                 x86_64            2.4.0-3.amzn2023.0.3                amazonlinux          113 k
java-17-amazon-corretto-devel        noarch            1:17.0.15b6-1.amzn2023.1            amazonlinux          142 k
jcl-over-slf4j                       noarch            1.7.32-3.amzn2023.0.4                amazonlinux          25 k
jsoup                                noarch            1.16.1-0.amzn2023.0.2                amazonlinux          433 k
jsr-305                              noarch            3.0.2-5.amzn2023.0.4                amazonlinux          32 k
maven-amazon-corretto17              noarch            1:8.0-4-3.amzn2023.0.5                amazonlinux          9.4 k
maven-lib                            noarch            1:3.8.4-3.amzn2023.0.5                amazonlinux          1.5 M
maven-resolver                       noarch            1:1.7.3-3.amzn2023.0.4                amazonlinux          557 k
maven-shared-utils                   noarch            3.3.4-4.amzn2023.0.3                amazonlinux          152 k
maven-wagon                          noarch            3.4.2-6.amzn2023.0.4                amazonlinux          113 k
plexus-cipher                        noarch            1.0-3.amzn2023.0.3                amazonlinux          27 k
plexus-classworlds                   noarch            2.6.0-10.amzn2023.0.4                amazonlinux          61 k
plexus-containers-component-annotatio noarch            2.1.0-9.amzn2023.0.4                amazonlinux          19 k
plexus-interpolation                 noarch            1.26-10.amzn2023.0.4                amazonlinux          80 k
plexus-sec-dispatcher                noarch            2.0-3.amzn2023.0.3                amazonlinux          34 k
plexus-utils                          noarch            3.3.0-9.amzn2023.0.4                amazonlinux          254 k
publicsuffix-list                    noarch            20240212-61.amzn2023                amazonlinux          89 k
sisu                                  noarch            1:0.3.4-9.amzn2023.0.4                amazonlinux          510 k
slf4j                                 noarch            1.7.32-3.amzn2023.0.4                amazonlinux          70 k
Transaction Summary
=====
Dependencies resolved.

```

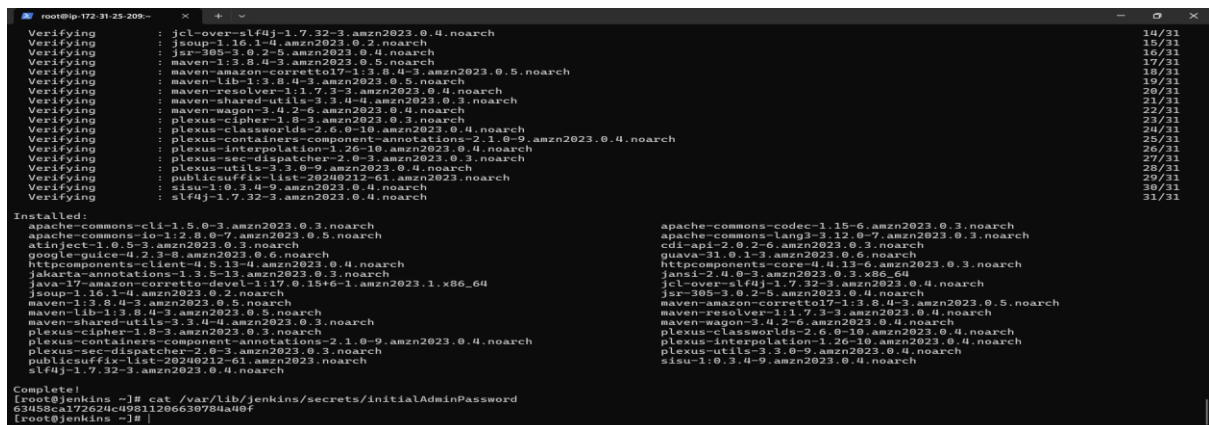
Maven package is installed.



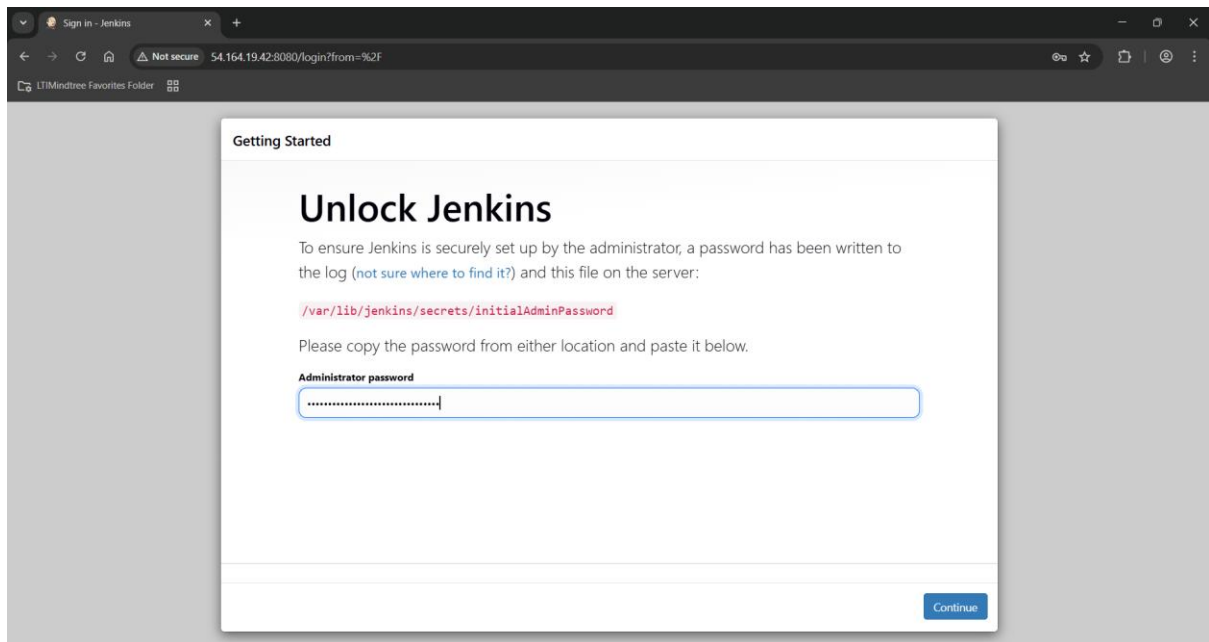
Jenkins public IP is copied.



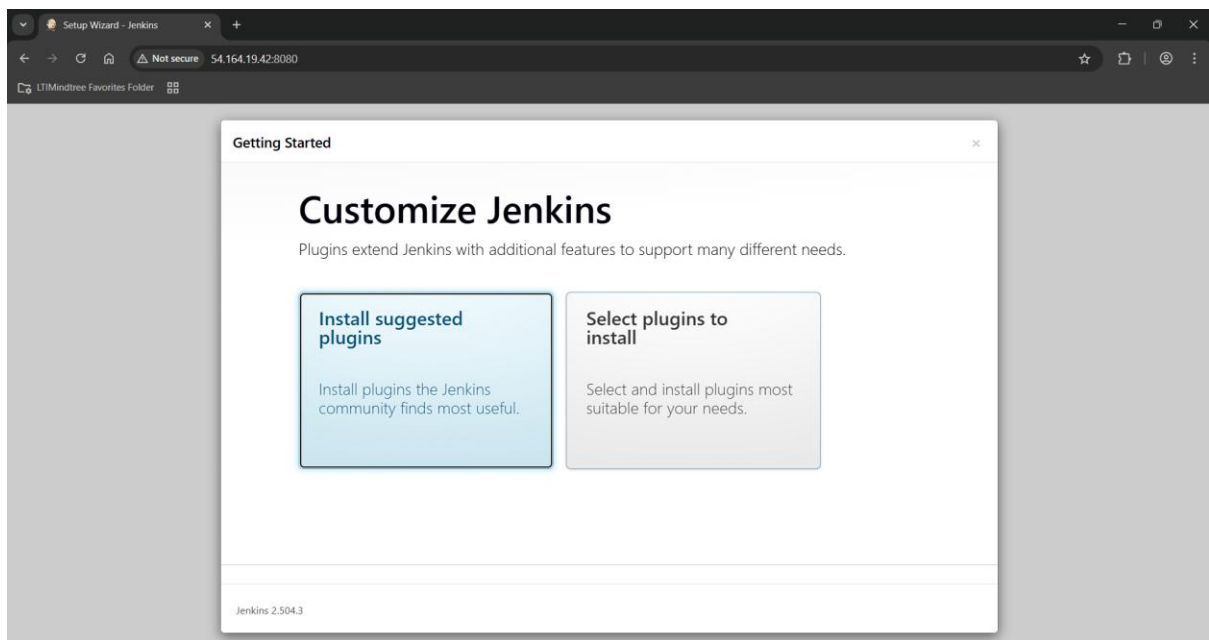
Through Jenkins public IP and port 8080, Jenkins dashboard is opened.



Administrator password is taken from Jenkins machine.



Pasted in Jenkins dashboard to access it.



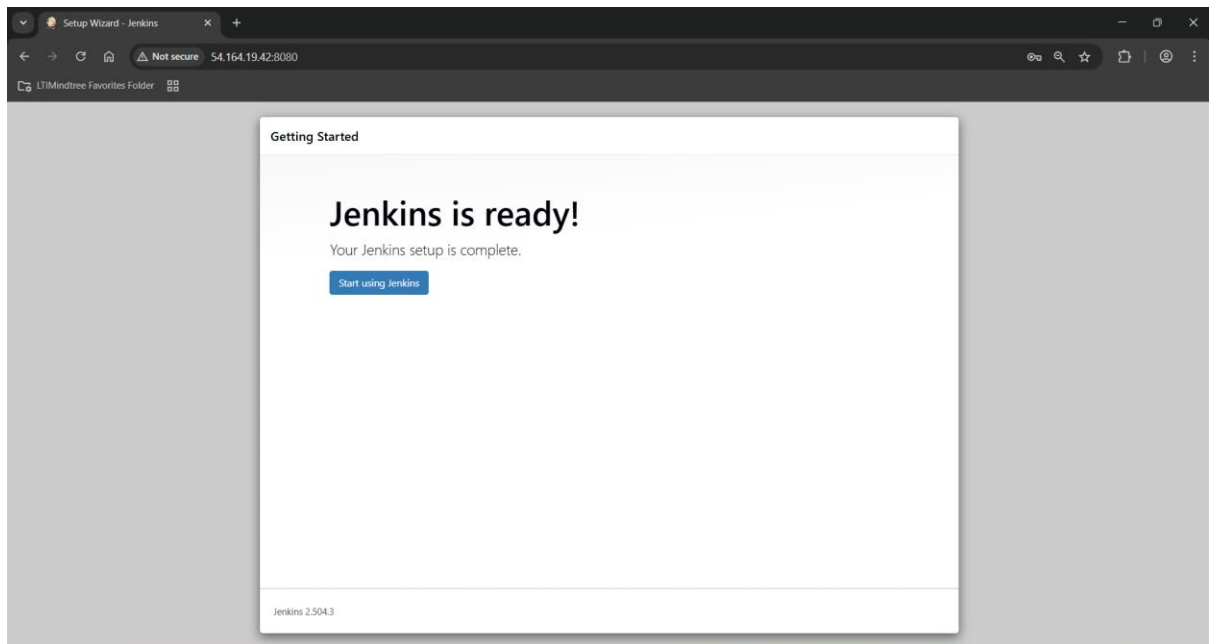
Install suggested plugins.

The screenshot shows the 'Getting Started' section of the Jenkins Setup Wizard. The main heading is 'Create First Admin User'. Below this, there are five input fields: 'Username' (containing 'admin'), 'Password' (masked with four dots), 'Confirm password' (masked with four dots), 'Full name' (containing 'Hariram'), and 'E-mail address' (containing 'hariramdevops@outlook.com'). At the bottom left, it says 'Jenkins 2.504.3'. At the bottom right, there are two buttons: 'Skip and continue as admin' and 'Save and Continue'.

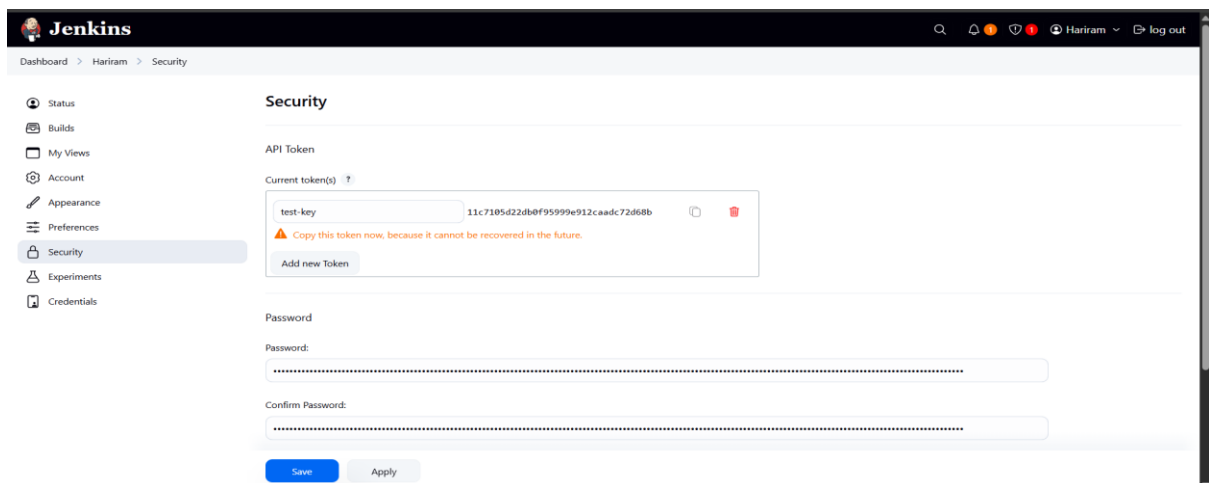
Created admin user with username: admin and password: admin.

The screenshot shows the 'Getting Started' section of the Jenkins Setup Wizard. The main heading is 'Instance Configuration'. Below this, there is a 'Jenkins URL:' label followed by an input field containing 'http://54.164.19.42:8080/'. Below the input field, there is a paragraph of text explaining the Jenkins URL and its importance. At the bottom left, it says 'Jenkins 2.504.3'. At the bottom right, there are two buttons: 'Not now' and 'Save and Finish'.

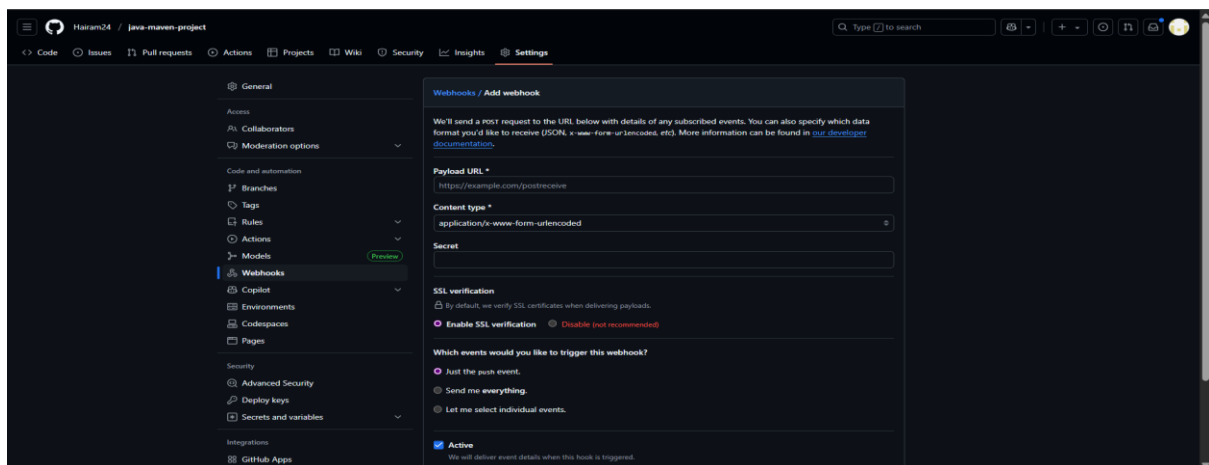
This is the Jenkins url.



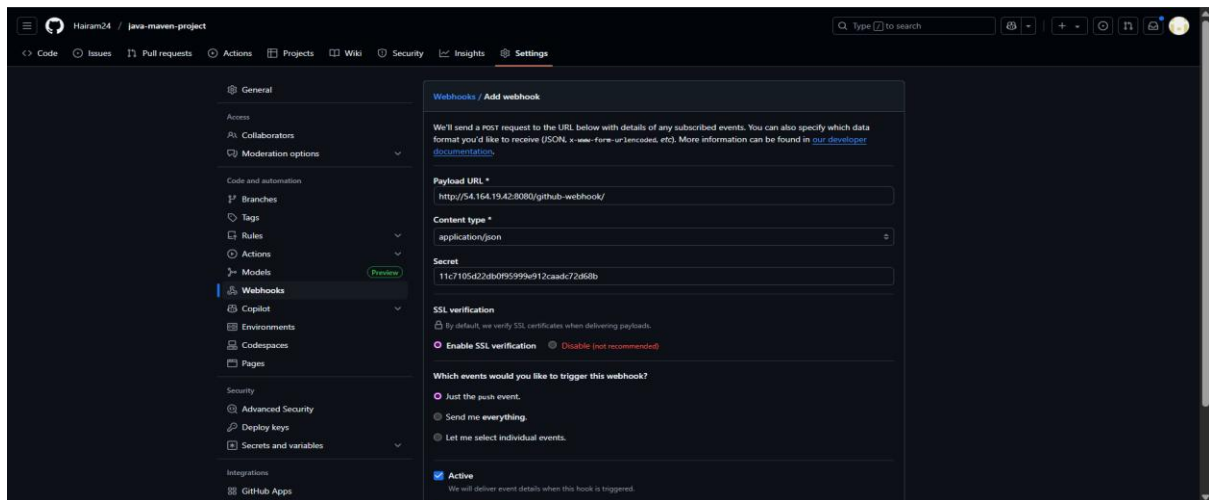
Jenkins is ready to use.



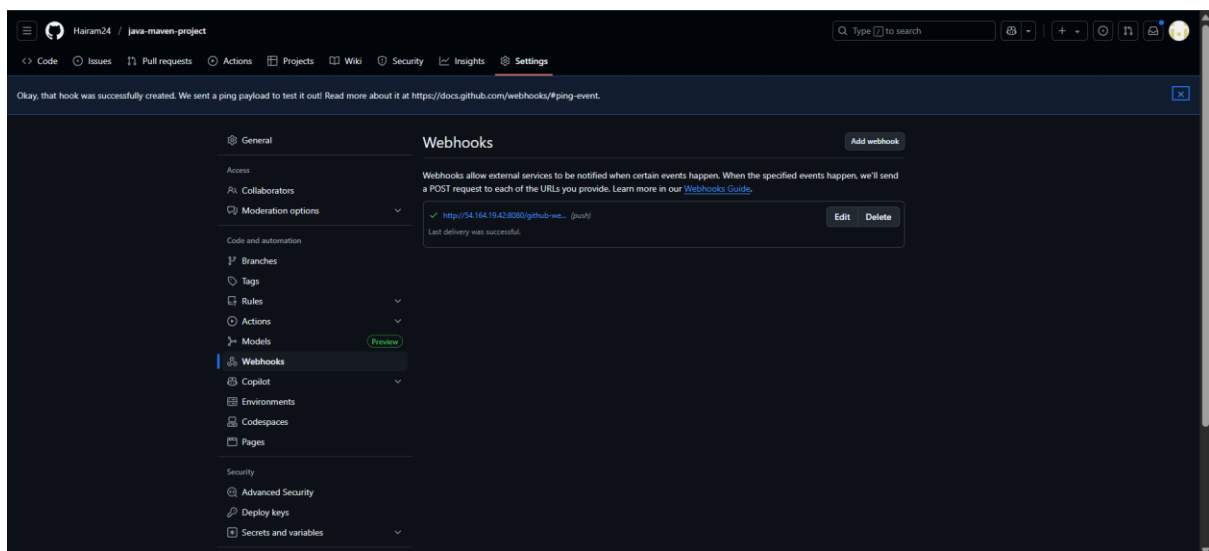
In dashboard > profile > security , create a API token.



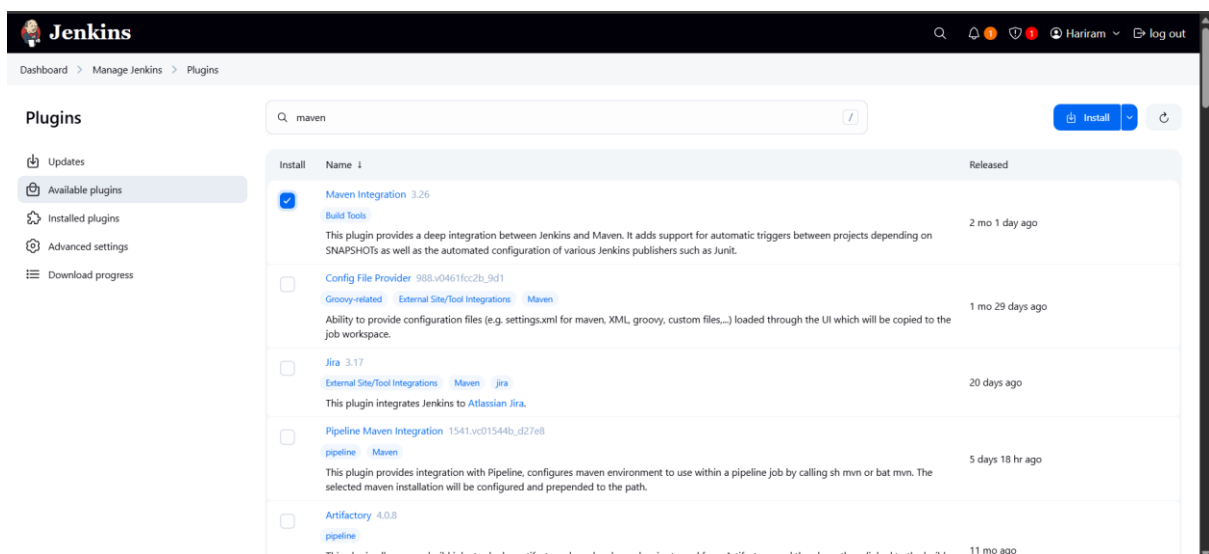
API key pasted in repo setting > webhook.



Connection type : application/json is selected.



Webhook is created successfully.



Maven plugin is installed.

Jenkins

Dashboard > Manage Jenkins > Plugins

Plugins

Search: github in

Install

Install	Name	Released
☒	GitHub Integration 0.7.2 emailtext Build Triggers GitHub Integration Plugin for Jenkins	5 mo 13 days ago

REST API Jenkins 2.504.3

GitHub integration plugin is installed.

Jenkins

Dashboard > Manage Jenkins > Plugins

Plugins

Search: deploy

Install

Install	Name	Released
☒	Deploy to container 1.17 Artifact Uploaders This plugin allows you to deploy a war to a container after a successful build. Glassfish 3.x remote deployment	1 mo 14 days ago
☐	Docker Pipeline 621.v9a_73f881d9232 pipeline DevOps Deployment docker Build and use Docker containers from pipelines.	27 days ago
☐	Artifactory 4.0.8 pipeline This plugin allows your build jobs to deploy artifacts and resolve dependencies to and from Artifactory, and then have them linked to the build job that created them. The plugin includes a vast collection of features, including a rich pipeline API library and release management for Maven and Gradle builds with Staging and Promotion.	11 mo ago
☐	Ansible 524.v9fa_a_4c989224 pipeline External Site/Tool Integrations DevOps Build Tools Deployment Invoke Ansible Ad-Hoc commands and playbooks.	3 mo 10 days ago
☐	Office 365 Connector / Power Automate workflows 5.1.0 Build Notifiers Sends message from Jenkins job to Microsoft Teams using Power Automate workflows (previously Office 365-connector). This version	4 mo 13 days ago

Deploy to container plugin is installed.

Dashboard > Manage Jenkins > Plugins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

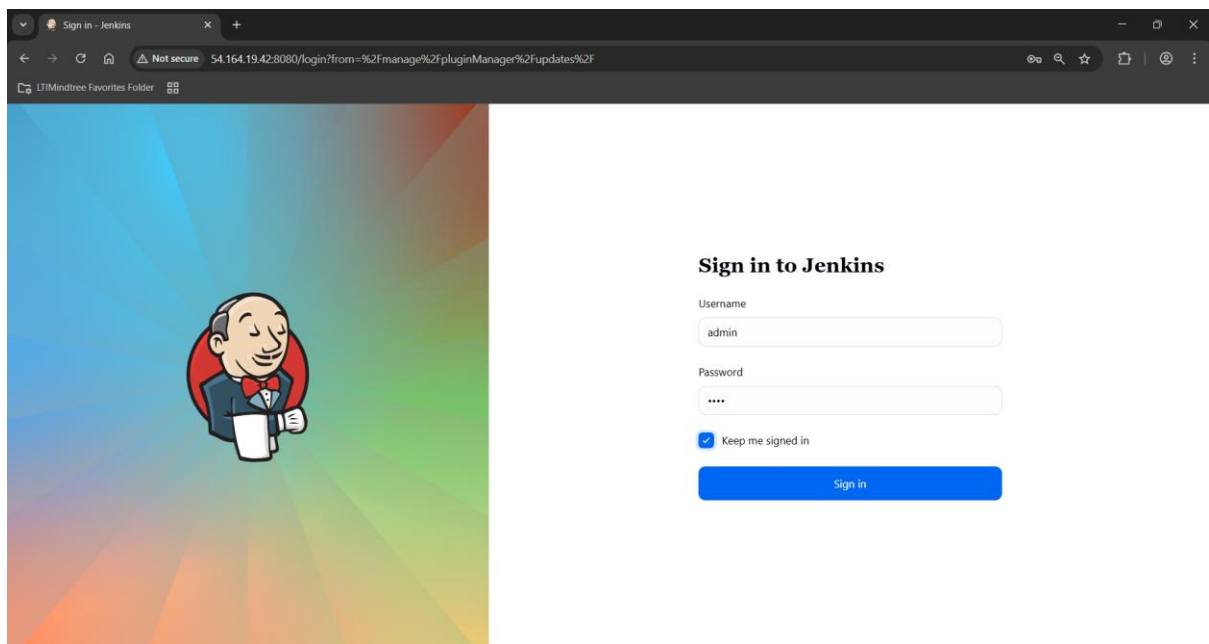
Plugin	Status
EXTERNAL AUTHENTICATION STRATEGY	Success
PAM Authentication	Success
LDAP	Success
jsoup API	Success
Email Extension	Success
Mailer	Success
Theme Manager	Success
Dark Theme	Success
Loading plugin extensions	Success
Javadoc	Success
Dev Tools Symbols API	Success
JSch dependency	Success
Maven Integration	Success
Loading plugin extensions	Success
GitHub Integration	Success
Loading plugin extensions	Success
Deploy to container	Success
Loading plugin extensions	Success

→ [Go back to the top page](#)
(you can start using the installed plugins right away)

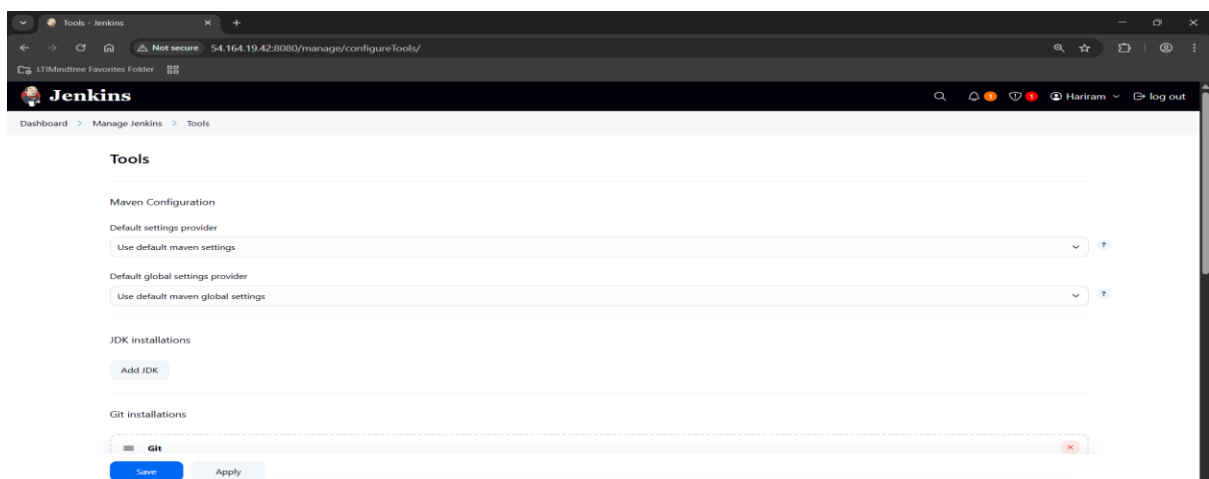
→ ☐ Restart Jenkins when installation is complete and no jobs are running

REST API Jenkins 2.504.3

Restarted the Jenkins.



Logged in to the Jenkins.



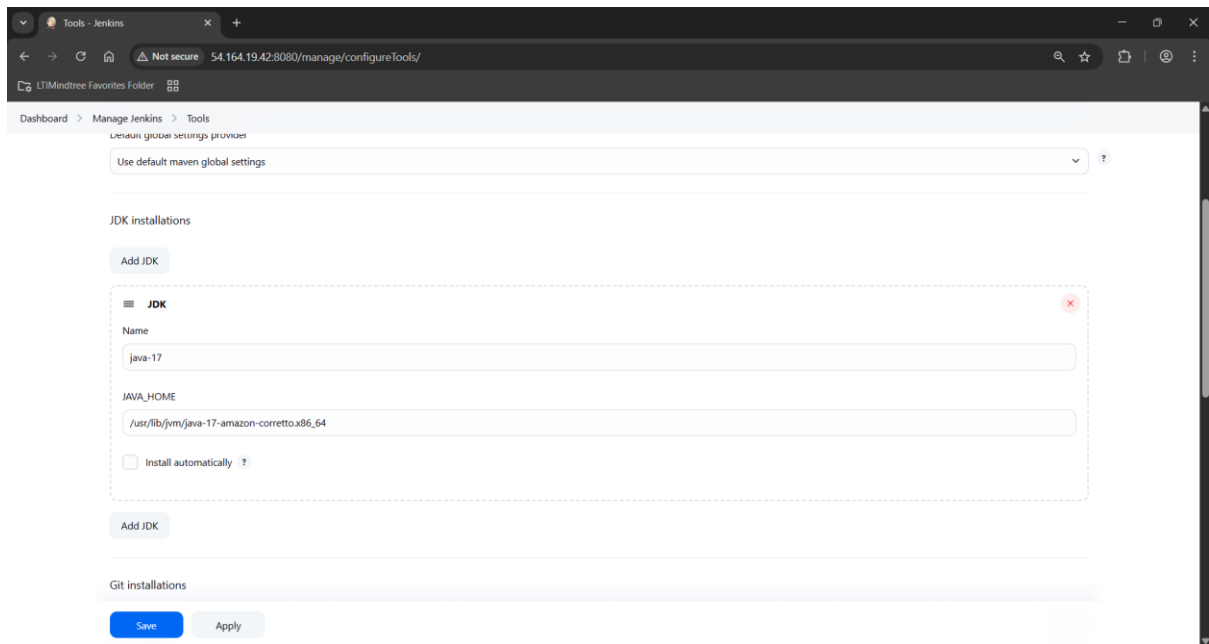
In manage Jenkins > tools add the JDK and Maven path.

```
root@ip-172-31-25-209:~# mvn -v
Verifying : maven-resolver-1:1.7.3-3.amzn2023.0.4.noarch
Verifying : maven-shared-utils-3.3.4-4.amzn2023.0.3.noarch
Verifying : maven-wagon-3.4.2-6.amzn2023.0.4.noarch
Verifying : plexus-cipher-1.8-3.amzn2023.0.3.noarch
Verifying : plexus-classworlds-2.6.0-10.amzn2023.0.4.noarch
Verifying : plexus-containers-component-annotations-2.1.0-9.amzn2023.0.4.noarch
Verifying : plexus-interpolation-1.26-10.amzn2023.0.4.noarch
Verifying : plexus-sec-dispatcher-2.0-3.amzn2023.0.3.noarch
Verifying : plexus-utils-3.3.0-9.amzn2023.0.4.noarch
Verifying : publicsuffix-list-20240212-61.amzn2023.noarch
Verifying : sisu-1:0.3.4-9.amzn2023.0.4.noarch
Verifying : slf4j-1.7.32-3.amzn2023.0.4.noarch

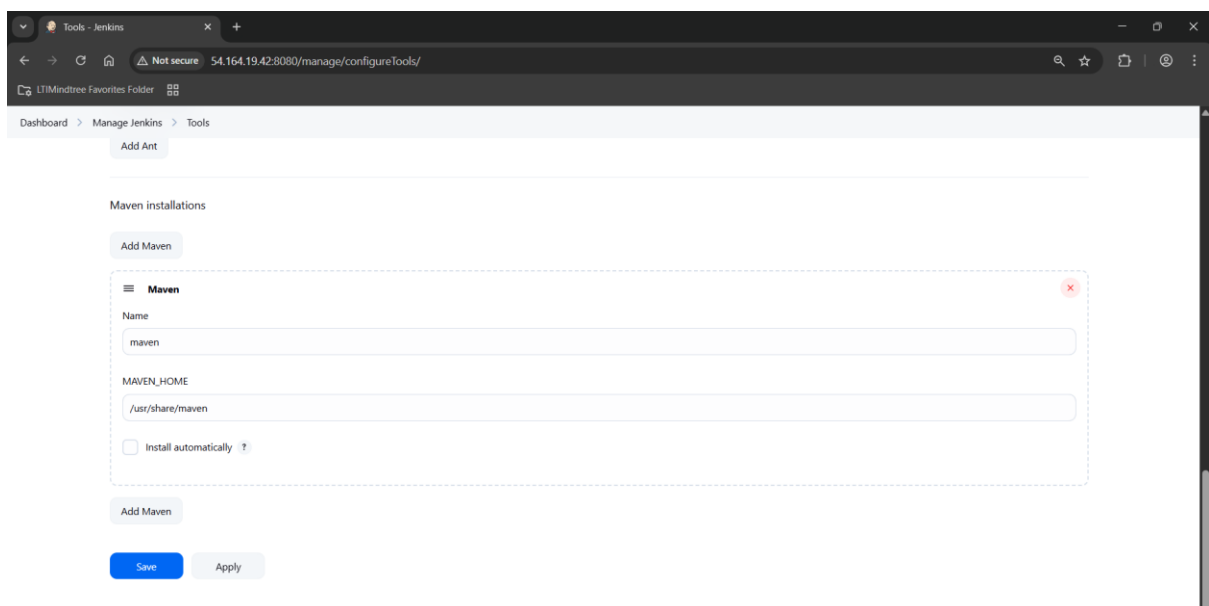
Installed:
apache-commons-cli-1.5.0-3.amzn2023.0.3.noarch
apache-commons-io-1:2.8.0-7.amzn2023.0.5.noarch
atinject-1.0.5-3.amzn2023.0.3.noarch
google-guice-4.2.3-0.amzn2023.0.6.noarch
httpcomponents-client-4.5.13-4.amzn2023.0.4.noarch
jakarta-annotations-1.3.5-13.amzn2023.0.3.noarch
java-17-amazon-corretto-devel-1:17.0.15+6-1.amzn2023.1.x86_64
jsoup-1.16.1-4.amzn2023.0.2.noarch
maven-1:3.8.4-3.amzn2023.0.5.noarch
maven-lib-1:13.6-4-3.amzn2023.0.5.noarch
maven-shared-utils-3.3.4-4.amzn2023.0.3.noarch
plexus-cipher-1.8-3.amzn2023.0.3.noarch
plexus-containers-component-annotations-2.1.0-9.amzn2023.0.4.noarch
plexus-sec-dispatcher-2.0-3.amzn2023.0.3.noarch
publicsuffix-list-20240212-61.amzn2023.noarch
slf4j-1.7.32-3.amzn2023.0.4.noarch

Complete!
[root@jenkins ~]# cat /var/lib/jenkins/secrets/initialAdminPassword
63458ca172624c49811206630784a40f
[root@jenkins ~]# mvn -v
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
Java version: 17.0.15, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.141-155.222.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@jenkins ~]#
```

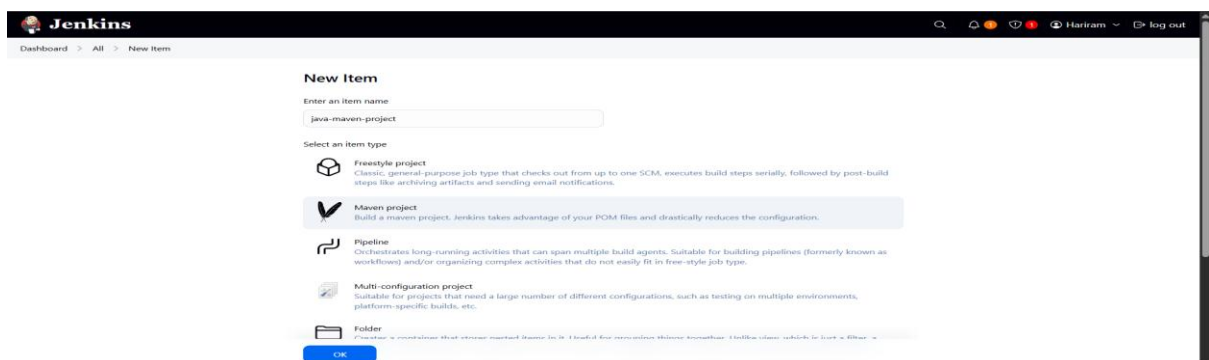
Using mvn -v java and maven path is copied.



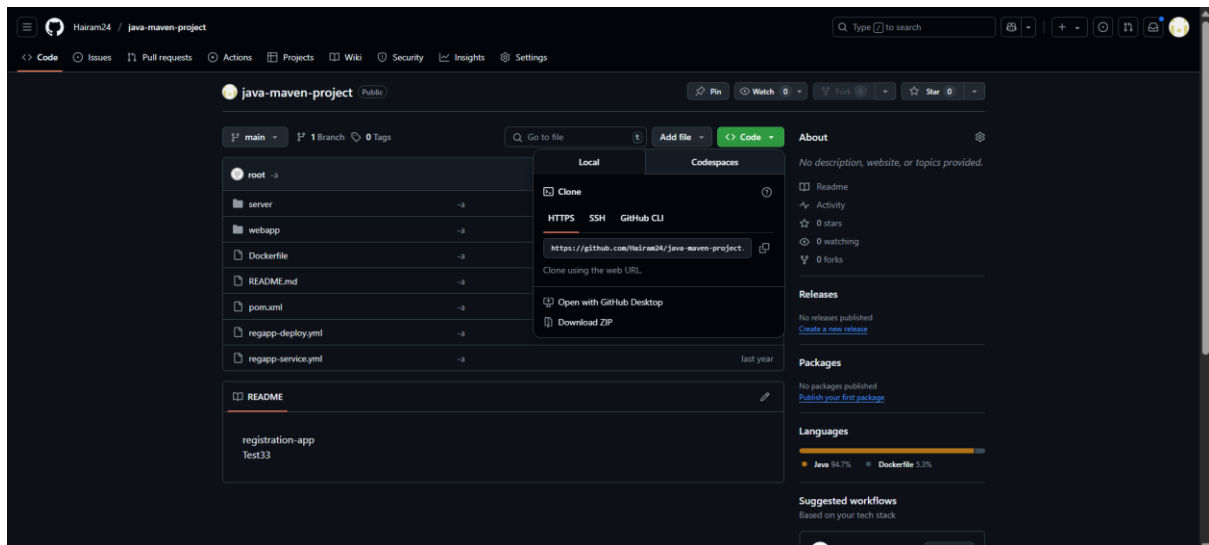
JDK path is added.



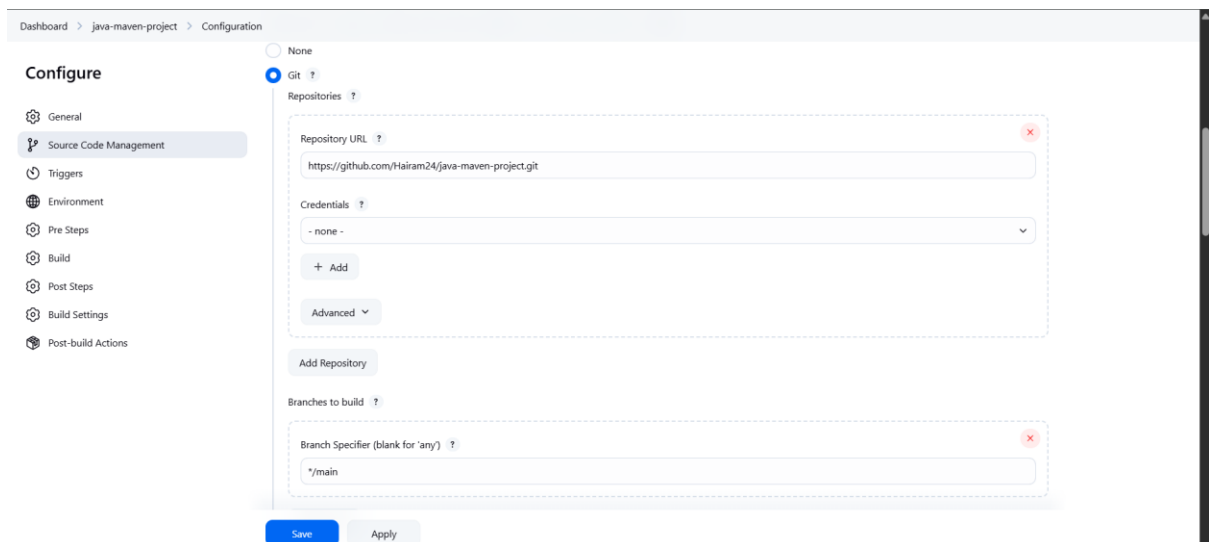
Maven path is added.



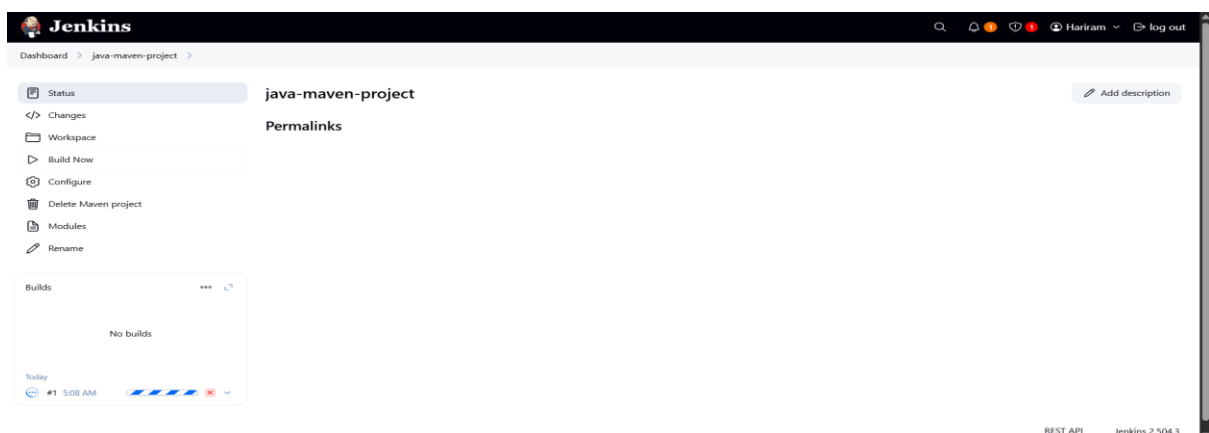
Create a new maven project.



Copy the https repo link.



Add the git repository.



Build now to check.

Jenkins

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Hariram

🚪 log out

Dashboard > java-maven-project > #1 > Console Output

Status

</> Changes

📄 Console Output

📋 Edit Build Information

🗑️ Delete build '#1'

🕒 Timings

🔗 Git Build Data

🔄 Redeploy Artifacts

📄 Test Result

🖨️ See Fingerprints

✅ Console Output

Download

Copy

View as plain text

Started by user Hariram

Running as SYSTEM

Building in workspace /var/lib/jenkins/workspace/java-maven-project

The recommended git tool is: NONE

No credentials specified

Cloning the remote Git repository

Cloning repository <https://github.com/Hairam24/java-maven-project.git>

> git init /var/lib/jenkins/workspace/java-maven-project # timeout=10

Fetching upstream changes from <https://github.com/Hairam24/java-maven-project.git>

> git --version # timeout=10

> git fetch --tags --force --progress -- <https://github.com/Hairam24/java-maven-project.git> +refs/heads/*:refs/remotes/origin/* # timeout=10

> git config remote.origin.url <https://github.com/Hairam24/java-maven-project.git> # timeout=10

> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10

Avoid second fetch

> git rev-parse refs/remotes/origin/main^{commit} # timeout=10

Checking out Revision 03262507ba776ce0949bb2010179ecd06ef9756 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f 03262507ba776ce0949bb2010179ecd06ef9756 # timeout=10

Commit message: "-a"

First time build. Skipping changelog.

Parsing POMs

Discovered a new module com.example.maven-project:maven-project Maven Project

Discovered a new module com.example.maven-project:server Server

Build successfully.

Dashboard > java-maven-project > #1 > Console Output

```
[INFO] --- maven-install-plugin:2.4:install (default-install) @ webapp ---
[INFO] Installing /var/lib/jenkins/workspace/java-maven-project/webapp/target/webapp.war to /var/lib/jenkins/.m2/repository/com/example/maven-project/webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.war
[INFO] Installing /var/lib/jenkins/workspace/java-maven-project/webapp/pom.xml to /var/lib/jenkins/.m2/repository/com/example/maven-project/webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.pom
[INFO] -----
[INFO] Reactor Summary for Maven Project 1.0-SNAPSHOT:
[INFO]
[INFO] Maven Project ..... SUCCESS [ 1.747 s]
[INFO] Server ..... SUCCESS [ 5.060 s]
[INFO] Webapp ..... SUCCESS [ 2.690 s]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 12.037 s
[INFO] Finished at: 2025-06-27T05:08:29Z
[INFO] -----
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving /var/lib/jenkins/workspace/java-maven-project/webapp/pom.xml to com.example.maven-project/webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.pom
[JENKINS] Archiving /var/lib/jenkins/workspace/java-maven-project/webapp/target/webapp.war to com.example.maven-project/webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.war
[JENKINS] Archiving /var/lib/jenkins/workspace/java-maven-project/server/pom.xml to com.example.maven-project/server/1.0-SNAPSHOT/server-1.0-SNAPSHOT.pom
[JENKINS] Archiving /var/lib/jenkins/workspace/java-maven-project/server/target/server.jar to com.example.maven-project/server/1.0-SNAPSHOT/server-1.0-SNAPSHOT.jar
[JENKINS] Archiving /var/lib/jenkins/workspace/java-maven-project/pom.xml to com.example.maven-project/maven-project/1.0-SNAPSHOT/maven-project-1.0-SNAPSHOT.pom
channel stopped
Finished: SUCCESS
```

REST API Jenkins 2.504.3

Build finished success.

Jenkins

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Hariram

🚪 log out

Dashboard > java-maven-project > #1

Status

</> Changes

📄 Console Output

📋 Edit Build Information

🗑️ Delete build '#1'

🕒 Timings

🔗 Git Build Data

🔄 Redeploy Artifacts

📄 Test Result

🖨️ See Fingerprints

✅ #1 (Jun 27, 2025, 5:08:07 AM)

Add description

Keep this build forever

🕒 Started by user Hariram

Started 9 min 45 sec ago
Took 21 sec

🕒 This run spent:

- 33 ms waiting;
- 21 sec build duration;
- 21 sec total from scheduled to completion.

🔗 Revision: 03262507ba776ce0949bb2010179ecd06ef9756

Repository: <https://github.com/Hairam24/java-maven-project.git>

- refs/remotes/origin/main

📄 Test Result (no failures)

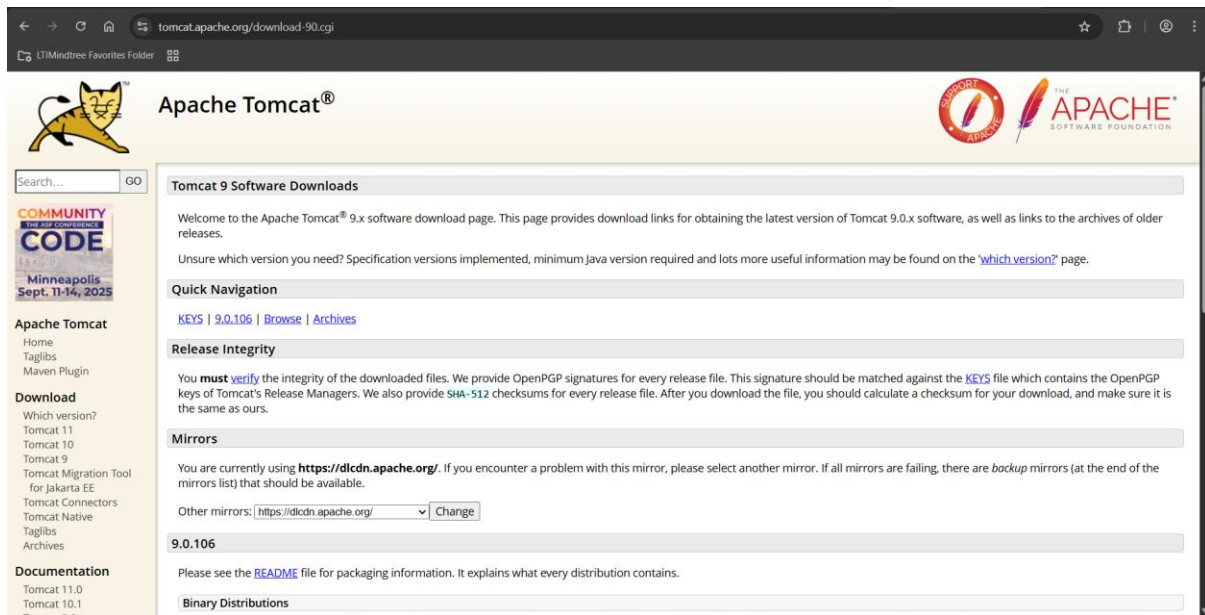
Module Builds

✅ Maven Project 2.2 sec

✅ Server 5.4 sec

✅ Webapp 3.1 sec

REST API Jenkins 2.504.3



Copy the tomcat 9 tar file link.

```
[root@tomcat ~]# wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.106/bin/apache-tomcat-9.0.106.tar.gz
--2025-06-27 05:26:35-- https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.106/bin/apache-tomcat-9.0.106.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13097554 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.106.tar.gz'

apache-tomcat-9.0.106.tar.gz 100%[=====] 12.40M --.-KB/s in 0.09s

2025-06-27 05:26:35 (131 MB/s) - 'apache-tomcat-9.0.106.tar.gz' saved [13097554/13097554]

[root@tomcat ~]#
```

Download the tar file.

```
root@ip-172-31-17-39:~
[root@tomcat ~]# tar -zxvf apache-tomcat-9.0.106.tar.gz
apache-tomcat-9.0.106/conf/
apache-tomcat-9.0.106/conf/catalina.policy
apache-tomcat-9.0.106/conf/catalina.properties
apache-tomcat-9.0.106/conf/context.xml
apache-tomcat-9.0.106/conf/jaspic-providers.xml
apache-tomcat-9.0.106/conf/jaspic-providers.xsd
apache-tomcat-9.0.106/conf/logging.properties
apache-tomcat-9.0.106/conf/server.xml
apache-tomcat-9.0.106/conf/tomcat-users.xml
apache-tomcat-9.0.106/conf/tomcat-users.xsd
apache-tomcat-9.0.106/conf/web.xml
apache-tomcat-9.0.106/bin/
apache-tomcat-9.0.106/lib/
apache-tomcat-9.0.106/logs/
apache-tomcat-9.0.106/temp/
apache-tomcat-9.0.106/webapps/
apache-tomcat-9.0.106/webapps/ROOT/
apache-tomcat-9.0.106/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.106/webapps/docs/
apache-tomcat-9.0.106/webapps/docs/META-INF/
apache-tomcat-9.0.106/webapps/docs/WEB-INF/
apache-tomcat-9.0.106/webapps/docs/WEB-INF/jsp/
apache-tomcat-9.0.106/webapps/docs/annotationapi/
apache-tomcat-9.0.106/webapps/docs/api/
apache-tomcat-9.0.106/webapps/docs/appdev/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/docs/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/src/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/src/mypackage/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/web/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/web/WEB-INF/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/web/images/
apache-tomcat-9.0.106/webapps/docs/architecture/
```

Unzip the tar file.

```
[root@tomcat ~]# cd apache-tomcat-9.0.106
[root@tomcat apache-tomcat-9.0.106]# cd bin
[root@tomcat bin]# chmod +x startup.sh
[root@tomcat bin]# chmod +x shutdown.sh
[root@tomcat bin]# ln -s /opt/apache-tomcat-9.0.106/bin/startup.sh /usr/local/bin/tomcatup
[root@tomcat bin]# ln -s /opt/apache-tomcat-9.0.106/bin/shutdown.sh /usr/local/bin/tomcatdown
```

Tomcat installation is started.

```
[root@tomcat ~]# cd apache-tomcat-9.0.106
[root@tomcat apache-tomcat-9.0.106]# cd bin
[root@tomcat bin]# ./startup.sh
Using CATALINA_BASE:   /root/apache-tomcat-9.0.106
Using CATALINA_HOME:   /root/apache-tomcat-9.0.106
Using CATALINA_TMPDIR: /root/apache-tomcat-9.0.106/temp
Using JRE_HOME:         /usr
Using CLASSPATH:        /root/apache-tomcat-9.0.106/bin/bootstrap.jar:/root/apache-tomcat-9.0.106/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@tomcat bin]#
```

Tomcat is started.

```
root@ip-172-31-17-39:~/apac x + v
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/examples/META-INF/context.xml
```

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
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 contributor license agreements. See the NOTICE file distributed with
 this work for additional information regarding copyright ownership.
 The ASF licenses this file to You under the Apache License, Version 2.0
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 distributed under the license is distributed on an "AS IS" BASIS,
 WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
 See the License for the specific language governing permissions and
 limitations under the License.
-->
<Context>
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
    sameSiteCookies="strict" />
  <!-- <Valve className="org.apache.catalina.valves.RemoteAddrValve"
    allow="127.\.d+\.d+\.d+\.d+:::1|0:0:0:0:0:0:0:1" /> -->
</Context>
--
--
--
--
--
--
--
--
-- INSERT --
```

Commented the IP part.

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
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 contributor license agreements. See the NOTICE file distributed with
 this work for additional information regarding copyright ownership.
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 distributed under the license is distributed on an "AS IS" BASIS,
 WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
 See the License for the specific language governing permissions and
 limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
    sameSiteCookies="strict" />
  <!-- <Valve className="org.apache.catalina.valves.RemoteAddrValve"
    allow="127.\.d+\.d+\.d+\.d+:::1|0:0:0:0:0:0:0:1" /> -->
  <Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|string)|org\.apache\.catalina\.filters\.Cs
    rPreventionFilter|$ruCache(?:\s1)?|java\.util\.(?:Linked)?HashMap"/>
</Context>
--
--
--
--
--
--
--
--
-- INSERT --
```

Commented the IP part.


```
root@ip-172-31-17-39:~/apac x + v
<?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements. See the NOTICE file distributed with
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The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
    sameSiteCookies="strict" />
  <!-- <Valve className="org.apache.catalina.valves.RemoteAddrValve"
    allow="127\.\d+\.\d+\.\d+|::1|0:0:0:0:0:0:0:1" /> -->
  <Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|string)|org\.apache\.catalina\.filters\.Cs
rfPreventionFilter|\$LruCache(?:\$1)?|java\.util\.(?:Linked)?HashMap"/>
</Context>
-- INSERT --
```

Commented the IP part.

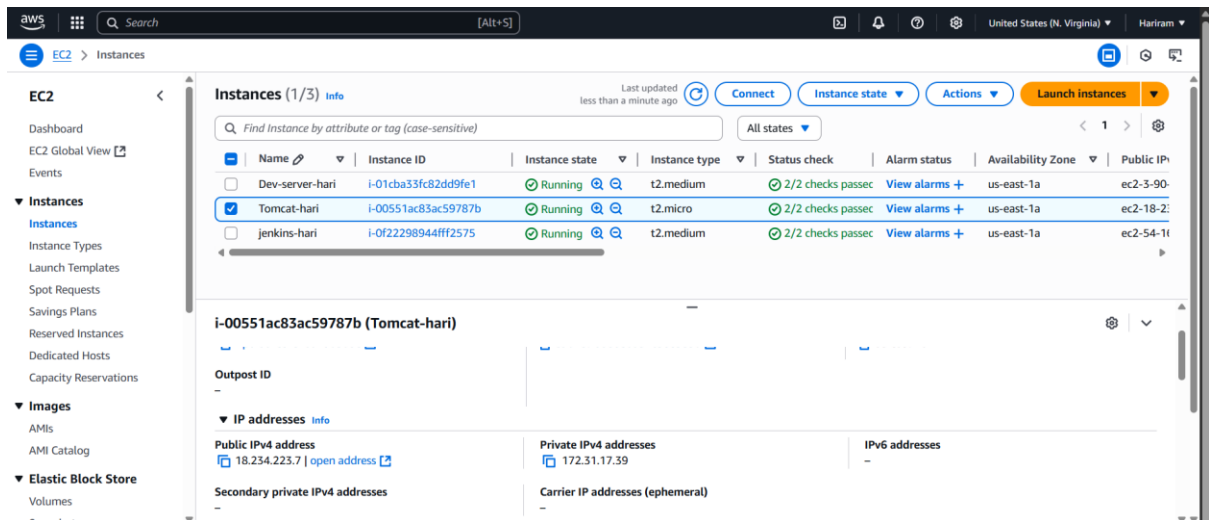
```
root@ip-172-31-17-39:~/apac x + v
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/examples/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/host-manager/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/manager/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]#
```

```
root@ip-172-31-17-39:~/apac x + v
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/examples/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/host-manager/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/manager/META-INF/context.xml
[root@tomcat apache-tomcat-9.0.106]# cd conf
[root@tomcat conf]# ll
total 228
drwxr-x---. 3 root root   23 Jun 27 05:40 Catalina
-rw-----. 1 root root 12953 Jun  5 19:02 catalina.policy
-rw-----. 1 root root  7654 Jun  5 19:02 catalina.properties
-rw-----. 1 root root  1400 Jun  5 19:02 context.xml
-rw-----. 1 root root  1149 Jun  5 19:02 jaspic-providers.xml
-rw-----. 1 root root  2313 Jun  5 19:02 jaspic-providers.xsd
-rw-----. 1 root root  4003 Jun  5 19:02 logging.properties
-rw-----. 1 root root  8022 Jun  5 19:02 server.xml
-rw-----. 1 root root  2756 Jun  5 19:02 tomcat-users.xml
-rw-----. 1 root root  2558 Jun  5 19:02 tomcat-users.xsd
-rw-----. 1 root root 173782 Jun  5 19:02 web.xml
[root@tomcat conf]# vi tomcat-users.xml
```

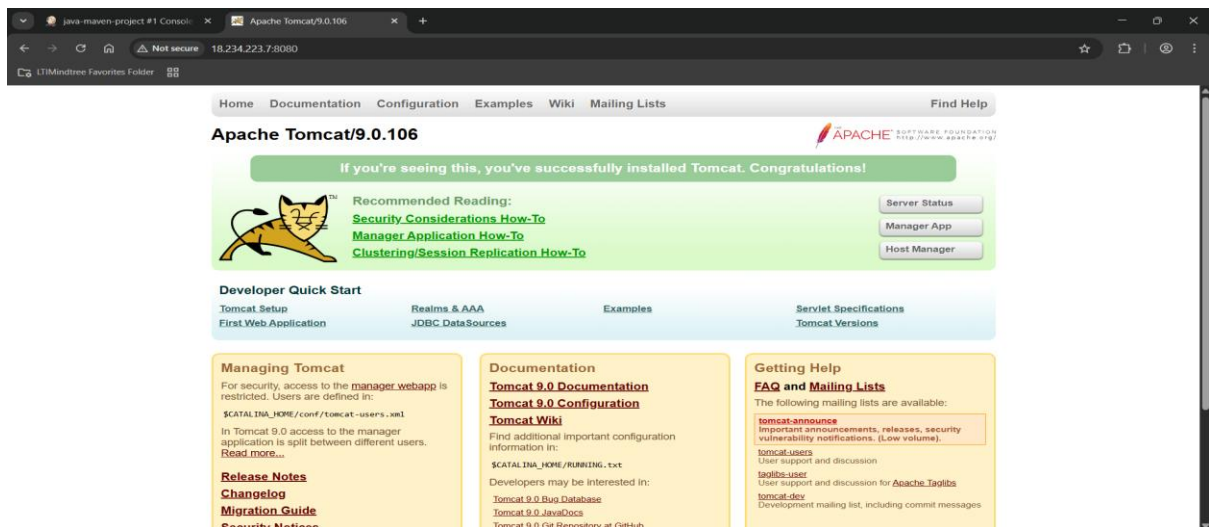
```
root@ip-172-31-17-39:~/a x + v
- manager-status - allows access to the status pages only

The users below are wrapped in a comment and are therefore ignored. If you
wish to configure one or more of these users for use with the manager web
application, do not forget to remove the <!-- ... --> that surrounds them. You
will also need to set the passwords to something appropriate.
-->
<!--
<user username="admin" password="<must-be-changed>" roles="manager-gui"/>
<user username="robot" password="<must-be-changed>" roles="manager-script"/>
-->
<!--
The sample user and role entries below are intended for use with the
examples web application. They are wrapped in a comment and thus are ignored
when reading this file. If you wish to configure these users for use with the
examples web application, do not forget to remove the <!-- ... --> that surrounds
them. You will also need to set the passwords to something appropriate.
-->
<!--
<role rolename="tomcat"/>
<role rolename="role1"/>
<user username="tomcat" password="<must-be-changed>" roles="tomcat"/>
<user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
<user username="role1" password="<must-be-changed>" roles="role1"/>
-->
<role rolename="manager-gui"/>
<role rolename="manager-script"/>
<role rolename="manager-jmx"/>
<role rolename="manager-status"/>
<user username="admin" password="admin" roles="manager-gui,manager-script,manager-jmx,manager-status"/>
<user username="deployer" password="deployer" roles="manager-script"/>
<user username="tomcat" password="secret" roles="manager-gui"/>
-->
-- INSERT --
```

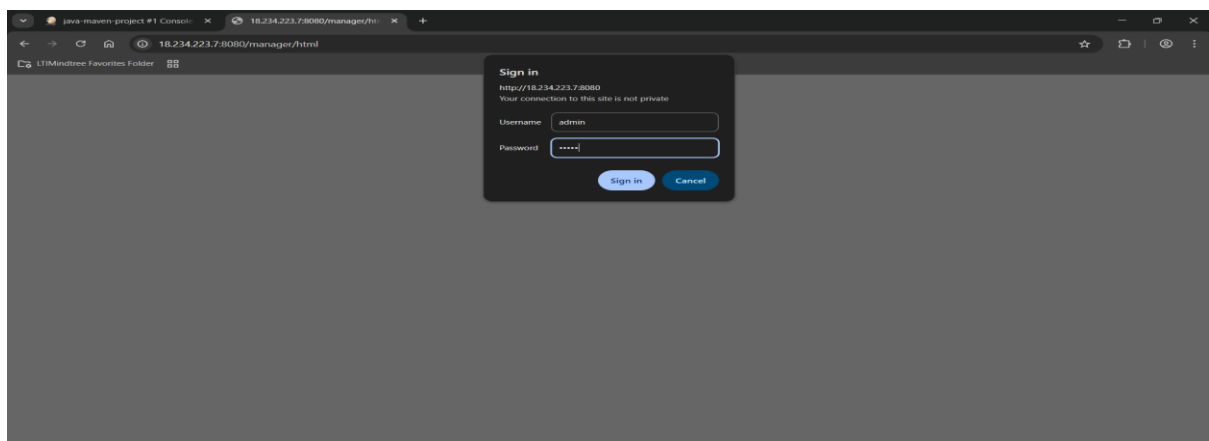
Roles are added.



Tomcat machine public IP is copied.



Tomcat dashboard is accessible.



Login through role credentials.

Tomcat Web Application Manager

Message: OK

Manager

List Applications HTML Manager Help Manager Help Server Status

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Deploy

Tomcat setup is done.

Jenkins

Dashboard > Manage Jenkins > Credentials

Credentials

Type	ID	Name

Stores scoped to Jenkins

Store	Domains	Name
System	(global)	

REST API Jenkins 2.504.3

In Jenkins add global credentials.

Jenkins

Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted)

New credentials

Kind: Username with password

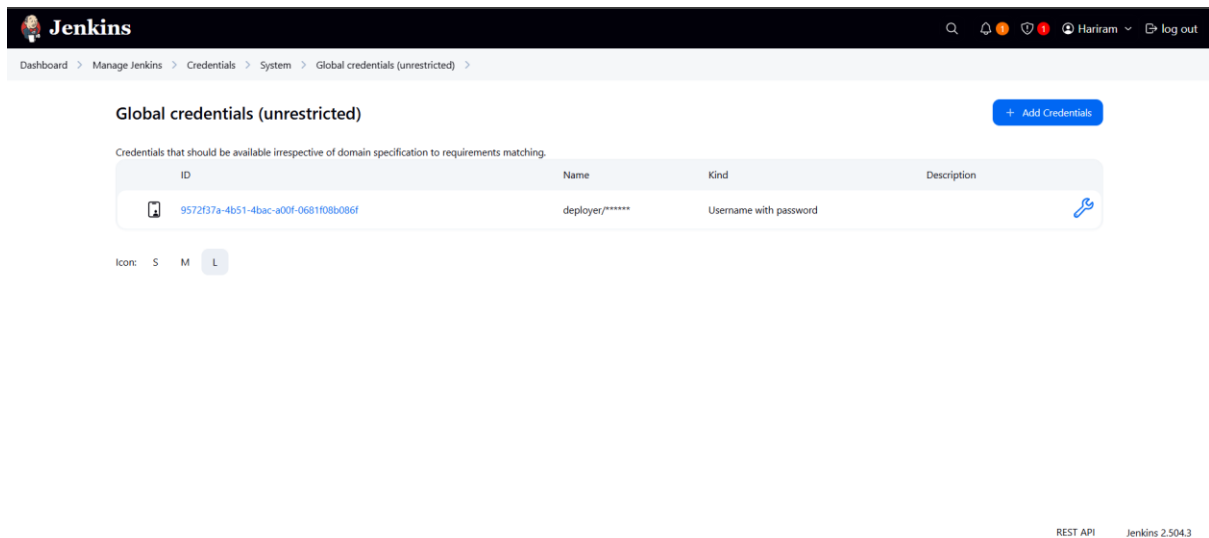
Scope: Global (Jenkins, nodes, items, all child items, etc)

Username: deployer

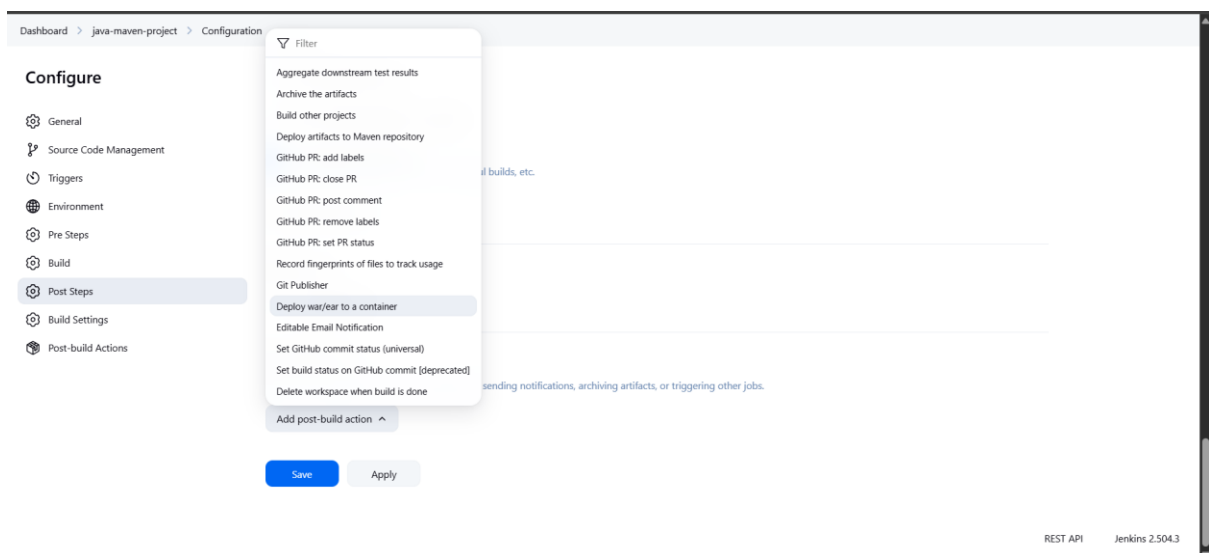
Password: deployer

Create

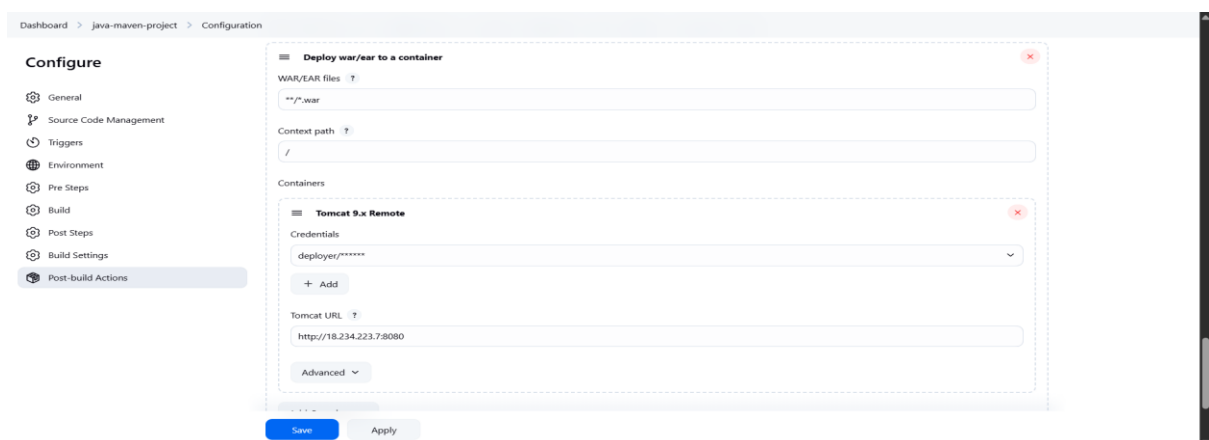
Username: deployer , Password: deployer.



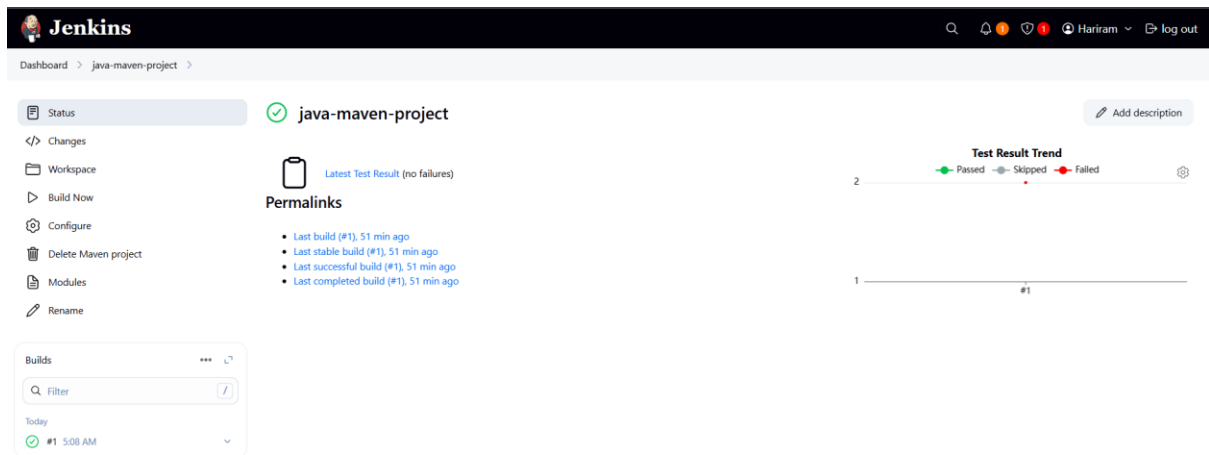
Credential is added successfully.



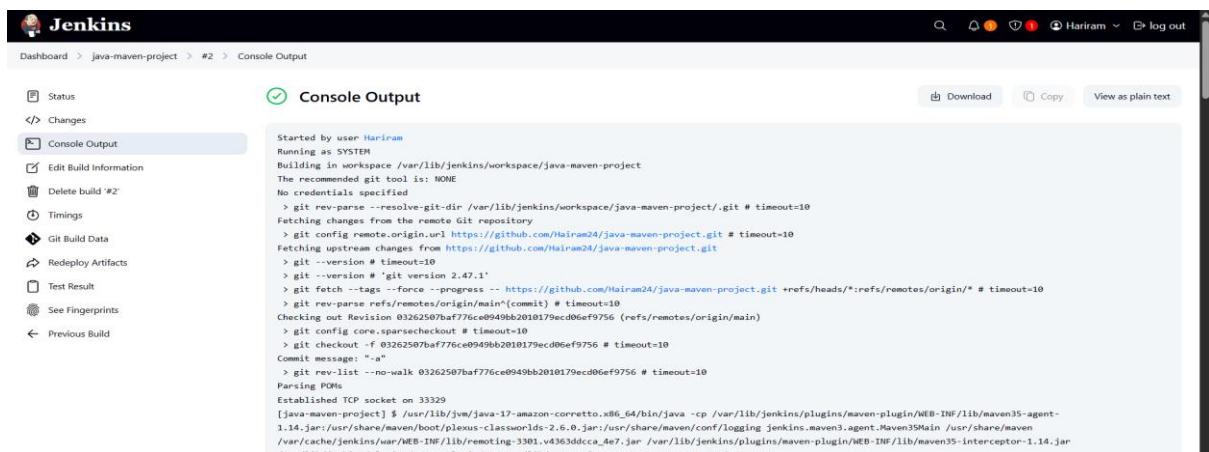
Add post build action. Use deploy war/ear to a container.



Entered the required details.



Build now.



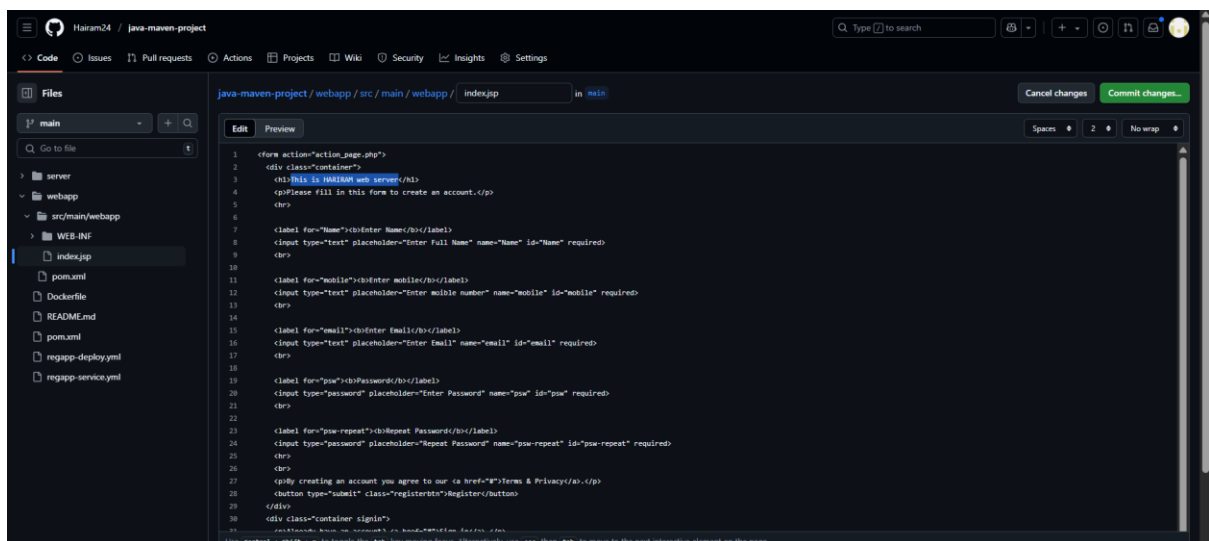
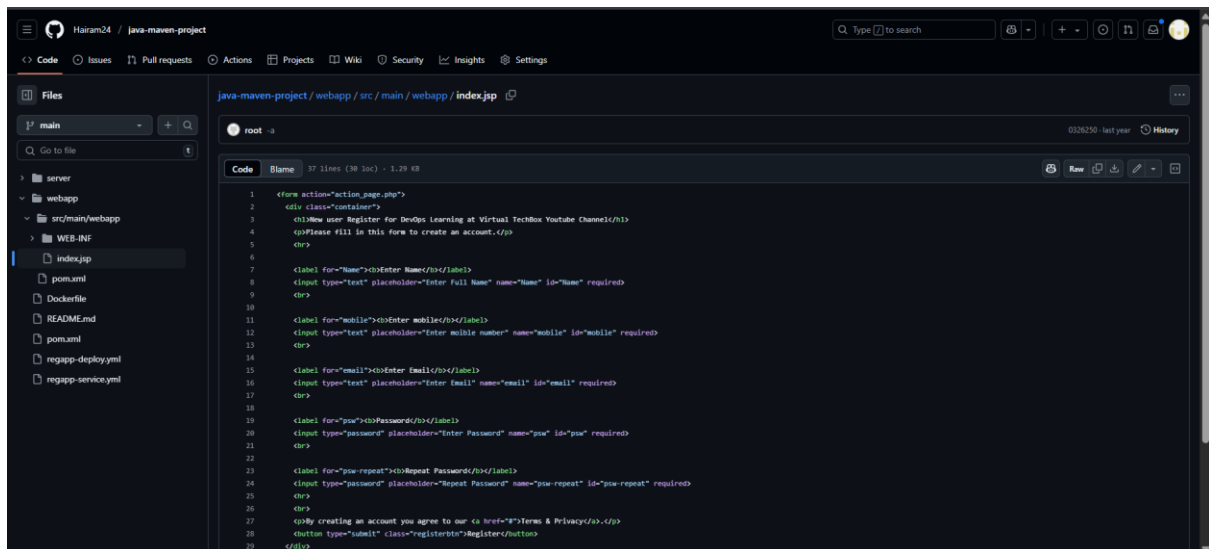
Build successfull.

The screenshot shows a web browser window with a registration form. The form is titled 'New user Register for DevOps Learning at Virtual TechBox Youtube Channel'. It includes fields for 'Enter Name', 'Enter mobile', 'Enter Email', 'Password', and 'Repeat Password'. Below the form, there is a 'Register' button and a link to 'Terms & Privacy'. The browser's address bar shows the URL '18.234.223.78080'.

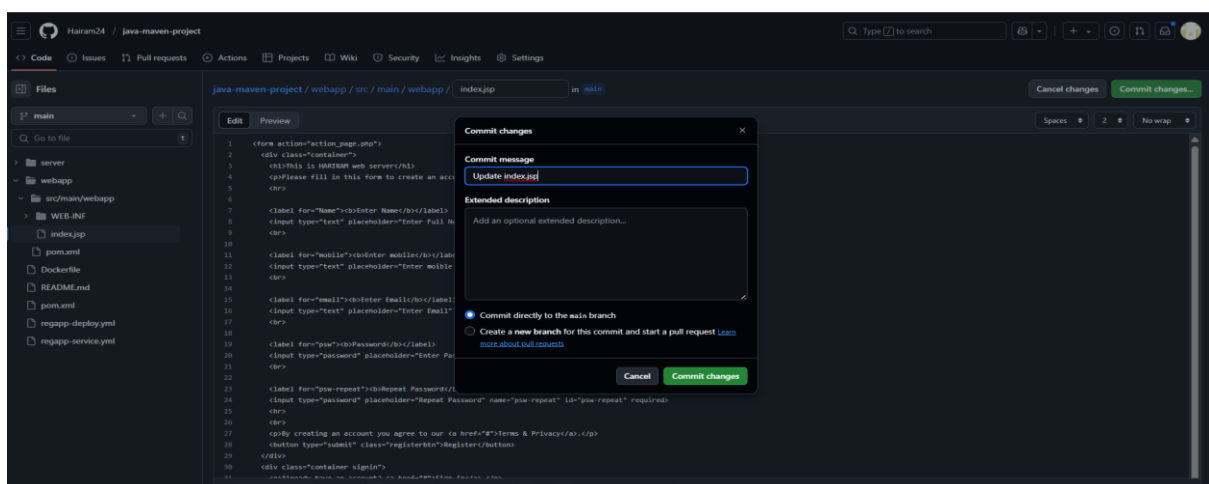
Thank You, Happy Learning

See You Again

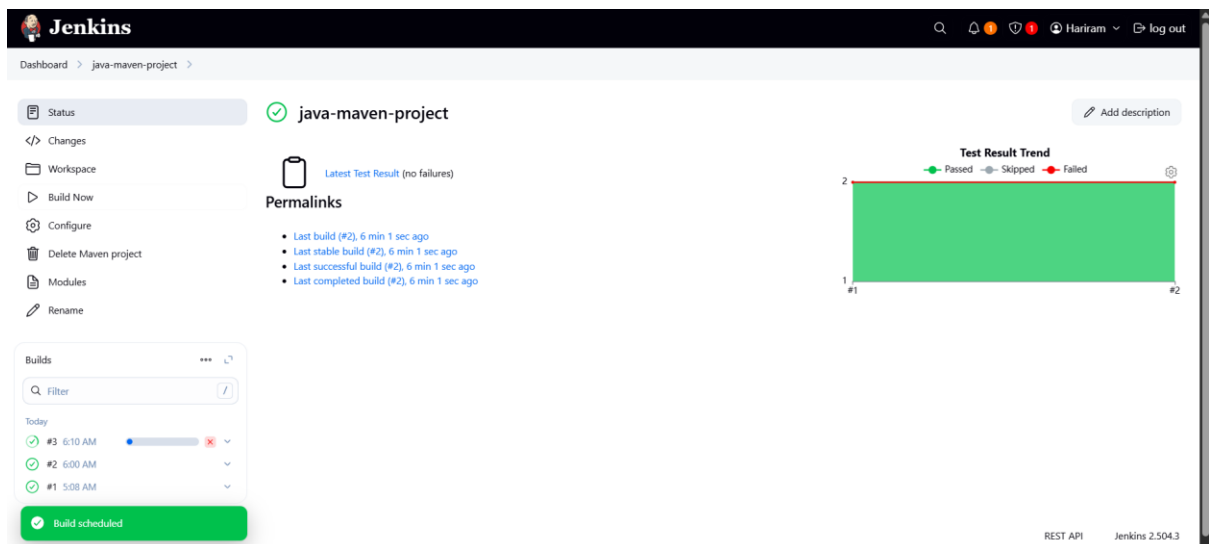
Web page is visible in tomcat.



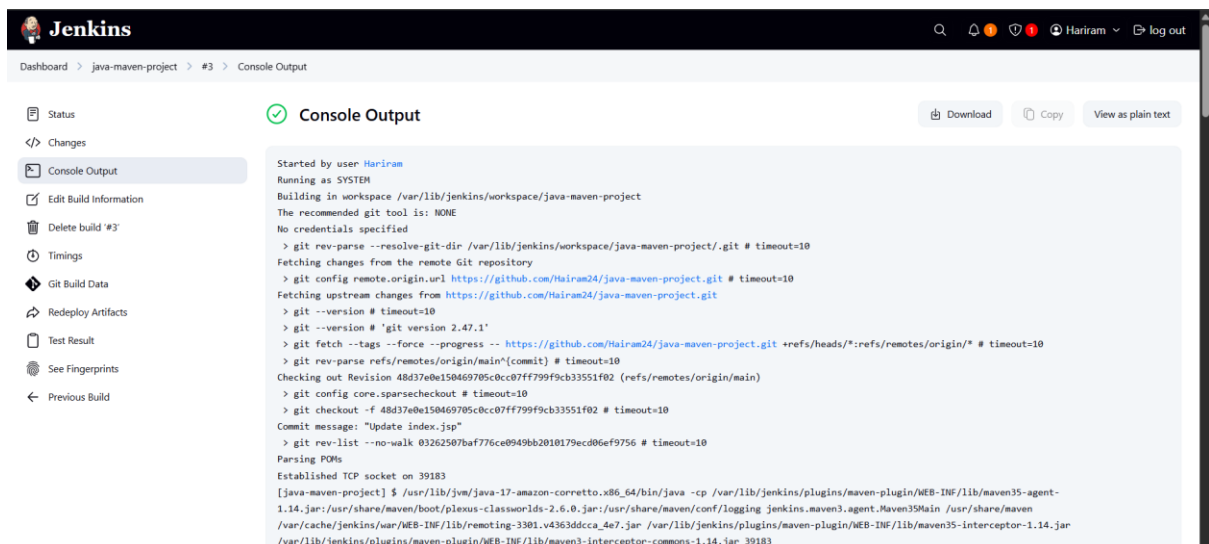
Made change in project index code.



Commit the changes.



Manual Build.



Build successfully.

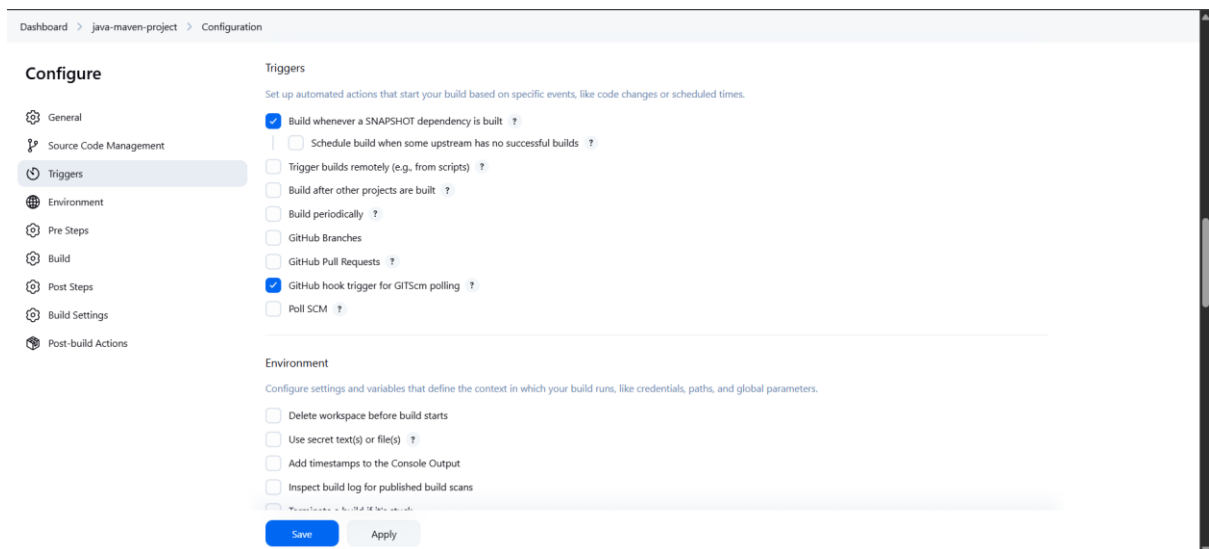
The screenshot shows the Jenkins web interface. At the top, there's a navigation bar with the Jenkins logo, a search icon, and a user profile for 'Hariram' with a 'log out' button. Below the navigation bar, a breadcrumb trail reads 'Dashboard > java-maven-project > #3 > Changes'. On the left, a sidebar contains links to 'Status', 'Changes' (which is highlighted), 'Console Output', 'Edit Build Information', 'Delete build #3', 'Timings', 'Git Build Data', 'Redeploy Artifacts', 'Test Result', 'See Fingerprints', and 'Previous Build'. The main content area is titled 'Changes' with a green checkmark icon. Under 'Summary', it lists '1. Update index.jsp (details)'. A commit box shows 'Commit 48d37e0e150469705c0cc07f799f9cb33551f02 by noreply' with the update 'index.jsp'. Below this, a link 'webapp/src/main/webapp/index.jsp (diff)' is visible. At the bottom right, it says 'REST API' and 'Jenkins 2.504.3'.

Changes made was updated.

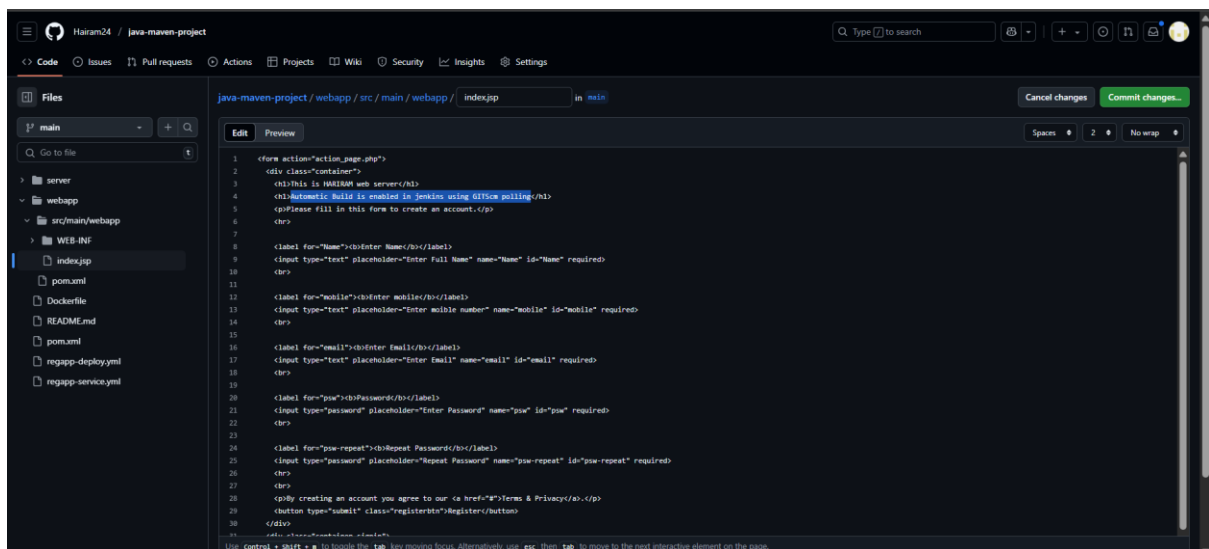
The screenshot shows a web browser window with the address '18.234.223.78080'. The page title is 'This is HARIRAM web server'. Below the title, it says 'Please fill in this form to create an account.' There is a form with the following fields: 'Enter Name' (with a placeholder 'Enter Full Name'), 'Enter mobile' (with a placeholder 'Enter mobile number'), 'Enter Email' (with a placeholder 'Enter Email'), 'Password' (with a placeholder 'Enter Password'), and 'Repeat Password' (with a placeholder 'Repeat Password'). Below the form, it says 'By creating an account you agree to our [Terms & Privacy](#).' There is a 'Register' button. Below the button, it says 'Already have an account? [Sign in](#).' At the bottom, it says 'Thank You, Happy Learning' and 'See You Again'.

Reflected in the web page.

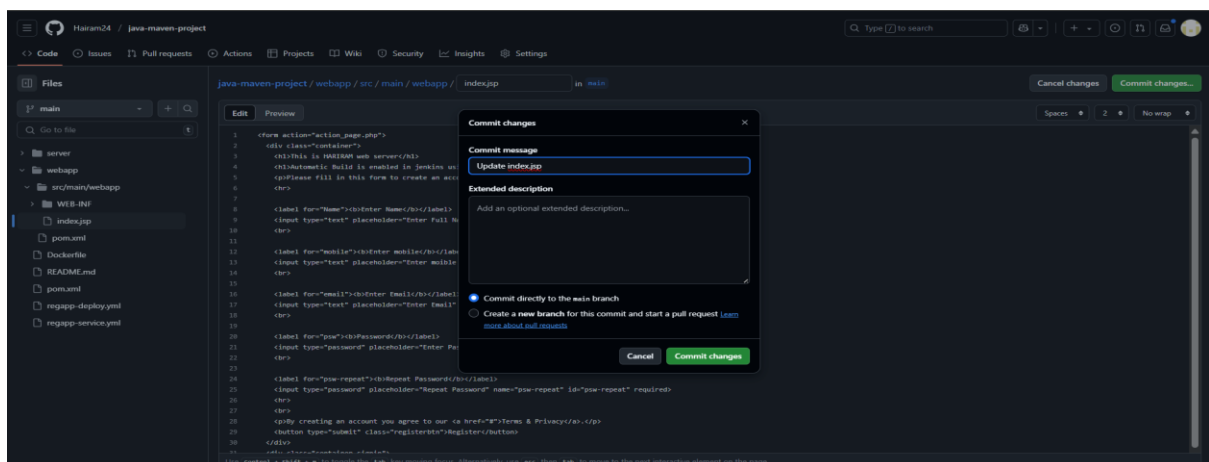
The screenshot shows the Jenkins 'Configuration' page for the 'java-maven-project'. The breadcrumb trail is 'Dashboard > java-maven-project > Configuration'. On the left, a sidebar contains links to 'Configure', 'General', 'Source Code Management', 'Triggers' (which is highlighted), 'Environment', 'Pre Steps', 'Build', 'Post Steps', 'Build Settings', and 'Post-build Actions'. The main content area is titled 'Configure' and has a sub-section 'Triggers'. It says 'Set up automated actions that start your build based on specific events, like code changes or scheduled times.' There are several checkboxes: 'Build whenever a SNAPSHOT dependency is built' (checked), 'Schedule build when some upstream has no successful builds', 'Trigger builds remotely (e.g., from scripts)', 'Build after other projects are built', 'Build periodically', 'GitHub Branches', 'GitHub Pull Requests', 'GitHub hook trigger for GITScm polling', and 'Poll SCM'. Below the 'Triggers' section, there is an 'Environment' section with the text 'Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.' It has checkboxes for 'Delete workspace before build starts', 'Use secret text(s) or file(s)', 'Add timestamps to the Console Output', and 'Inspect build log for published build scans'. At the bottom, there are 'Save' and 'Apply' buttons.



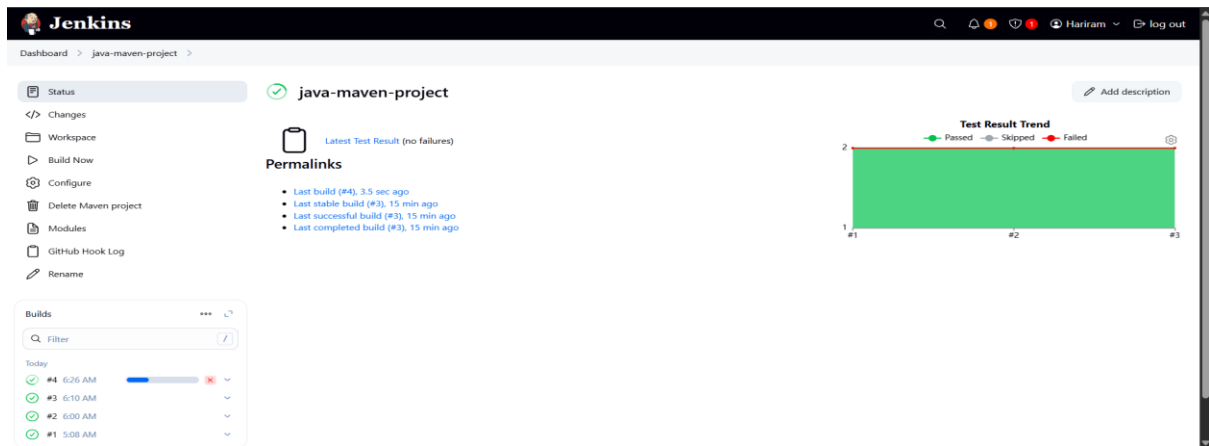
To automate build when code is changed, using GITscm polling.



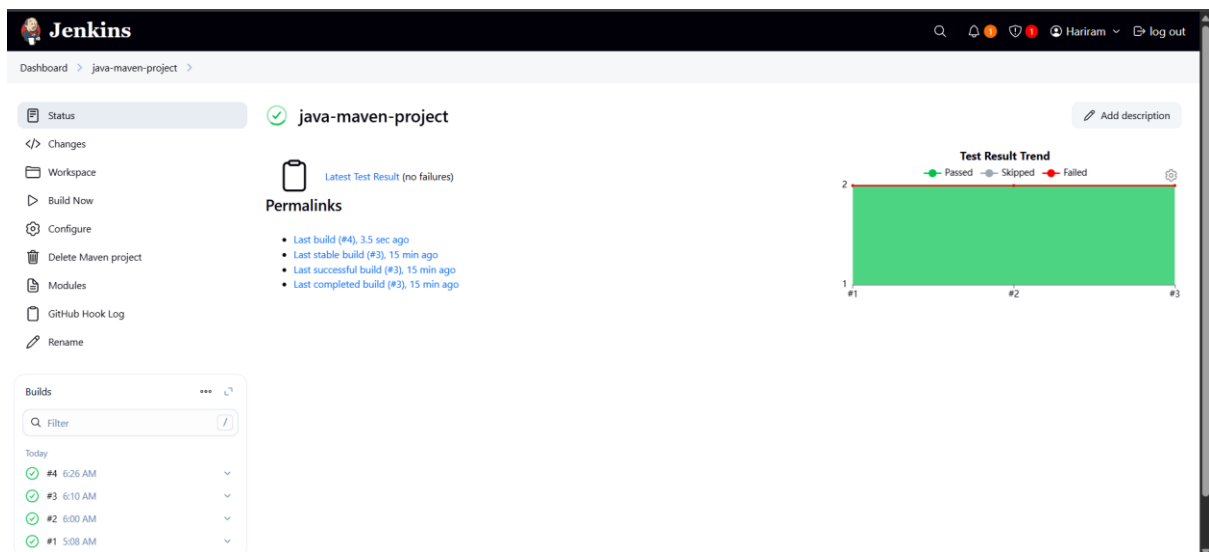
Made change in index code.



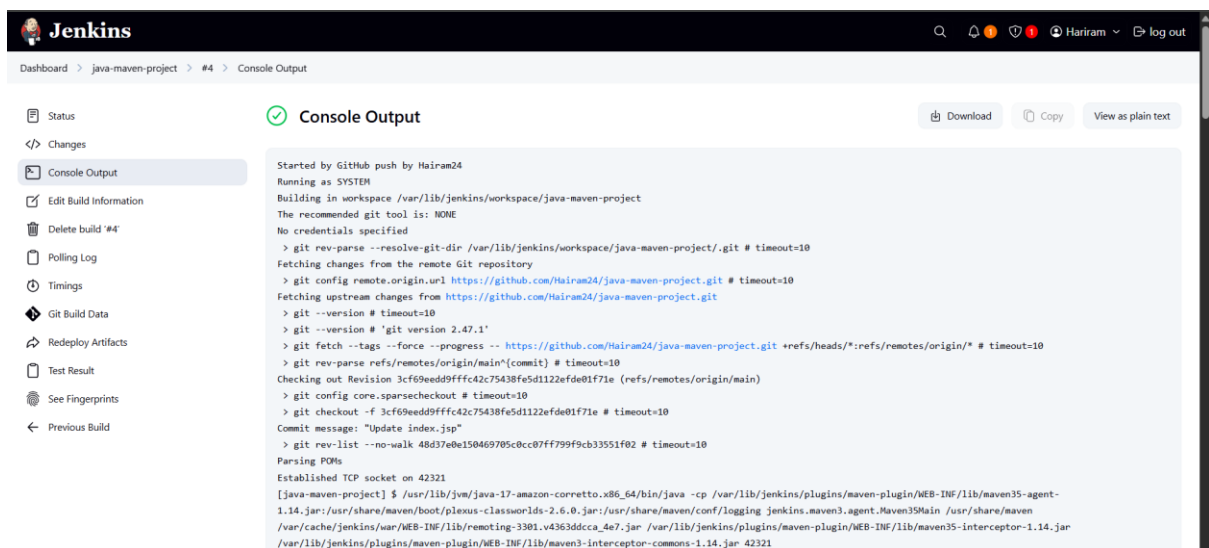
Commit the changes.



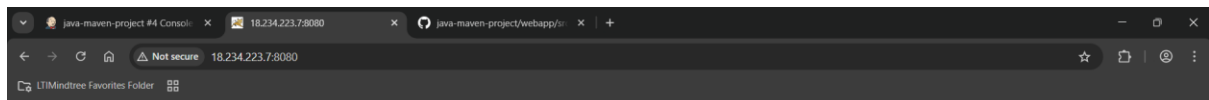
Automatically build is done.



Build successfully.



Console output.



This is HARIRAM web server

Automatic Build is enabled in jenkins using GITScm polling

Please fill in this form to create an account.

Enter Name

Enter mobile

Enter Email

Password

Repeat Password

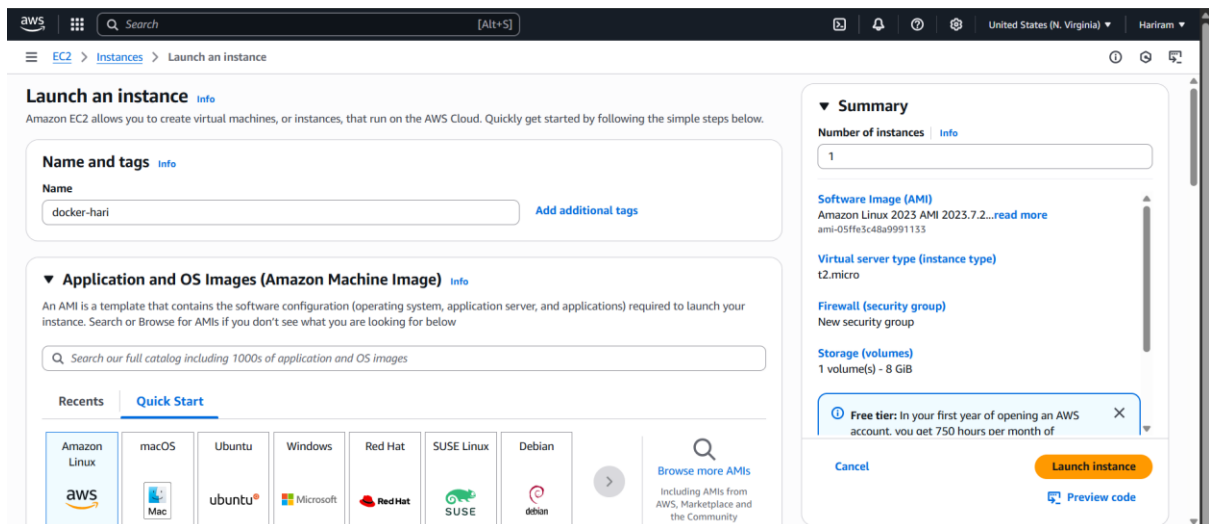
By creating an account you agree to our [Terms & Privacy](#).

Already have an account? [Sign in](#).

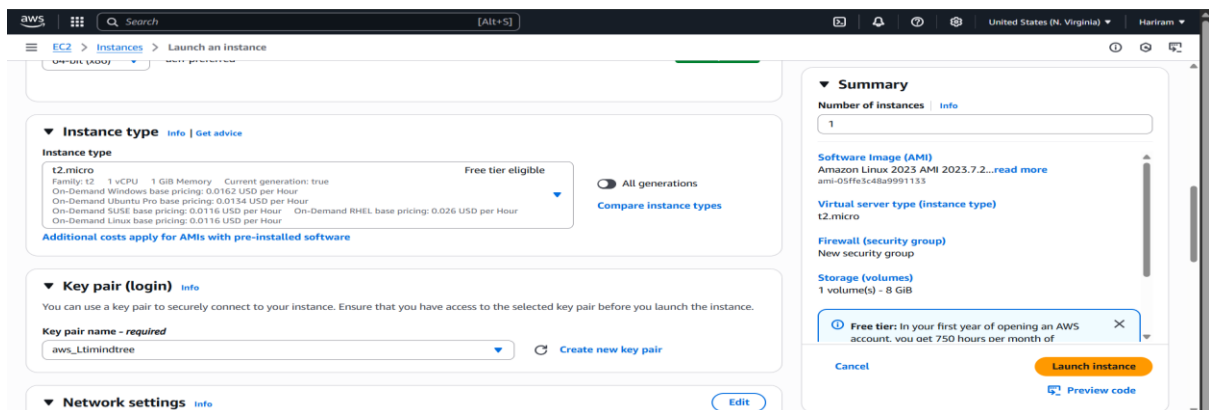
Thank You, Happy Learning

See You Again

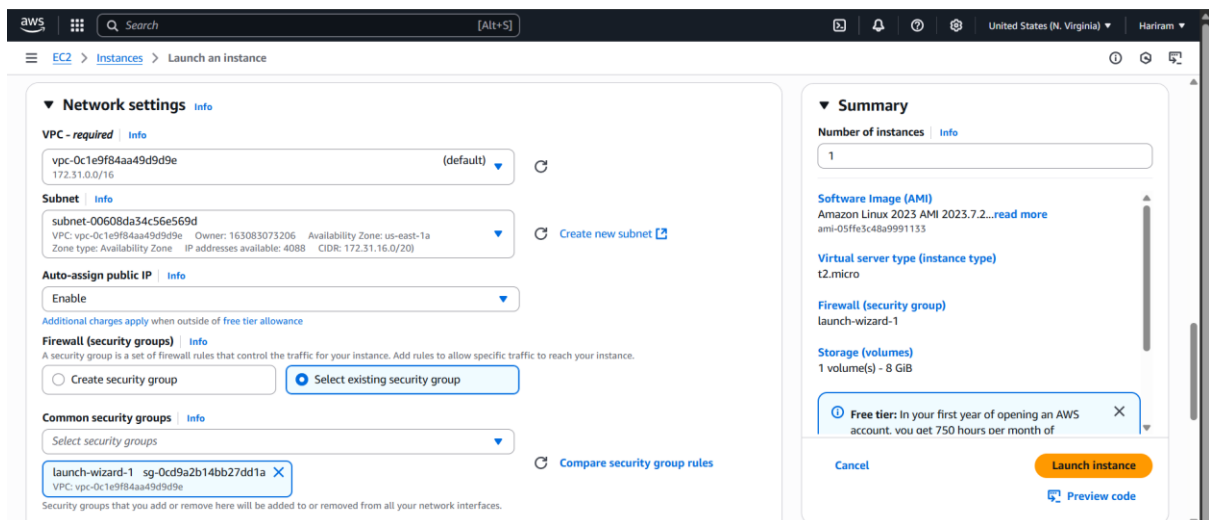
Reflected in we page.



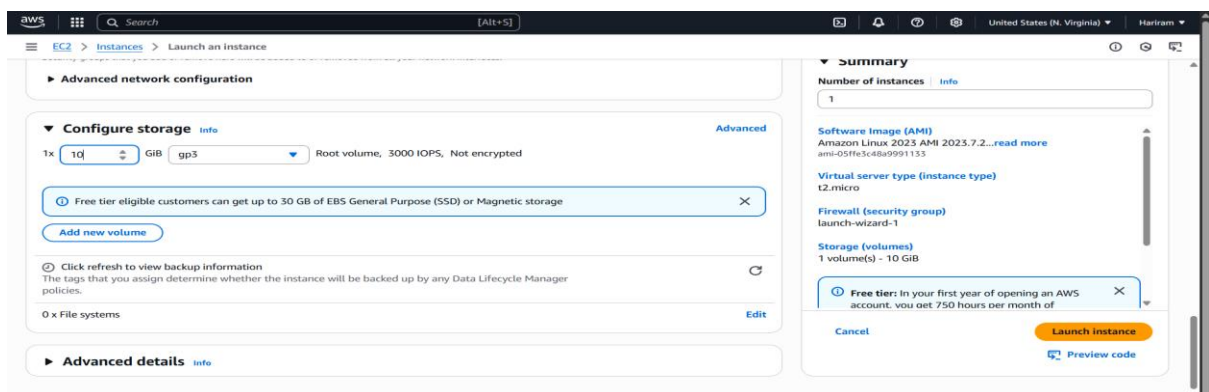
Docker machine is created with amazon linux.



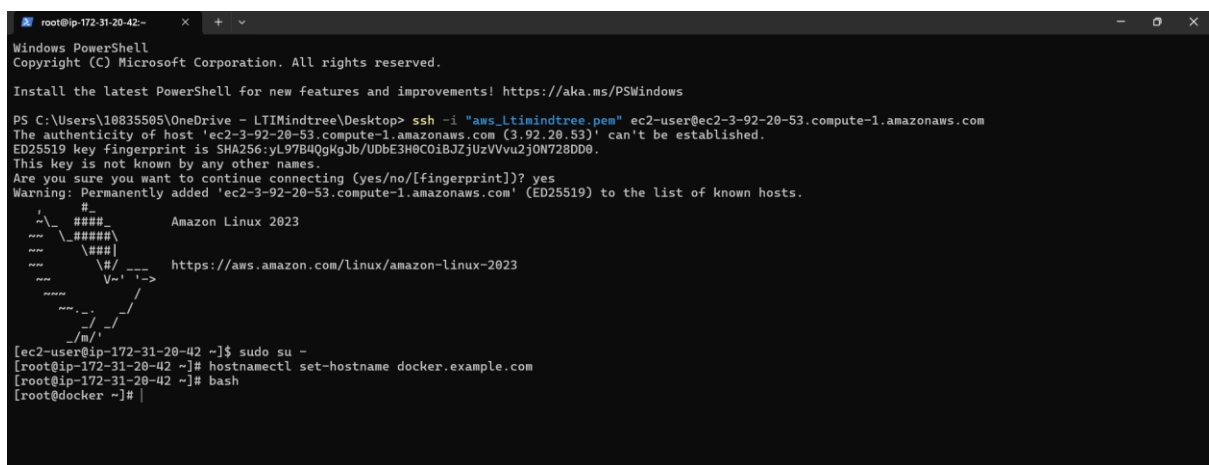
T2.micro is selected. Existing key pair is selected.



Existing security group is selected.



10 GB storage is selected.



Docker server connected and hostname is changed.

```
root@ip-172-31-20-42- [root@dockey ~]# yum install docker*
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Installing:
docker x86_64 25.0.8-1.amzn2023.0.4 amazonlinux 44 M
Installing dependencies:
containerd-selinux noarch 3.2.233-0-1.amzn2023 amazonlinux 55 k
containerd x86_64 2.0.5-1.amzn2023.0.1 amazonlinux 25 M
iptables-libs x86_64 1.8.8-3.amzn2023.0.2 amazonlinux 401 k
iptables-nft x86_64 1.8.8-2.amzn2023.0.2 amazonlinux 183 k
libcgroup x86_64 3.0-1.amzn2023.0.1 amazonlinux 75 k
libnetfilter_conntrack x86_64 1.0.8-2.amzn2023.0.2 amazonlinux 58 k
libnftnl x86_64 1.0.1-19.amzn2023.0.2 amazonlinux 39 k
libnftnl x86_64 1.2.2-2.amzn2023.0.2 amazonlinux 84 k
pigz x86_64 2.5-1.amzn2023.0.3 amazonlinux 83 k
runc x86_64 1.2.4-2.amzn2023.0.1 amazonlinux 3.4 M
Transaction Summary
=====
Install 11 Packages
Total download size: 74 M
Installed size: 200 M
Is this ok [y/N]: y
Downloading Packages:
(C1/11): container-selinux-2.233.0-1.amzn2023.noarch.rpm 495 kB/s | 55 kB 00:00
(C2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm 7.9 MB/s | 401 kB 00:00
(C3/11): iptables-nft-1.8.8-2.amzn2023.0.2.x86_64.rpm 3.5 MB/s | 183 kB 00:00
(C4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm 2.0 MB/s | 75 kB 00:00
(C5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm 1.4 MB/s | 58 kB 00:00
(C6/11): libnftnl-1.0.1-19.amzn2023.0.2.x86_64.rpm 782 kB/s | 39 kB 00:00
(C7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm 2.0 MB/s | 84 kB 00:00
(C8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm 2.1 MB/s | 83 kB 00:00
(C9/11): runc-1.2.4-2.amzn2023.0.1.x86_64.rpm 9.5 MB/s | 3.4 MB 00:00
(C10/11): containerd-2.0.5-1.amzn2023.0.1.x86_64.rpm 22 MB/s | 25 MB 00:01
(C11/11): docker-25.0.8-1.amzn2023.0.4.x86_64.rpm 25 MB/s | 44 MB 00:01
Total 42 MB/s | 74 MB 00:01
```

Docker package is installed.

```
root@jenkins- [root@jenkins ~]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:eqIEW/GnAcGb9VR+LbiCqFIHpQpSkqOQqAb1B2mezI root@jenkins.example.com
The key's randomart image is:
+--[RSA 3072]-----+
|o+o.o...|
|+o+... ..|
|o== o. o o|
|+Bo*.... o o|
|o *E+.o S o|
|o++ o o|
|o o o|
|.. o|
+-----[SHA256]-----+
[root@jenkins ~]#
```

SSH key is generated.

```
root@ip-172-31-20-42- [root@dockey ~]# ip a s
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group default qlen 1000
    link/ether 0a:ff:da:92:e3:a9 brd ff:ff:ff:ff:ff:ff
    altname eni-02dd003258b8ad7
    altname device-number-0 0
    inet 172.31.28.42/20 metric 512 brd 172.31.31.255 scope global dynamic enX0
        valid_lft 3392sec preferred_lft 3392sec
    inet6 fe80::8ff:daff:fe92:e3a9/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
[root@dockey ~]# passwd root
Changing password for user root.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@dockey ~]#
```

Root password is set.

```
root@dockey- #LogLevel INFO
# Authentication:
#LoginGraceTime 2m
#PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
#PubkeyAuthentication yes
# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile .ssh/authorized_keys
#AuthorizedPrincipalsFile none
# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes
# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no
# Change to no to disable s/key passwords
#ModuliInteractiveAuthentication yes
# Kerberos options
#KerberosAuthentication no
#KerberosDialectPaswd yes
#KerberosTicketCleanup yes
-- INSERT --
64, 27 36%
```

```
root@docker:~  
#LogLevel INFO  
  
# Authentication:  
#LoginGraceTime 2m  
#PermitRootLogin yes  
#StrictModes yes  
#MaxAuthTries 6  
#MaxSessions 10  
  
#PubkeyAuthentication yes  
  
# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2  
# but this is overridden so installations will only check .ssh/authorized_keys  
AuthorizedKeysFile .ssh/authorized_keys  
  
#AuthorizedPrincipalsFile none  
  
# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts  
#HostbasedAuthentication no  
# Change to yes if you don't trust ~/.ssh/known_hosts for  
# HostbasedAuthentication  
#IgnoreUserKnownHosts no  
# Don't read the user's ~/.rhosts and ~/.shosts files  
#IgnoreRhosts yes  
  
# Explicitly disable PasswordAuthentication. By presetting it, we  
# avoid the cloud-init set_passwords module modifying sshd_config and  
# restarting sshd in the default instance launch configuration.  
PasswordAuthentication yes  
PermitEmptyPasswords no  
  
# Change to no to disable s/key passwords  
#KbdInteractiveAuthentication yes  
  
# Kerberos options  
#KerberosAuthentication no  
#KerberosOrLocalPasswd yes  
#KerberosTicketCleanup yes  
-- INSERT --
```

```
root@docker:~  
[root@docker ~]# vim /etc/ssh/sshd_config  
[root@docker ~]# systemctl restart sshd  
[root@docker ~]# systemctl enable sshd  
[root@docker ~]# |
```

Sshd configuration is edited and sshd is restarted.

```
[root@jenkins ~]# ssh-copy-id root@172.31.20.42  
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"  
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed  
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys  
root@172.31.20.42's password:  
  
Number of key(s) added: 1  
  
Now try logging into the machine, with: "ssh 'root@172.31.20.42'"  
and check to make sure that only the key(s) you wanted were added.  
  
[root@jenkins ~]# |
```

Ssh public key is added to Jenkins

```
root@jenkins:~  
#Logging  
#SyslogFacility AUTH  
#LogLevel INFO  
  
# Authentication:  
#LoginGraceTime 2m  
#PermitRootLogin yes  
#StrictModes yes  
#MaxAuthTries 6  
#MaxSessions 10  
  
#PubkeyAuthentication yes  
  
# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2  
# but this is overridden so installations will only check .ssh/authorized_keys  
AuthorizedKeysFile .ssh/authorized_keys  
  
#AuthorizedPrincipalsFile none  
  
# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts  
#HostbasedAuthentication no  
# Change to yes if you don't trust ~/.ssh/known_hosts for  
# HostbasedAuthentication  
#IgnoreUserKnownHosts no  
# Don't read the user's ~/.rhosts and ~/.shosts files  
#IgnoreRhosts yes  
  
# Explicitly disable PasswordAuthentication. By presetting it, we  
# avoid the cloud-init set_passwords module modifying sshd_config and  
# restarting sshd in the default instance launch configuration.  
PasswordAuthentication yes  
PermitEmptyPasswords no  
  
# Change to no to disable s/key passwords  
#KbdInteractiveAuthentication yes  
  
# Kerberos options  
-- INSERT --
```

```
root@jenkins:~  
[root@jenkins ~]# vim /etc/ssh/sshd_config  
[root@jenkins ~]# systemctl start sshd  
[root@jenkins ~]# systemctl enable sshd  
[root@jenkins ~]# |
```

Jenkins sshd config is edited.

Dashboard > Manage Jenkins > System >

SSH Servers

SSH Server

Name ?

jenkins

Hostname ?

172.31.25.209

Username ?

root

Remote Directory ?

/

☐ Avoid sending files that have not changed ?

Advanced ^

Edited

☒ Use password authentication, or use a different key ?

Passphrase / Password ?

....

Save

Apply

Dashboard > Manage Jenkins > System >

Username ?

root

Remote Directory ?

/

☐ Avoid sending files that have not changed ?

Advanced ^

Edited

☒ Use password authentication, or use a different key ?

Passphrase / Password ?

....

Path to key ?

Key ?

Save

Apply

Jenkins ssh server is added.

Dashboard > Manage Jenkins > System >

Proxy password

Success

Test Configuration

Add

Advanced v

GitHub Pull Requests

Published Jenkins URL

☐ Actualise local repo on factory creation

Save

Apply

Dashboard > Manage Jenkins > System

SSH Server

Name ?

docker

Hostname ?

172.31.20.42

Username ?

root

Remote Directory ?

/

☐ Avoid sending files that have not changed ?

Advanced ^

Edited

☒ Use passphrase authentication, or use a different key ?

Passphrase / Password ?

Path to key ?

Key ?

Save

Apply

Dashboard > Manage Jenkins > System

Proxy user ?

Proxy password

Success

Test Configuration

Add

Advanced v

GitHub Pull Requests

Published Jenkins URL

☐ Actualise local repo on factory creation

Save

Apply

Docker ssh server is added.

Dashboard > java-maven-project > Configuration

Configure

General

Source Code Management

Triggers

Environment

Pre Steps

Build

Post Steps

Build Settings

Post-build Actions

SSH Server

Name ?

jenkins

Advanced v

Transfers

Transfer Set

Source files ?

Remove prefix ?

Remote directory ?

Exec command ?

rsync -avh /var/lib/jenkins/workspace/java-maven-project root@172.31.25.209:/opt

Save

Apply

Jenkins post build is added with exec commands.

aws [Search] [Alt+S] United States (N. Virginia) Hariram

Amazon ECR > Private registry > Repositories > Create private repository

Create private repository

General settings

Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.
163083073206.dkr.ecr.us-east-1.amazonaws.com/my-ecr-hari

11 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters _-./.

Image tag mutability
Choose the tag mutability setting.

☒ **Mutable**
Image tags can be overwritten.

☐ **Immutable**
Image tags can't be overwritten.

Encryption settings [Info](#)

The encryption settings for a repository can't be changed once the repository is created.

Encryption configuration
By default, repositories use the industry standard Advanced Encryption Standard (AES) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the images in your repository.

☒ **AES-256**
Industry standard Advanced Encryption Standard (AES) encryption

Create a ECR.

aws [Search] United States (N. Virginia) Hariram

Amazon ECR > Private registry > Repositories

Amazon Elastic Container Registry

- Private registry
 - Repositories
 - Summary
 - Images
 - Permissions
 - Lifecycle Policy
 - Repository tags
 - Features & Settings
- Public registry
 - Repositories
 - Settings

ECR public gallery [AWS](#)

Amazon ECS [AWS](#)

Amazon EKS [AWS](#)

Push commands for my-ecr-hari

[macOS / Linux](#) Windows

Make sure that you have the latest version of the AWS CLI and Docker installed. For more information, see [Getting Started with Amazon ECR](#).

Use the following steps to authenticate and push an image to your repository. For additional registry authentication methods, including the Amazon ECR credential helper, see [Registry Authentication](#).

- Retrieve an authentication token and authenticate your Docker client to your registry. Use the AWS CLI:
`aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 163083073206.dkr.ecr.us-east-1.amazonaws.com`
Note: If you receive an error using the AWS CLI, make sure that you have the latest version of the AWS CLI and Docker installed.
- Build your Docker image using the following command. For information on building a Docker file from scratch see the instructions [here](#). You can skip this step if your image is already built:
`docker build -t my-ecr-hari .`
- After the build completes, tag your image so you can push the image to this repository:
`docker tag my-ecr-hari:latest 163083073206.dkr.ecr.us-east-1.amazonaws.com/my-ecr-hari:latest`
- Run the following command to push this image to your newly created AWS repository:
`docker push 163083073206.dkr.ecr.us-east-1.amazonaws.com/my-ecr-hari:latest`

[Close](#)

Dashboard > java-maven-project > Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings
- Post-build Actions

Name [?](#)

docker

Advanced [?](#)

Transfers

Transfer Set

Source files [?](#)

Remove prefix [?](#)

Remote directory [?](#)

Exec command [?](#)

```
cd /opt
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 163083073206.dkr.ecr.us-east-1.amazonaws.com
docker build -t my-ecr-hari
docker tag my-ecr-hari:latest 163083073206.dkr.ecr.us-east-1.amazonaws.com/my-ecr-hari:latest
docker push 163083073206.dkr.ecr.us-east-1.amazonaws.com/my-ecr-hari:latest
```

[Save](#) [Apply](#)

```
root@tomcat-  x  +  -
#LoginGraceTime 2m
#PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes


# Kerberos options
-- INSERT --
```

60,18 37%

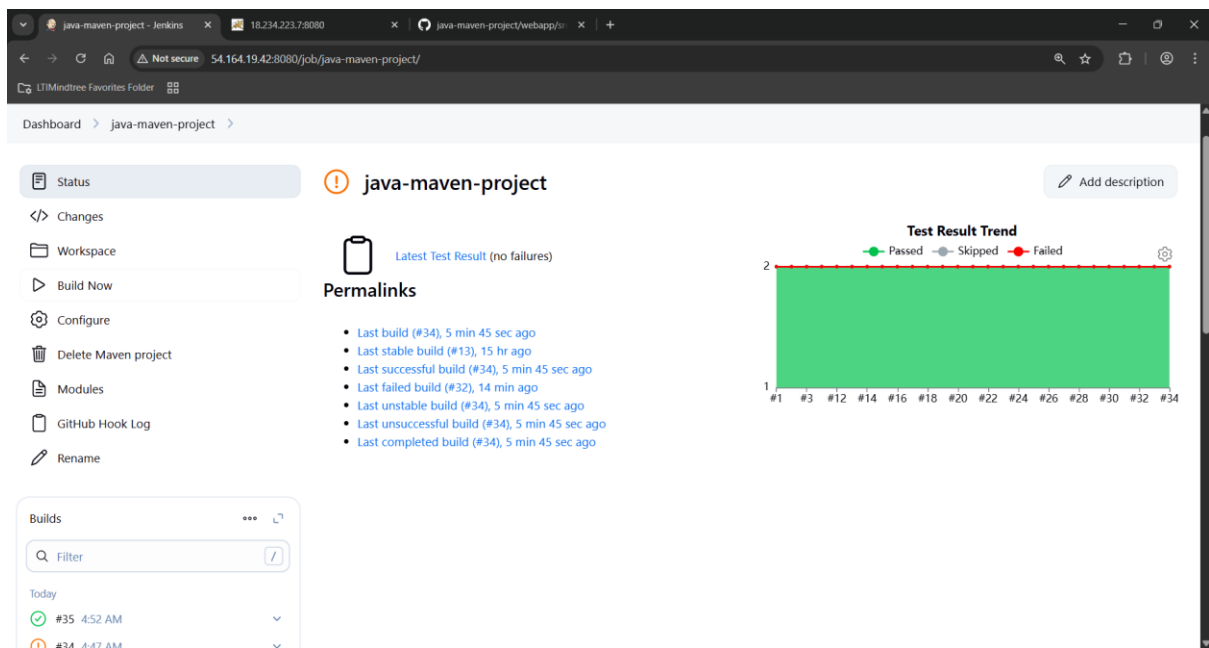
```
root@tomcat-  x  +  -
[root@tomcat ~]# vim /etc/ssh/sshd_config
[root@tomcat ~]# systemctl restart sshd
[root@tomcat ~]# systemctl enable sshd
```

```
root@tomcat:~/ssh
no-port-forwarding,no-agent-forwarding,no-X11-forwarding,command="echo 'Please login as the user \"ec2-user\" rather than the user \"root\".';echo;sleep 10;exit 142" ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDowCDZw93pIkZ5Zyr4FM50jcgHTQnJdrf8gRXklMU5mZUBLAqJABcR5LsyZodN+NiAX9Yee4zd3f+0meqX4SzXTxzDRNHGTS0m7FbdB5Z4dw/IUIjsRcSOICVPvJQmKqB6FcrmlMaa3yguwvt2pBWJcggfvD0rIyisqKOGwztIUoHuDWCrlag8Lh+gXBLZx5FRF4mpglg67MRNP22dzE+hd5TiBe2DKQvSjU38FiidFRVm5X+EkA0VnqJBxVjqUspolzK/z3W25cVLqh0/NrBQRdn+0J8YNPtYzo2Np5ghhKv3ie4z0tQcNv8Aa5ZWjFLptqmcIvg9NzvdM25QDE/ aws_Ltimindtree

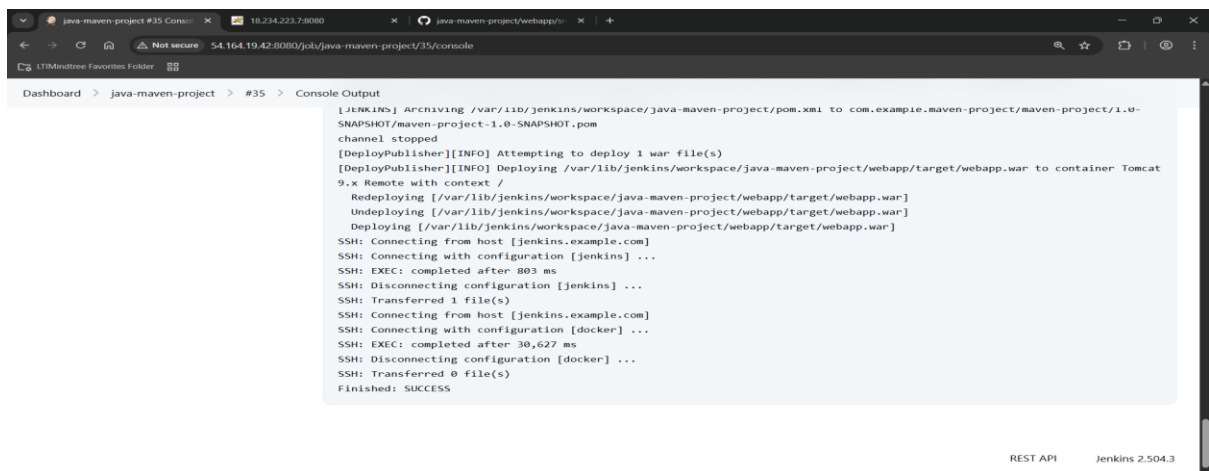
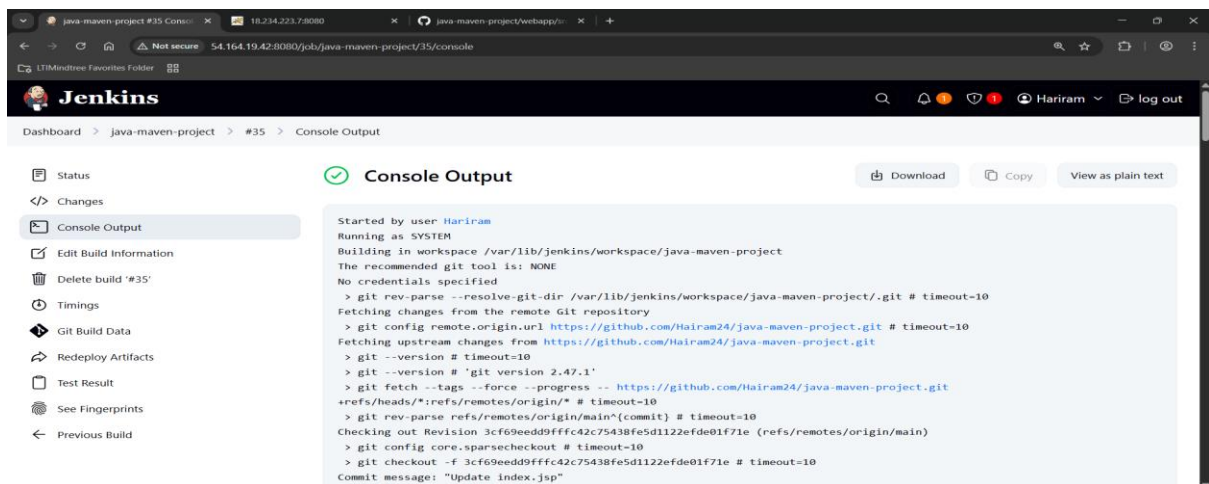
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDQC5Y0IjG0TByF5rAD0TAHB4U4WmJmloJctqnyUx2ua6oVcQRxJkVwXkhBbgMt5oVznfgwR8sXSGs1l5y800hdLAhdUcTLaWnV5pabz35wakqaT96hEYOLVF5gjbPvRMcFoW77YVZVN79U5zTs4+IxorA/EU8+0T81CeP8rbFnCdLNUGbxQBk1x7i7qAPUZTbU/02XxhkFhfQ3eRU+ZApip0JzDLTuf6pqaamfWGSfpL1BY4334qthfZnbA846srtLR5zdBdW5B7U5P27MA5qh/nJkmSfv8pPHQIOWAXPDKFG2fPUTCgR1jQhZ6tuzCBBFJLbfCgfbAn4izGd4ygvN0qXS9axUxpnJbYpoI+y08cNhpvfIuXXLIjbtKf+g7aHODGTevhQ4a0jWuQJ3fLsbom/yis0ChncRFM7h1bUPnUzTL+HJRHCzy9cwb4V5fv5n+S7PNEU18bj4YneqNgzf5WYfgiz9JG1Zfe9Tn/hkkmB6n3LRSJPJ5VDqkWywNWeU= root@jenkins.example.com

-- INSERT --
```

Jenkins pub key is added to the tomcat authorized_keys.



Build successful.



Console output build successfully.

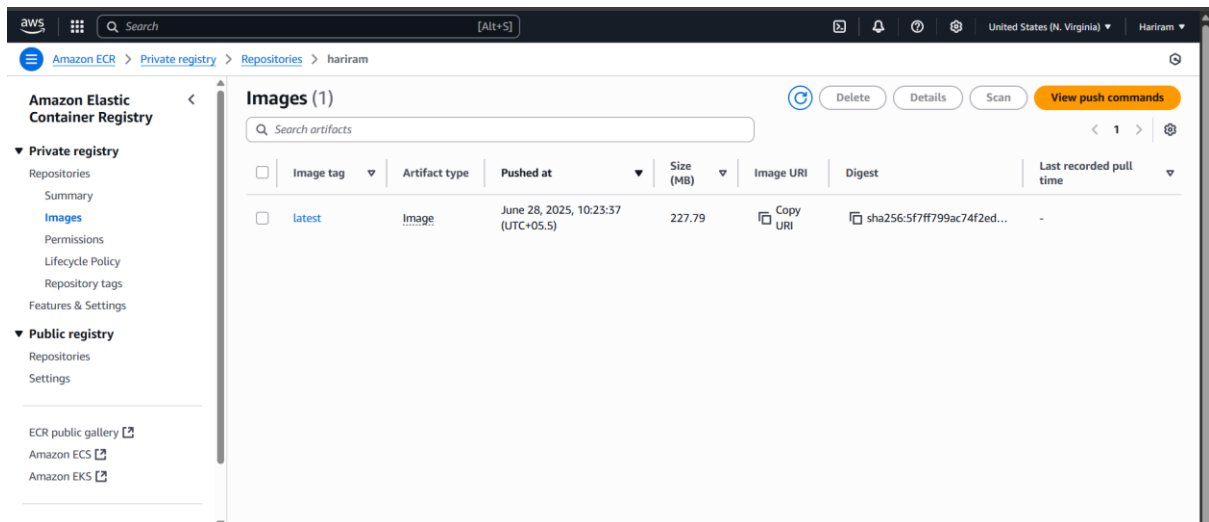
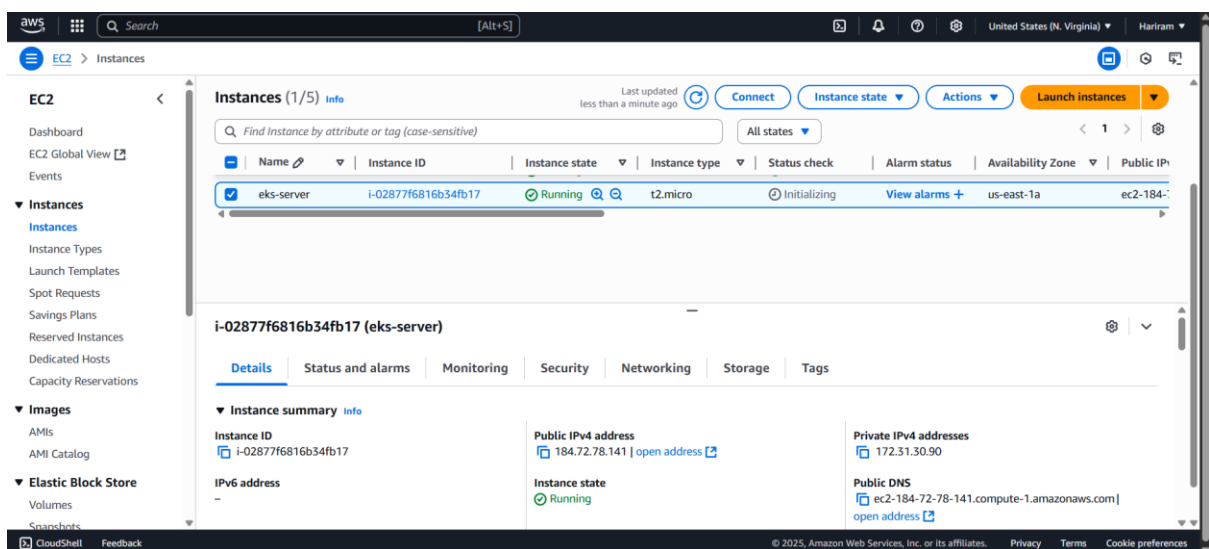
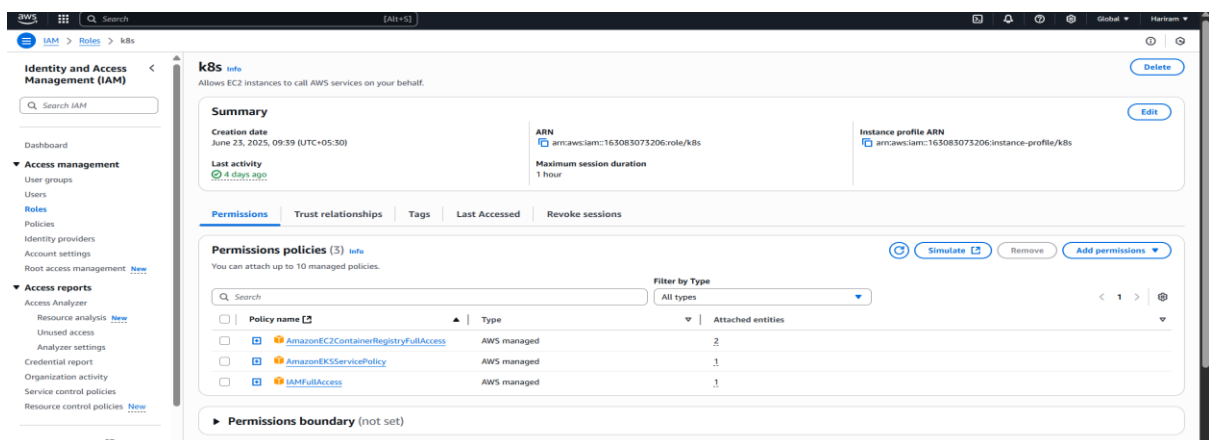


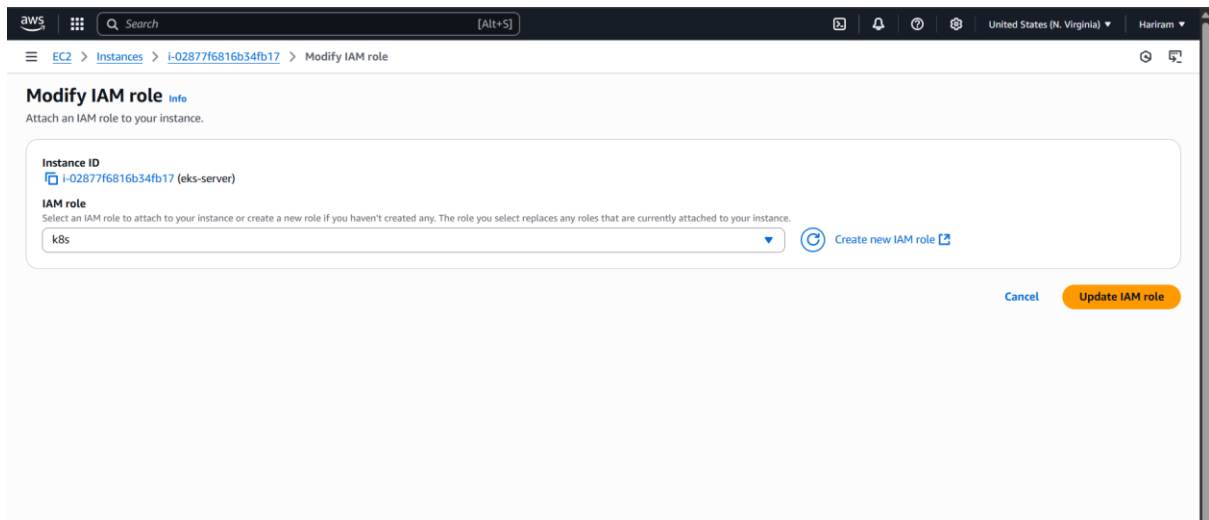
Image is created in ECR.



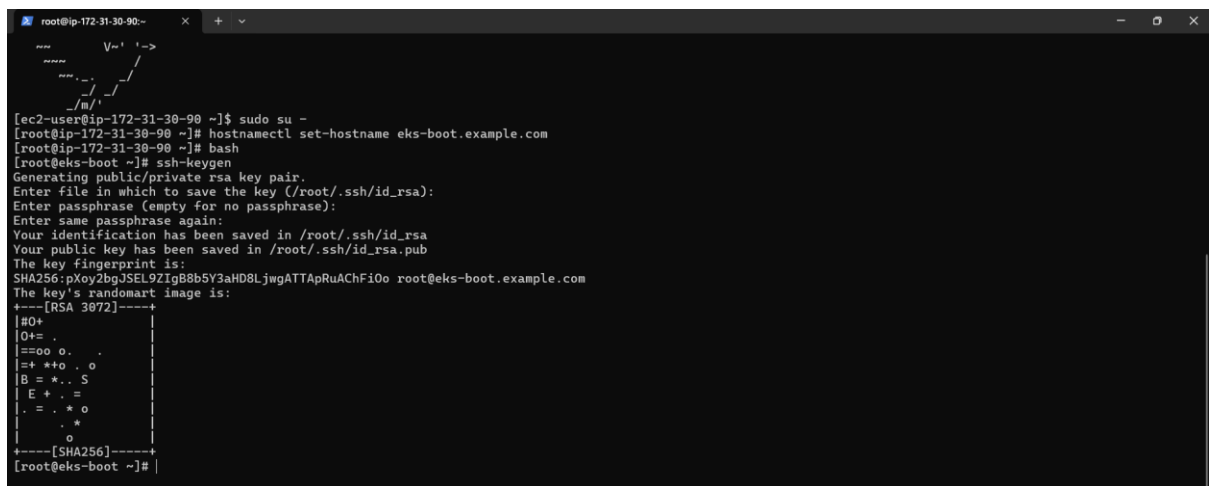
EKS server is created with amazon linux and existing security group k8s-sg with all k8s necessary ports allowed.



Roles is created with necessary permissions.



Roles is added to the eks-server.



EKS-server is started. Hostname is changed. Ssh key is generated.



Eks-server pub key is added to Jenkins.

```
root@ip-172-31-30-90:~/.ssh x + v
no-port-forwarding,no-agent-forwarding,no-X11-forwarding,command="echo 'Please login as the user \"ec2-user\" rather than the user \"root\".';echo;sleep 10;
exit 142" ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDowCD2w93pIkZ5Zyr4FM50jcgHTQnJdrf8gRXklMU5mZUBLAqJABcR5lsyzodN+N1iAX9Yee4zd3f+0meqX452xTxzDRNKGt50m7FbdBSZ4d
w/IUJsRcSOICVPVjQwKqB6FcmIMaa3yguwvt2pBMJcggfvd0r1yisqK0GwztIUoHUDWCrLag8Lh+gXBLZx5FRF4mpg1g67WRNP22dzE+hD5T1Be2DKQvSjU38F11DcFRVn5X+EKA0VnqJ8xVjqUspolZK/z
3W25cVLqH0/NrBQrdn+0J8YNptYzo2Np5ghhKV3ie4z0tQcNY8Aa5ZwJF1ptqmc1vg9NzvdM25QDE/ aws_Ltimindtree

ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDQC5oIJGoTByF5rAD0TAHB44WmJmloJctqnyUx2ua6oVcQRxJkVwXkhBbgNt5oVZnfgwR8sXSGs115yB00hdLAhdUcTLaWnV5pabz35wakqaT96hEYOLVFS
gjbPVrMcfoW77YZVN79UsZts4+Ixora/EUB+0T81CeP8rbFnCdLNUGbXQ8k1x7i7qAPUzTbU/02XXhKFhfQ3eRU+ZAip0JzDLTuf6pqaamFWGSpPPL1BY4334qthfZnbA846stRL5zd58DwSB7U5P27MA5q
h/nJkmSfv8pPHQIOWAXPDKFG2fPUTCgRiJqMz6tuzCB8FJlbfCgF8An4izGd4ygVn8qX59axUxpnJbYpoI+y08cNHpvfiuXXLIjbtKf+g7aHODGTeVhQ4a0jWuQJ3F1sbom/yis0ChncRFM7h1b1UPnUZtRL
+HJRHcy9cwB4V5vfSn+S7PNEU18Bj4YneqNgzf5WYfgiz9JGizFe9Tn/hkkmB6n3LRSJP3J5VDqkWywNWeU= root@jenkins.example.com

-- INSERT --
```

Jenkins pub key is added to eks-server.

```
root@ip-172-31-30-90:~ x + v
#HostKey /etc/ssh/ssh_host_ed25519_key
# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile .ssh/authorized_keys

#AuthorizedPrincipalsFile none

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

-- INSERT --
```

```
root@ip-172-31-30-90:~ x + v
[root@eks-boot ~]# vim /etc/ssh/sshd_config
[root@eks-boot ~]# systemctl restart sshd
[root@eks-boot ~]# systemctl enable sshd
```

Eks-server sshd config is edited.

```
root@ip-172-31-30-90:~# yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
root@eks-boot ~# curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 63.1M 100 63.1M    0     0 68.1M    0 --:--:-- --:--:-- --:--:-- 68.0M
root@eks-boot ~# unzip awscliv2.zip
Archive:  awscliv2.zip
  creating: aws/
  inflating: aws/dist/
  inflating: aws/install
  inflating: aws/README.md
  inflating: aws/THIRD_PARTY_LICENSES
  creating: aws/dist/awscli/
  creating: aws/dist/docutils/
  creating: aws/dist/lib-dynload/
  inflating: aws/dist/aws
  inflating: aws/dist/aws_completer
  inflating: aws/dist/libpython3.11.so.1.0
  inflating: aws/dist/_awscli.abi3.so
  inflating: aws/dist/_ruamel_yaml.cpython-311-x86_64-linux-gnu.so
  inflating: aws/dist/libz.so.1
  inflating: aws/dist/liblzma.so.5
  inflating: aws/dist/libbz2.so.1
  inflating: aws/dist/libbrotli.so.1
  inflating: aws/dist/libuuid.so.1
  inflating: aws/dist/libreadline.so.8
  inflating: aws/dist/libtinfo.so.6
  inflating: aws/dist/libsqlite3.so.0
  inflating: aws/dist/base_library.zip
  inflating: aws/dist/lib-dynload/_datetime.cpython-311-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_unicodedata.cpython-311-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_csv.cpython-311-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_statistics.cpython-311-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_contextvars.cpython-311-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_decimal.cpython-311-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_pickle.cpython-311-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_hashlib.cpython-311-x86_64-linux-gnu.so
```

Amazon Cli package is installed.

```
root@ip-172-31-30-90:~#
  inflating: aws/dist/awscli/botocore/data/auditmanager/2017-07-25/endpoint-rule-set-1.json
  inflating: aws/dist/awscli/botocore/data/auditmanager/2017-07-25/paginators-1.json
  creating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/paginators-1.json
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/endpoint-rule-set-1.json
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/service-2.json
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/completions-1.json
  creating: aws/dist/awscli/botocore/.changes/next-release/
  inflating: aws/dist/awscli/botocore/.changes/next-release/api-change-connect-59117.json
  inflating: aws/dist/awscli/topics/topic-tags.json
  inflating: aws/dist/awscli/topics/config-vars.rst
  inflating: aws/dist/awscli/topics/s3-config.rst
  inflating: aws/dist/awscli/topics/db-expressions.rst
  inflating: aws/dist/awscli/topics/return-codes.rst
  inflating: aws/dist/awscli/topics/s3-faq.rst
  inflating: aws/dist/awscli/data/cli.json
  inflating: aws/dist/awscli/data/metadata.json
  inflating: aws/dist/awscli/data/ac.index
  creating: aws/dist/awscli/customizations/sso/
  creating: aws/dist/awscli/customizations/wizard/
  creating: aws/dist/awscli/customizations/wizard/wizards/
  creating: aws/dist/awscli/customizations/wizard/wizards/configure/
  creating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/
  creating: aws/dist/awscli/customizations/wizard/wizards/events/
  creating: aws/dist/awscli/customizations/wizard/wizards/iam/
  creating: aws/dist/awscli/customizations/wizard/wizards/lambda/
  inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
  inflating: aws/dist/awscli/customizations/sso/index.html
root@eks-boot ~# sudo ./aws/install
You can now run: /usr/local/bin/aws --version
root@eks-boot ~# aws configure
AWS Access Key ID [None]: AKIASLGECG23OZQMXIHS
AWS Secret Access Key [None]: FUGHLJLLWkiIxnkHYD7yWtwXPMVkh4CDN2DgO7G8Y
Default region name [None]: us-east-1
Default output format [None]: table
root@eks-boot ~#
```

Aws configuration credentials are given from the IAM user.

```
root@ip-172-31-30-90:~#
  inflating: aws/dist/awscli/topics/db-expressions.rst
  inflating: aws/dist/awscli/topics/return-codes.rst
  inflating: aws/dist/awscli/topics/s3-faq.rst
  inflating: aws/dist/awscli/data/cli.json
  inflating: aws/dist/awscli/data/metadata.json
  inflating: aws/dist/awscli/data/ac.index
  creating: aws/dist/awscli/customizations/sso/
  creating: aws/dist/awscli/customizations/wizard/
  creating: aws/dist/awscli/customizations/wizard/wizards/
  creating: aws/dist/awscli/customizations/wizard/wizards/configure/
  creating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/
  creating: aws/dist/awscli/customizations/wizard/wizards/events/
  creating: aws/dist/awscli/customizations/wizard/wizards/iam/
  creating: aws/dist/awscli/customizations/wizard/wizards/lambda/
  inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
  inflating: aws/dist/awscli/customizations/sso/index.html
root@eks-boot ~# sudo ./aws/install
You can now run: /usr/local/bin/aws --version
root@eks-boot ~# aws configure
AWS Access Key ID [None]: AKIASLGECG23OZQMXIHS
AWS Secret Access Key [None]: FUGHLJLLWkiIxnkHYD7yWtwXPMVkh4CDN2DgO7G8Y
Default region name [None]: us-east-1
Default output format [None]: table
root@eks-boot ~# curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_${uname -s}_amd64.tar.gz" | tar xz -C /tmp
root@eks-boot ~# mv /tmp/eksctl /usr/local/bin
root@eks-boot ~# eksctl version
0.210.0
root@eks-boot ~# curl -LO https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-release/release/stable.txt)/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 53.7M 100 53.7M    0     0 85.7M    0 --:--:-- --:--:-- --:--:-- 85.8M
root@eks-boot ~# sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
root@eks-boot ~# kubectl version --client
Client Version: v1.31.0
Kustomize Version: v5.4.2
root@eks-boot ~#
```

Kubernetes package is installed.


```
root@ip-172-31-30-90/opt
[root@eks-boot ~]# eksctl create cluster --name milestone-2 --region us-east-1 --version 1.32 --vpc-public-subnets subnet-00608da34c56e569d,subnet-0541b7ff15bd92fd9 --without-nodesgroup
2025-06-28 10:09:31 [i] eksctl version 0.210.0
2025-06-28 10:09:31 [i] using region us-east-1
2025-06-28 10:09:31 [i] using existing VPC (vpc-c1e9f84aa49d9d9e) and subnets (private:map[] public:map[us-east-1a:[subnet-00608da34c56e569d us-east-1a 172.31.16.0/20 0] us-east-1b:[subnet-0541b7ff15bd92fd9 us-east-1b 172.31.32.0/20 0]])
2025-06-28 10:09:32 [i] custom VPC/subnets will be used; if resulting cluster doesn't function as expected, make sure to review the configuration of VPC/subnets
2025-06-28 10:09:32 [i] using Kubernetes version 1.32
2025-06-28 10:09:32 [i] creating EKS cluster "milestone-2" in "us-east-1" region with
2025-06-28 10:09:32 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=milestone-2'
2025-06-28 10:09:32 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "milestone-2" in "us-east-1"
2025-06-28 10:09:32 [i] CloudWatch logging will not be enabled for cluster "milestone-2" in "us-east-1"
2025-06-28 10:09:32 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=milestone-2'
2025-06-28 10:09:32 [i] default add-ons coredns, metrics-server, vpc-cni, kube-proxy were not specified, will install them as EKS add-ons
2025-06-28 10:09:32 [i]
2 sequential tasks: { create cluster control plane "milestone-2",
  2 sequential sub-tasks: {
    1 task: { create add-ons },
    wait for control plane to become ready,
  }
}
2025-06-28 10:09:32 [i] building cluster stack "eksctl-milestone-2-cluster"
2025-06-28 10:09:32 [i] deploying stack "eksctl-milestone-2-cluster"
2025-06-28 10:09:32 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:09:32 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:11:32 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:12:32 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:13:32 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:14:33 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:15:33 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:16:33 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:16:33 [i] creating add-on: coredns
2025-06-28 10:16:34 [i] successfully created add-on: coredns
2025-06-28 10:16:34 [i] creating add-on: metrics-server
2025-06-28 10:16:35 [i] successfully created add-on: metrics-server
2025-06-28 10:16:35 [i] recommended policies were found for "vpc-cni" add-on, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" add-on is via pod identity associations; after add-on creation is completed, add all recommended policies to the config file, under 'addon.PodIdentityAssociations', and run 'eksctl update add-on'
2025-06-28 10:16:35 [i] creating add-on: vpc-cni
2025-06-28 10:16:36 [i] successfully created add-on: vpc-cni
2025-06-28 10:16:36 [i] creating add-on: kube-proxy
2025-06-28 10:16:36 [i] successfully created add-on: kube-proxy
2025-06-28 10:16:36 [i] waiting for the control plane to become ready
2025-06-28 10:18:37 [i] saved kubeconfig as "/root/.kube/config"
2025-06-28 10:18:37 [i] no tasks
2025-06-28 10:18:37 [i] all EKS cluster resources for "milestone-2" have been created
2025-06-28 10:18:38 [i] kubectctl command should work with "/root/.kube/config", try 'kubectctl get nodes'
2025-06-28 10:18:38 [i] EKS cluster "milestone-2" in "us-east-1" region is ready
```

Cluster is created.

```
root@ip-172-31-30-90/opt
2025-06-28 10:09:32 [i] deploying stack "eksctl-milestone-2-cluster"
2025-06-28 10:10:02 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:10:32 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:11:32 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:12:32 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:13:32 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:14:33 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:15:33 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:16:33 [i] waiting for CloudFormation stack "eksctl-milestone-2-cluster"
2025-06-28 10:16:33 [i] creating add-on: coredns
2025-06-28 10:16:34 [i] successfully created add-on: coredns
2025-06-28 10:16:34 [i] creating add-on: metrics-server
2025-06-28 10:16:35 [i] successfully created add-on: metrics-server
2025-06-28 10:16:35 [i] recommended policies were found for "vpc-cni" add-on, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" add-on is via pod identity associations; after add-on creation is completed, add all recommended policies to the config file, under 'addon.PodIdentityAssociations', and run 'eksctl update add-on'
2025-06-28 10:16:35 [i] creating add-on: vpc-cni
2025-06-28 10:16:35 [i] successfully created add-on: vpc-cni
2025-06-28 10:16:36 [i] creating add-on: kube-proxy
2025-06-28 10:16:36 [i] successfully created add-on: kube-proxy
2025-06-28 10:16:36 [i] waiting for the control plane to become ready
2025-06-28 10:18:37 [i] saved kubeconfig as "/root/.kube/config"
2025-06-28 10:18:37 [i] no tasks
2025-06-28 10:18:37 [i] all EKS cluster resources for "milestone-2" have been created
2025-06-28 10:18:38 [i] kubectctl command should work with "/root/.kube/config", try 'kubectctl get nodes'
2025-06-28 10:18:38 [i] EKS cluster "milestone-2" in "us-east-1" region is ready
[root@eks-boot ~]# kubectctl get pods
No resources found in default namespace.
[root@eks-boot ~]# cd /opt/
[root@eks-boot opt]# vim regapp-deploy.yml
"regapp-deploy.yml" [New] 29L, 530B written
[root@eks-boot opt]# vim regapp-service.yml
[root@eks-boot opt]# ll
total 8
drwxr-xr-x.  # root root 33 Jun 20 06:56 aws
-rw-r--r--. 1 root root 530 Jun 28 10:20 regapp-deploy.yml
-rw-r--r--. 1 root root 196 Jun 28 10:20 regapp-service.yml
[root@eks-boot opt]#
```

Deployment and service yaml file is added in /opt.

Dashboard > Manage Jenkins > System >

SSH Server

Name ?

eks-boot-strap

Hostname ?

172.31.30.90

Username ?

root

Remote Directory ?

/

☐ Avoid sending files that have not changed ?

Advanced ^

Edited

☒ Use password authentication, or use a different key ?

Passphrase / Password ?

Save

Apply

Dashboard > Manage Jenkins > System >

Timeout (ms) ?
300000

☐ Disable exec ?

Proxy type ?
▼

Proxy host ?

Proxy port ?

Proxy user ?

Proxy password

Success

Test Configuration

Save Apply

Eks-server ssh is added to Jenkins system.

```
Windows PowerShell
[root@eks-boot ~]#
[root@eks-boot ~]# eksctl create nodegroup \
--cluster milestone-2 \
--region us-east-1 \
--name my-node-group \
--node-ami-family Ubuntu2204 \
--node-type t2.small \
--subnet-ids subnet-00608da34c56e569d,subnet-0541b7ff15bd92fd9 \
--nodes 3 \
--nodes-min 2 \
--nodes-max 4 \
--ssh-access \
--ssh-public-key /root/.ssh/id_rsa.pub
2025-06-28 18:46:46 [I] will use version 1.32 for new nodegroup(s) based on control plane version
2025-06-28 18:46:47 [I] nodegroup "my-node-group" will use "ami-887657d77f589280e" (Ubuntu2204/1.32)
2025-06-28 18:46:47 [I] using SSH public key "/root/.ssh/id_rsa.pub" as "eksctl-milestone-2-nodegroup-my-node-group-36:b4:21:bc:2f:62:53:d8:a7:6c:bc:52:25:21:86"
2025-06-28 18:46:47 [I] 1 nodegroup (my-node-group) was included (based on the include/exclude rules)
2025-06-28 18:46:47 [I] will create a CloudFormation stack for each of 1 managed nodegroups in cluster "milestone-2"
2025-06-28 18:46:47 [I]
2 sequential tasks: { fix cluster compatibility, 1 task: { 1 task: { create managed nodegroup "my-node-group" } } }
2025-06-28 18:46:47 [I] checking cluster stack for missing resources
2025-06-28 18:46:47 [I] cluster stack has all required resources
2025-06-28 18:46:48 [I] building managed nodegroup stack "eksctl-milestone-2-nodegroup-my-node-group"
2025-06-28 18:46:48 [I] deploying stack "eksctl-milestone-2-nodegroup-my-node-group"
2025-06-28 18:46:48 [I] waiting for CloudFormation stack "eksctl-milestone-2-nodegroup-my-node-group"
2025-06-28 18:47:18 [I] waiting for CloudFormation stack "eksctl-milestone-2-nodegroup-my-node-group"
2025-06-28 18:49:23 [I] waiting for CloudFormation stack "eksctl-milestone-2-nodegroup-my-node-group"
2025-06-28 18:50:38 [I] waiting for CloudFormation stack "eksctl-milestone-2-nodegroup-my-node-group"
2025-06-28 18:51:37 [I] waiting for CloudFormation stack "eksctl-milestone-2-nodegroup-my-node-group"
2025-06-28 18:51:37 [I] no tasks
2025-06-28 18:51:37 [I] created 0 nodegroup(s) in cluster "milestone-2"
2025-06-28 18:51:37 [I] nodegroup "my-node-group" has 3 node(s)
2025-06-28 18:51:37 [I] node "ip-172-31-28-156.ec2.internal" is ready
2025-06-28 18:51:37 [I] node "ip-172-31-31-165.ec2.internal" is ready
2025-06-28 18:51:37 [I] node "ip-172-31-34-46.ec2.internal" is ready
2025-06-28 18:51:37 [I] waiting for at least 2 node(s) to become ready in "my-node-group"
2025-06-28 18:51:37 [I] node "ip-172-31-28-156.ec2.internal" is ready
2025-06-28 18:51:37 [I] node "ip-172-31-31-165.ec2.internal" is ready
2025-06-28 18:51:37 [I] node "ip-172-31-34-46.ec2.internal" is ready
2025-06-28 18:51:37 [I] created 1 managed nodegroup(s) in cluster "milestone-2"
2025-06-28 18:51:38 [I] checking security group configuration for all nodegroups
2025-06-28 18:51:38 [I] all nodegroups have up-to-date cloudformation templates
```

Node group is created.

Dashboard > java-maven-project > Configuration

Configure

General

Source Code Management

Triggers

Environment

Pre Steps

Build

Post Steps

Build Settings

Post-build Actions

SSH Server

Name ?
eks-boot-strap

Advanced ▼

Transfers

Transfer Set

Source files ?

Remove prefix ?

Remote directory ?
/opt

Exec command ?
cd /opt
kubectl delete -f regapp-deploy.yml
kubectl apply -f regapp-deploy.yml
kubectl apply -f regapp-service.yml

Save Apply

Post build eks-server is added with exec commands.

Jenkins

Dashboard

>

java-maven-project

>

#42

Status

Changes

Console Output

Edit Build Information

Delete build #42

Polling Log

Timings

Git Build Data

Redeploy Artifacts

Test Result

See Fingerprints

Previous Build

#42 (Jun 28, 2025, 12:46:03 PM)

Add description

Keep this build forever

Started by GitHub push by Hairam24

Started 29 sec ago
Took 16 sec

This run spent:

- 7.5 sec waiting
- 16 sec build duration
- 24 sec total from scheduled to completion.

Revision: d57ab07121c0c4b11603c40d26987bd350b002c

Repository: <https://github.com/Hairam24/java-maven-project.git>

- refs/remotes/origin/main

Test Result (no failures)

Module Builds

Maven Project

0.88 sec

Server

2.5 sec

Webapp

1.3 sec

REST APIJenkins 2.504.3

Jenkins

Dashboard

>

java-maven-project

>

#42

>

Console Output

Status

Changes

Console Output

Edit Build Information

Delete build #42

Polling Log

Timings

Git Build Data

Redeploy Artifacts

Test Result

See Fingerprints

Previous Build

Console Output

Download

Copy

View as plain text

Started by GitHub push by Hairam24

Running as SYSTEM

Building in workspace /var/lib/jenkins/workspace/java-maven-project

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/java-maven-project/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url https://github.com/Hairam24/java-maven-project.git # timeout=10

Fetching upstream changes from https://github.com/Hairam24/java-maven-project.git

> git --version # timeout=30

> git --version # 'git version 2.47.1'

> git fetch --tags --force --progress -- https://github.com/Hairam24/java-maven-project.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/main (commit) # timeout=10

Checking out Revision d57ab07121c0c4b11603c40d26987bd350b002c (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f d57ab07121c0c4b11603c40d26987bd350b002c # timeout=10

Commit message: "Update index.jsp"

> git rev-list --no-walk 28ee1145328b5ac8930fc2844276ad4e90215 # timeout=10

Parsing POMs

Established TCP socket on 33041

[java-maven-project] \$ /usr/lib/jvm/java-17-amazon-corretto.x86_64/bin/java -cp /var/lib/jenkins/plugins/maven-plugin/WEB-INF/lib/maven35-agent-1.14.jar:/usr/share/maven/boot/plexus-classworlds-2.6.0.jar:/usr/share/maven/conf/logging-jenkins.maven3.agent.Maven35Main:/usr/share/maven/var/cache/jenkins/war/WEB-INF/lib/remoting-3801.v4363ddcca_4e7.jar:/var/lib/jenkins/plugins/maven-plugin/WEB-INF/lib/maven35-interceptor-1.14.jar:/var/lib/jenkins/plugins/maven-plugin/WEB-INF/lib/maven35-interceptor-commons-1.14.jar 33041

<--[JENKINS REMOTING CAPACITY]-->channel started

Executing Maven: -B -f /var/lib/jenkins/workspace/java-maven-project/pom.xml install

[INFO] Scanning for projects...

[WARNING]

[WARNING] Some problems were encountered while building the effective model for com.example.maven-project:server:jar:1.0-SNAPSHOT

Jenkins

Dashboard

>

java-maven-project

>

#42

>

Console Output

Status

Changes

Console Output

Edit Build Information

Delete build #42

Polling Log

Timings

Git Build Data

Redeploy Artifacts

Test Result

See Fingerprints

Previous Build

Console Output

Download

Copy

View as plain text

[INFO] Finished at: 2025-06-28T12:46:12Z

[INFO] -----

Waiting for Jenkins to finish collecting data

[JENKINS] Archiving /var/lib/jenkins/workspace/java-maven-project/webapp/pom.xml to com.example.maven-project:webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.pom

[JENKINS] Archiving /var/lib/jenkins/workspace/java-maven-project/webapp/target/webapp.war to com.example.maven-project:webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.war

[JENKINS] Archiving /var/lib/jenkins/workspace/java-maven-project/server/pom.xml to com.example.maven-project:server/1.0-SNAPSHOT/server-1.0-SNAPSHOT.pom

[JENKINS] Archiving /var/lib/jenkins/workspace/java-maven-project/server/target/server.jar to com.example.maven-project:server/1.0-SNAPSHOT/server-1.0-SNAPSHOT.jar

[JENKINS] Archiving /var/lib/jenkins/workspace/java-maven-project/pom.xml to com.example.maven-project:maven-project/1.0-SNAPSHOT/maven-project-1.0-SNAPSHOT.pom

channel stopped

[DeployPublisher][INFO] Attempting to deploy 1 war file(s)

[DeployPublisher][INFO] Deploying /var/lib/jenkins/workspace/java-maven-project/webapp/target/webapp.war to container Tomcat 9.x Remote with context /

Redeploying [/var/lib/jenkins/workspace/java-maven-project/webapp/target/webapp.war]

Undeploying [/var/lib/jenkins/workspace/java-maven-project/webapp/target/webapp.war]

Deploying [/var/lib/jenkins/workspace/java-maven-project/webapp/target/webapp.war]

SSH: Connecting from host [jenkins.example.com]

SSH: Connecting with configuration [jenkins] ...

SSH: EXEC: completed after 885 ms

SSH: Disconnecting configuration [jenkins] ...

SSH: Transferred 1 file(s)

SSH: Connecting from host [jenkins.example.com]

SSH: Connecting with configuration [docker] ...

SSH: EXEC: completed after 3,203 ms

SSH: Disconnecting configuration [docker] ...

SSH: Transferred 0 file(s)

SSH: Connecting from host [jenkins.example.com]

SSH: Connecting with configuration [eks-boot-strap] ...

SSH: EXEC: completed after 2,603 ms

SSH: Disconnecting configuration [eks-boot-strap] ...

SSH: Transferred 0 file(s)

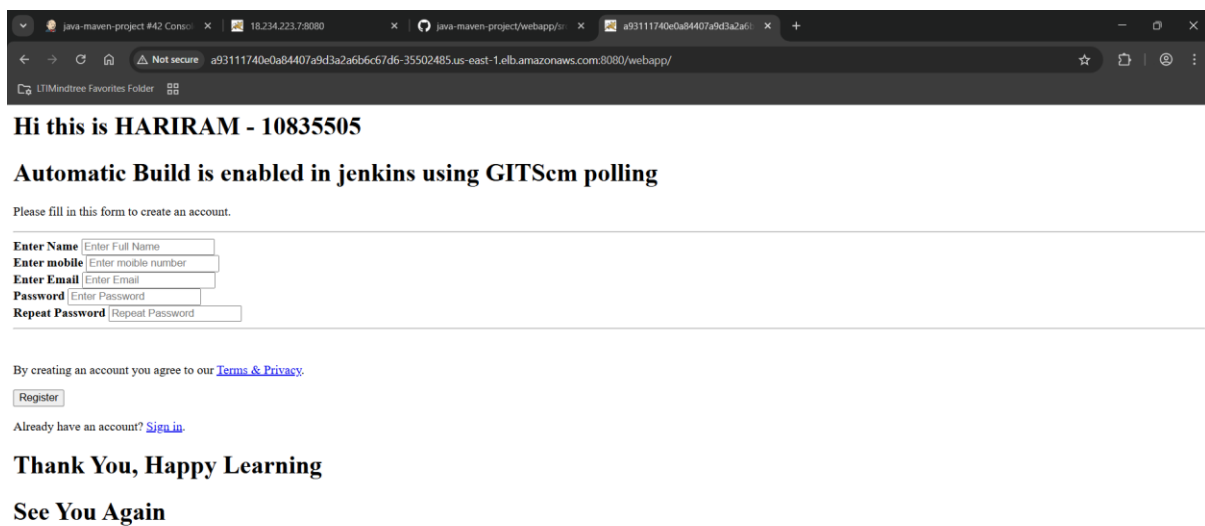
Finished: SUCCESS

REST APIJenkins 2.504.3

Build successfully.

```
root@eks-boot-~# kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
regapp-deployment-5fc55cbd8d-mhmk1 1/1     Running   0           38m
regapp-deployment-5fc55cbd8d-ssxtk 1/1     Running   0           38m
root@eks-boot-~# kubectl get svc
NAME      TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes ClusterIP   10.100.0.1       <none>            443/TCP          51m
regapp-service LoadBalancer 10.100.125.236  a93111740e0a84407a9d3a2a6b6c67d6-35502485.us-east-1.elb.amazonaws.com 8080:31288/TCP 38m
root@eks-boot-~# |
```

Two pods are running after the build successful.



Final output is displayed in server ip:8080/webapp .